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Effets d'une mutation sur la valeur sélective d'une bactérie

Estimation théorique et implémentation numérique

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Mots clés : estimation nonparamétrique, problèmes inverses

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EFFETS D'UNE MUTATION SUR LA VALEUR SÉLECTIVE D'UNE BACTÉRIE
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Résumé

French version

Mots clés : estimation nonparamétrique, problèmes inverses

Abstract

English version

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Tous les organismes subissent des mutations. Une mutation est une altération spontanée ou provoquée du support de l'information génétique. C'est un mécanisme qui joue un rôle clé dans l'histoire de la vie ; en particulier, il explique l'apparition de nouveaux caractères héréditaires au sein d'une population.

Ces nouveaux caractères peuvent modifier la *valeur sélective* d'un individu. La valeur sélective, appelée aussi *fitness*, est un concept fondamental en biologie évolutive. Introduit par Darwin en 1859 dans son ouvrage *On the Origin of Species* [5], ce terme désigne la capacité d'un individu ayant un certain génome à survivre et à se reproduire.

On distingue trois types de mutations :

- Certaines sont délétères voir létales, c'est-à-dire qu'elles réduisent le développement de l'individu ou le tuent.
- Certaines sont neutres et n'induisent aucun effet sur le développement.
- Certaines sont bénéfiques et favorisent le développement de l'individu.

Les effets d'une mutation sur la valeur sélective d'un individu sont plus ou moins marqués, ce qui forme ainsi un continuum d'effets.

La manière dont ces mutations affectent la valeur sélective d'un individu est une question centrale en biologie évolutive. En particulier, les biologistes s'intéressent à la densité de la répartition de ces effets (*DFE*, *Density Fitness Effect*).

Connaître la forme de la DFE est une question essentielle. Dans un article publié en 2007 [7], Eyre-Walker et al. donnent plusieurs exemples dans le but d'insister sur l'importance de son étude.

- Dans l'espèce humaine, le génome d'un individu possède plus d'une centaine de nouvelles mutations que l'on ne retrouve pas chez ses parents. L'étude de la DFE permet d'étudier leurs effets afin de savoir s'ils sont bénéfiques ou non.
- La DFE permet de comprendre et de quantifier la diversité génétique que l'on retrouve dans les maladies humaines ainsi que son évolution future.
- La DFE permet de prévoir les conséquences du maintien à petite taille d'une population d'animaux ou de plantes, comme dans les programmes d'élevage en captivité.

L'étude de la DFE est un passage obligé pour comprendre et prévoir la trajectoire évolutive d'une population et répondre à plusieurs questions qui restent encore en suspens.

Cependant, il est difficile d'obtenir une estimation de la DFE [2]. En particulier, jusqu'à présent les moyens techniques ne permettaient pas d'observer précisément l'apparition d'une mutation dans le génome.

Le point de départ de ce projet est un article de 2018 écrit par Lydia Robert et al. [19]. Dans cet article, les auteur.ices présentent un nouveau protocole expérimental qui permet de suivre en temps réel l'apparition de nouvelles mutations chez *E. coli*.

Cette nouvelle méthode expérimentale permet d'obtenir de nouvelles données afin d'estimer la DFE. En effet, il est possible grâce à ces données de savoir immédiatement quand une mutation apparaît. D'autres expériences permettent de regarder l'évolution de la valeur sélective

d'une cellule au cours du temps.

Dès lors, plusieurs méthodes sont possibles pour tenter d'estimation de la DFE à partir de ces observations expérimentales. Cependant il s'agit d'un problème inverse sévèrement mal posé.

Dans ce mémoire, nous proposons d'étudier une question particulière d'estimation :

Inférer la DFE à partir de mesures expérimentales de la valeur sélective au cours du temps au sein d'une lignée cellulaire.

Pour répondre à cette question, nous devons prendre en compte que les données expérimentales sont extrêmement bruitées. Cette question rejoint une problématique plus large qui a de nombreuses autres applications possibles.

Chapter 1

Introduction

1.1 Motivation

1.2 Elements of literature

Chapter 2

Description of the biological question

For over a century, researchers have tried to understand the impact of mutations on an individual's selective value. However, this understanding remained limited, in particular because experimental techniques did not allow the appearance of new mutations to be observed directly, or their effects to be measured precisely.

In a 2018 article, Robert and al. [19] propose to overcome this difficulty by describing a new experimental method that makes it possible to see the appearance of new mutations in real time. In the same article, they explain second experimental method for estimating the selective value of a cell line over time. These new techniques provide new data for understanding more precisely the impact of mutations on an individual's selective value.

In this section, we present the experimental techniques presented in [19] as well as the available data.

2.1 Detect mutations : *Mutation Visualisation experiment*

2.1.1 The Mismatch Repair

Deoxyribonucleic acid (DNA) is a double-helix molecule that carries genetic information. It is an extremely precise piece of machinery, which means that the slightest disturbance to one of its strands is likely to alter the behaviour, and therefore the ability, of the cell to survive.

To avoid such events, living cells have developed complex repair systems capable of recognising and correcting these disturbances. When these systems correctly eliminate the disturbance, there are no consequences for the cell. On the other hand, when the repair system fails to eliminate the disturbance, a mutation occurs.

2.1.1.1 DNA replication

During DNA replication, the two strands are separated by an enzyme: *DNA Polymerase*. These strands (matrices) enable two new identical DNA helices to be reconstructed by base

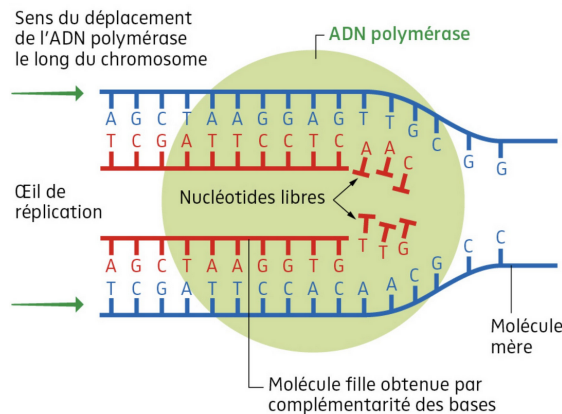


Figure 2.1 – Simplified representation of a replication fork. Note that the new strands are complementary to the old strands.

[\[Faire une image en anglais\]](#)

complementarity. The new strand is determined by the old strand: an A base on the mother strand will ensure a T base on the new strand, etc. (Figure ??).

Sometimes a replication error occurs. In this case, one of the bases of the new strand is not complementary to its associated base. This is known as a mismatch. (Figure ??).

2.1.1.2 Mismatch detection and repair

The DNA Mismatch repair (MMR) is monitoring mechanism used during replication [9]. It detects and repairs mismatches committed by the DNA polymerase during the replication process. The MMR system best known today is that of *Escherichia coli*, known as the MutHLS system, comprising the MutH, MutL and MutS proteins.

The repair system. The MutS protein recognises the mismatch and the MutL protein confirms the mismatch, then both bind to the DNA. The accumulation of MutS and MutL proteins activates the MutH protein, which cuts the neosynthesised strand. Resynthesis is then possible. (Figure ??)

2.1.1.3 The emergence of a mutation

A mutation occurs when the mismatch repair system fails to repair a mismatch. In this case, the newly formed DNA has two mismatched bases. During replication, the DNA polymerase will form two new strands, one of which will carry a mutation compared to the original genome.

2.1.2 Real-time mutation detection

In [19], the authors develop a new strategy to observe the appearance of new mutations in real time.

To begin with, they add a fluorescent protein (YFP) to the MutL protein. As the MutL proteins accumulate around the mismatch sites, it is possible to see the appearance of a new mutation using imaging techniques.

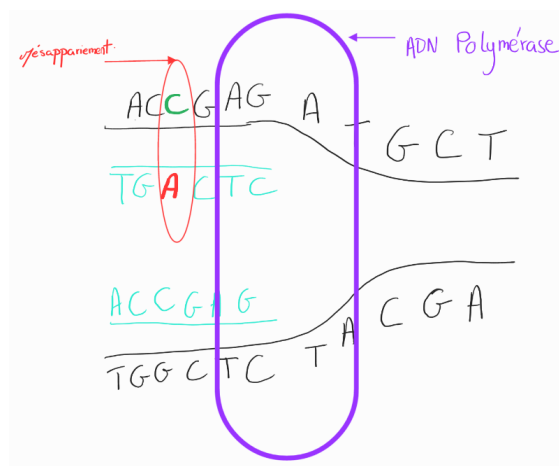


Figure 2.2 – Artistic representation of a mismatch. A base G has been transformed into a base A.

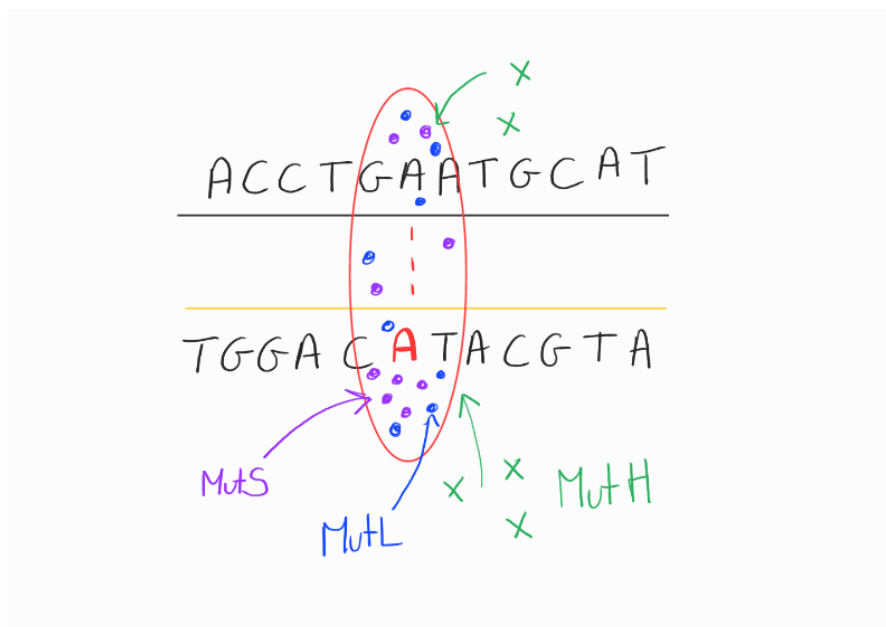


Figure 2.3 – Schematic representation of the action of the mutHLS system.

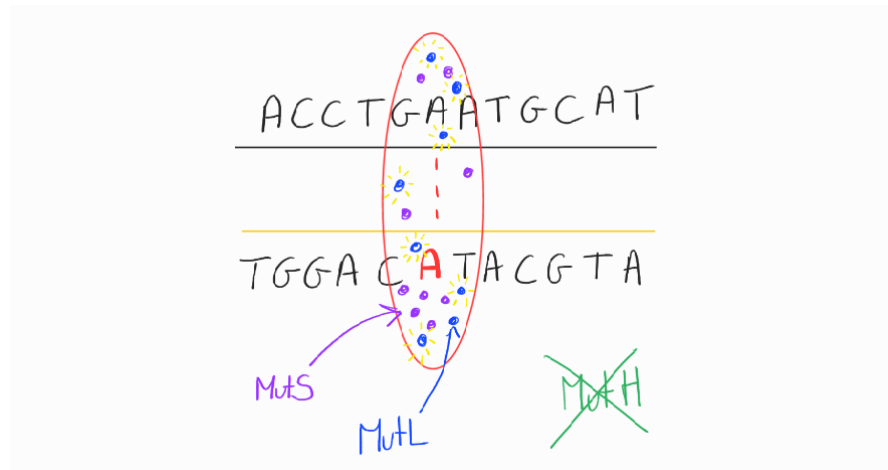


Figure 2.4 – Schematic representation of the MV experiment

In addition, the MutH endonuclease is deactivated to ensure that the mismatch is not repaired (Figure ??).

2.1.3 Discussion

Using the results obtained in [19], the authors argue that the emergence of new mutations follows a Poisson process.

2.2 Measuring fitness: *microfluidic Mutation Accumulation (μ MA) experiment*

2.2.1 Description of the experiment

Mutation accumulation (MA) is the classic method for directly studying the effects of mutation [14, 8]. Several colonies of cells are derived from a single common ancestor. When the colonies have reached a sufficient size, a cell is randomly drawn from each colony to form a new one. [Mettre une figure]. Several studies combine MA experiments with genome sequencing to estimate the rate and effects of mutations [12, 24] and MA experiment has been used to study mutations for several species [11, 16, 21, 1]. It is commonly assume that natural selection have no role in the evolution of colonies even if this method produces several selection biases that can skew the results [13].

Microfluidic methods can be used to avoid these biases. The Microfluidic Mutation Accumulation (μ MA) experiment is a technique developped in [19, 18] to study mutation rates, evolutionary dynamics and genetic stability in microorganisms. This experiment create a controlled environment where mutations can accumulate over successive generations without the selective pressures that typically eliminate deleterious mutations. By isolating and propagating single cells repeatedly, the μ MA experiment allows to directly observe the effects of accumulated mutations on fitness, genome stability, and evolutionary trajectories, providing insights into how mutations contribute to genetic diversity and evolution.

In [19], the authors track the evolution of 1476 cell lines using microfluidic methods. For each new generation, only one daughter cell is kept, regardless of its fitness. By filming the different cell lines over time, it is possible to measure changes in growth rates.

2.2.2 Description of the dataset

Part I

Theoretical Analysis

Chapter 3

Statistical modelling

3.1 Introduction

3.1.1 Motivation

In the domain of non-parametric statistics, the deconvolution and the decompounding problems are classical and have been widely studied [duval2013density, 4, 6, 10]. We give a brief description of the two problems:

Deconvolution

Deconvolution has applications in many fields, such as image processing [kundur1996blind], microscopy [swedlow2007quantitative], astronomy [starck2002deconvolution, pantin2017deconvolution], seismology [ulrych1971application, sacchi1997reweighting] and medicine [liu2009deconvolution, ralston2005deconvolution].

Let X and ε be two independent random variables with density f and f_ε . The deconvolution problem consists in estimating f from an *i.i.d.* sample of $Y = X + \varepsilon$, *i.e.* from noisy observations

$$Y_i = X_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where $(X_i)_{i=1}^n$ are *i.i.d.* random variables with density f and $(\varepsilon_i)_{i=1}^n$ are, *i.i.d.*, with density f_ε and independent of $(X_i)_{i=1}^n$.

This problem has been extensively studied in the literature. The most popular approach estimates f with Fourier estimators [carroll1988optimal, devroye1989consistent, stefanski1990deconvolving, neumann1997effect]. Other approaches have also been developed, for instance by using spline-based methods [averbuch2009spline], wavelet decomposition [johnstone2004wavelet, starck2003wavelets] and penalization methods [comte2006penalized, comte2007finite]. In most studies, the author assumes that the noise density f_ε is known, however this assumption is not necessary if an additional error sample is available [comte2011data, 10]. Many adaptive methods, which only rely on data-driven procedures, have also been developed [griffiths1977adaptive, 6].

Decompounding

In the literature, the term decompounding first appeared in the article of Buchmann and Grübel [4]. Decompounding has many applications in financial mathematics [embrechts2013modelling]

and queuing theory [**babai2011analysis**, **gomez2016retrial**].

Let $(X_k)_{k \in \mathbb{N}}$ be a family of *i.i.d.* random variable with density f and N be a homogeneous Poisson process with intensity $\lambda \in (0, \infty)$. Define $Y = (Y_t)_{t \geq 0}$ as the compound Poisson process

$$Y_t = \sum_{k=1}^{N_t} X_k, \quad t \geq 0.$$

The decompounding problem consists in estimating f from observations $(Y_{i\Delta}, i = 1, \dots, n)$ of the trajectory of Y over $[0, T]$ at a sampling rate $\Delta > 0$, *i.e.* at time $(i\Delta, i = 1, \dots, n)$. The decompounding problem has been the subject of many articles [**bogsted2010decompounding**, **van2004asymptotic**, 3]. Most of the time, f is estimated using its characteristic function and Fourier estimators [6]. The available data can be observed at high frequencies or at low frequencies [**coca2018efficient**, 6]. The decompounding problem is part of the more general framework of Levy processes that have been widely developed in recent years [**reiss2013testing**, **nickl2012donsker**, **applebaum2009levy**].

In this article, we study a model that combines both the deconvolution problem and the decompounding problem. In particular, we assume that we observe several compound Poisson processes but that all these observations are noisy due to measurement noise. More precisely, we consider a family of processes

$$Z_t^j = \left(\sum_{k=1}^{N_t^j} X_k^j \right) + \varepsilon_t^j, \quad t \geq 0, \quad j = 1, \dots, J.$$

We want to construct an estimator of the jump density of the process. Our motivation to study this type of model comes from evolutionary biology. Many studies aim to estimate the distribution of fitness effects (DFE) of cell. This probability density represents the effect of a new mutation on the fitness of a cell. An accurate determination of the DFE within a population would provide a better understanding of the evolutionary trajectory of the population [7].

In 2018, Robert et al. [19] developed new experimental methods and a probabilistic model to study the DFE in a population of *Escherichia coli* (*E. coli*). The effect of each mutation on the fitness of a cell is assumed to be drawn according to a random variable X of density f where f denotes the DFE of the cell. The authors conclude that the number of mutations in a cell follows a point Poisson process and they estimate the DFE using a method of moment.

More specifically, they observe $n = 1476$ cell lines using micro-fluidic methods [25]. For each lineage $j = 1, \dots, n$, it is assumed that mutations are deleterious and appear according to a Poisson point process $N^j = (N_t^j)_{t \geq 0}$ with intensity $\lambda > 0$. Each of these mutations modifies the selective value over time $(W_t^j)_{t \geq 0}$ of the j -lineage. Let us denote t_i^j ($i \in \mathbb{N}$) the time of occurrence of the $\{i\}$ -th mutation in lineage j . The quantity

$$s_i^j = \frac{W_{t_{i-1}^j}^j - W_{t_i^j}^j}{W_{t_{i-1}^j}^j}, \quad i > 0,$$

represents the relative effect of the $\{i\}$ -th mutation on the fitness of the individual. Assuming further that these mutations have independent and identically distributed (s_i) effects,

then

$$\frac{W_t^j}{W_0^j} = \prod_{i=1}^{N_t^j} (1 - s_i^j) .$$

By composing by the logarithm, the evolution of each lineage is controlled by a compound Poisson process, *i.e.*

$$Y_t^j = \ln W_t^j = \sum_{i=1}^{N_t^j} \ln(1 - s_i^j) .$$

Experimentally, lineages can only be observed at a sampling rate $\Delta > 0$ through discrete observations $(Y_{i\Delta}^j)_{i=1}^n$. Moreover, these observations are noisy due to measurement noise $(\varepsilon_i^j)_{i=1}^n$ on the i -th observation. In fact, it is only possible to observe the different lineages

$$Z_t^j = \left(\sum_{k=1}^{N_t^j} X_k^j \right) + \varepsilon_t^j , \quad t \geq 0 , \quad (3.1)$$

at sampling times $(i\Delta)_{i=1,\dots,n}$, with $X_k^j = \ln(1 - s_k^j)$. By abuse of notation, we consider that the noise ε_t^j exists for all t in each lineage $j = 1, \dots, 1476$.

The main goal of this article is to provide a nonparametric estimator that is adapted to estimate the probability density f from the sample of the noisy trajectories and study its asymptotic properties. Such an approach has several advantages. First, it offers a way to study the jump density of a noisy compound Poisson process when we observe several independent processes, which has, to our knowledge, been little studied in the literature. Second this method has the advantage of studying a model very close to the biological situation.

However, using a logarithm on the model to transform it into an 'additive' model only gives us information on $\log X$. While it is true that the two models are theoretically equivalent, the additive form forces us to make assumptions about the logarithm of the density we want to reconstruct, what is not easy to justify from modelling perspective. To overcome this, we use in `S:multipDecompounding` a method similar to the previous one to study the multiplicative decompounding problem.

In this setting, we discretely observe one trajectory of a multiplicative compound Poisson process

$$W_t = \prod_{i=1}^{N_t} (1 - s_i) ,$$

where $(N_t)_{t \geq 0}$ is a Poisson process with intensity λ independent of the *i.i.d.* random variables $(s_i)_i$ with density f .

The idea of studying 'multiplicative' models was developed in [**vardi1989multiplicative**] and [**vardi1992large**] where the authors studied a multiplicative deconvolution problem where the error U is uniformly distributed on $[0, 1]$. Such models can be studied using the Mellin transform, which can be seen as a multiplicative version of the Fourier transform. These estimators have been successfully used in [**miguel2021spectral**, **miguel2023multiplicative**] to study the multiplicative deconvolution problem with unknown noise.

3.1.2 Main results and organisation of the article

In S:decompounding, we discretely observe J trajectory of the process $Z = (Z_t)_{t \geq 0}$ defined in eq:main_process, i.e. $Z_t^j = \sum_{k=1}^{N_t^j} X_k^j + \varepsilon_t^j$, $t \geq 0$, $j = 1, \dots, J$, where $(X_k^j)_{k \in \mathbb{N}}$, $j = 1, \dots, J$ is a family of *i.i.d.* random variable with density f , $(N_t^j)_{t \geq 0}$, $j = 1, \dots, J$ are homogeneous Poisson processes with intensity $\lambda \in (0, \infty)$ and $(\varepsilon_t^j)_{t \geq 0}$, are *i.i.d.* centered random variable such that for every $t, s \geq 0$ and for every $i, j = 1, \dots, J$ the random variables ε_t^j and ε_s^i are independents. Let $\Delta > 0$. We suppose that we observe

$$(Z_{i\Delta}^j, i = 1, \dots, n, j = 1, \dots, J).$$

We aim at estimating f from these observations. In s:setting, we define an estimator $\hat{f}_{m,J}$, based on Fourier methods, that rely on two different times $t_1, t_2 \in \{i\Delta, i = 1, \dots, n\}$. We establish in thm:analysis_error_upper_bound_for_its_L2-risk

$$\begin{aligned} \mathbb{E} \left(\left\| \hat{f}_{m,J} - f \right\|^2 \right) &\leq \|f_m - f\|^{2 + \sum_{i=1}^2 \frac{4e^{4t_i}}{J(t_2 - t_1)^2} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2}} \\ &+ \frac{4K_{J,t_1,t_2}}{(t_2 - t_1)^2} \cdot \left(\frac{e[X_i^2]}{Jt_i} + \frac{e[\varepsilon^2]}{Jt_i^2} + 4 \frac{m}{(Jt_i)^2} \right), \end{aligned}$$

where $\|f\| = \left(\int_{\mathbb{R}} |f(x)|^2 dx \right)^{1/2}$, φ_ε denotes the characteristic function of ε and K_{J,t_1,t_2} is a constant that only depends on J, t_1 and t_2 . We discuss the rates of convergence in s:speed_vo_nedependingontheregular. In S:adaptive, we establish an adaptive procedure to automatically select the value of the threshold m according to the data.

In S:multipDecompounding, we discretely observe one trajectory of the multiplicative compound process

$$Y_t = \prod_{i=1}^{N_t} X_i, \quad (3.2)$$

where $(N_t)_{t \geq 0}$ is a Poisson process with constant intensity $\lambda > 0$, independent of the *i.i.d.* random variables $(X_j)_{j \in \mathbb{N}}$ with common density $f \in L^1([0, \infty)) \cap L^2([0, \infty))$. We suppose that we observe $(Y_{i\Delta}, i = 1, \dots, n)$. We aim at estimating f from these observations. In sec:mellinDEF, we recall some properties of the Mellin transform. In sec:mellin_estimator, we define the empirical Mellin estimator $\widehat{\mathcal{M}}_c(t) = \frac{1}{n} \sum_{k=1}^N X_k^{c-1+it}$, $t \in \mathbb{R}$ and we give some of this properties when it is well defined. In s:model, we define an estimator $\hat{f}_{m,\Delta}$ of f , based on the Mellin transform. In sec:risk_mellin, we establish in T : risk an upper bound risk

$$\mathbb{E} [\| \widehat{f}_{m,\Delta} - f \|_{\omega_1}^2] \leq \|f_m - f\|_{\omega_1}^2 + \frac{1}{2\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + \frac{25}{\pi} \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right).$$

where $\|f\|_{\omega_1} = \left(\int_0^\infty x |f(x)|^2 dx \right)^{1/2}$. In sec:mellin_adaptive, we introduce an adaptive procedure to select the optimal

We numerically illustrate our two methods on several examples in S:simu. The proofs are postponed to the sec:proof.

We rigorously define the distinguished logarithm which is a key element of our estimators in S:DistingLog and we describe some of its classical properties. We present some useful lemmas in

s:alem.

All of our computer code is available and documented on GitHub

<https://github.com/guimgarnier/decompounding-with-noise>

with several examples, making it easy to use on biological experimental data.

3.2 Decompounding with unknown noise

3.2.1 Notation

The empirical characteristic function

The characteristic function of a real random variable X is

$$\varphi_X(u) = \int_{\mathbb{R}} e^{iux} \mathbb{P}_X(dx) = \mathbb{E}[e^{iuX}] , \quad u \in \mathbb{R} .$$

If X has a density $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, then we recall the inversion formula

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi_X(u) e^{-iux} du , \quad x \in \mathbb{R} ,$$

and Parseval's identity

$$\int_{\mathbb{R}} |f(x)|^2 dx = \frac{1}{2\pi} \int_{\mathbb{R}} |\varphi_X(u)|^2 du .$$

Let $(X_1, \dots, X_N)$ be *i.i.d.* real-valued random variables with common characteristic function $\varphi_X(t)$. The function

$$\widehat{\varphi}_X^N(u) = \frac{1}{N} \sum_{k=1}^N e^{iuX_k} , \quad u \in \mathbb{R} ,$$

is called the empirical characteristic function associated with the sample $(X_1, \dots, X_N)$.

The empirical characteristic function is an unbiased estimator of φ_X *i.e.*

$$\mathbb{E}[\widehat{\varphi}_X^N(u)] = \varphi_X(u)$$

and

$$\mathbb{E}[|\widehat{\varphi}_X^N(u) - \varphi_X(u)|^2] = \frac{1}{N} (1 - |\varphi_X(u)|^2) . \quad (3.3)$$

For more detailed information, we refer to Ushakov's book [ushakov2011selected].

The distinguished logarithm

In the following, the notation $\log$ denotes the distinguished logarithm that we rigorously define in Appendix ??.

3.2.2 Statistical setting

Let $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Let $(X_i^j)_{i,j \geq 0}$ be *i.i.d.* real-valued random variables with density f and for some $J \in \mathbb{N}$, let $(N_t^j)_{t \geq 0, j = 1, \dots, J}$ be a family of *i.i.d.* Poisson point processes with intensity $\lambda \in (0, \infty)$, independent of $(X_i^j)_{i,j \geq 0}$.

We consider a family $(Y_t^j)_{t \geq 0, j = 1, \dots, J}$ of compound Poisson processes with intensity λ and jump size density f ,

$$Y_t^j = \sum_{k=1}^{N_t^j} X_k^j, \quad t \geq 0, \quad j = 1, \dots, J.$$

It is assumed that the observations are disturbed by a random noise independent of the observed process, *i.e.* we observe a process of the form

$$Z_t^j = Y_t^j + \varepsilon_t^j = \left(\sum_{k=1}^{N_t^j} X_k^j \right) + \varepsilon_t^j, \quad t \geq 0,$$

for every $i, j = 1, \dots, J$. For every reals $t, s \geq 0$, the random variables ε_t^j and ε_s^i are independents.

Also, we set the following assumptions:

Hypothèse.

(H₁) $\varphi_{X_1} \in L^1(\mathbb{R})$.

(H₂) $\mathbb{E}(X_1^2) < \infty$.

(H₃) $\forall t \in [0, \infty)$, $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t^2] < \infty$.

(H₄) $\forall m \in [0, \infty)$, $\exists c_m \in (0, \infty) : \forall u \in [0, m], |\varphi_\varepsilon(u)| \geq c_m$.

(H₅) $\forall u \in \mathbb{R}, |\varphi_\varepsilon(u)| > 0$.

For the sake of simplicity, we write ε instead of ε_t^j in the rest of this article.

Construction of the estimator.

We estimate the characteristic function φ_X of X . The main idea is to claim that if we have a good reconstruction of φ_X , by applying the inverse Fourier transform, we should have a good reconstruction of f .

For all $t \in [0, \infty)$, the characteristic function of the process on a single channel Z_t^j is given by

$$\varphi_{Z_t}(u) = e^{-\lambda t + \lambda t \varphi_X(u)} \cdot \varphi_\varepsilon(u), \quad \forall u \in \mathbb{R}. \quad (3.4)$$

To simplify notation, we consider in the rest of the article the particular case where $\lambda = 1$.

Consider two different times $0 < t_1 < t_2$, then

$$\frac{\varphi_{Z_{t_2}}}{\varphi_{Z_{t_1}}} = e^{-(t_2-t_1) + (t_2-t_1)\varphi_X(u)}, \quad \forall u \in \mathbb{R}.$$

Applying the distinguished logarithm, we obtain the explicit formula

$$\varphi_X(u) = 1 + \frac{1}{t_2 - t_1} \left[\log \varphi_{Z_{t_2}}(u) - \log \varphi_{Z_{t_1}}(u) \right], \quad \forall u \in \mathbb{R}.$$

This leads us to consider the estimator

$$\forall u \in \mathbb{R}, \quad \hat{\varphi}_X^J(u) = 1 + \frac{1}{t_2 - t_1} \left[\log \hat{\varphi}_{Z_{t_2}}^J(u) - \log \hat{\varphi}_{Z_{t_1}}^J(u) \right],$$

with for all $u \in \mathbb{R}$, $\tau \in \{t_1, t_2\}$

$$\hat{\varphi}_{Z_\tau}^J(u) = \frac{1}{J} \sum_{j=1}^J i Z_\tau^j e^{iu Z_\tau^j}, \quad \hat{\varphi}_{Z_\tau}^J(u) = \frac{1}{J} \sum_{j=1}^J e^{iu Z_\tau^j}, \quad \log \hat{\varphi}_{Z_\tau}^J(u) = \int_0^u \frac{\hat{\varphi}_{Z_\tau}^J(z)}{\hat{\varphi}_{Z_\tau}^J(z)} dz.$$

Since φ_X is a characteristic function, its modulus is bounded by 1. Nevertheless, this is not necessarily the case for $\hat{\varphi}_X^J$. We avoid this explosion problem taking

$$\tilde{\varphi}_X^J(u) = 1 + \frac{1}{t_2 - t_1} \left[\log \hat{\varphi}_{Z_{t_2}}^J(u) \cdot \mathbb{1}_{|\log \hat{\varphi}_{Z_{t_2}}^J(u)| \leq \ln(J)} - \log \hat{\varphi}_{Z_{t_1}}^J(u) \cdot \mathbb{1}_{|\log \hat{\varphi}_{Z_{t_1}}^J(u)| \leq \ln(J)} \right], \quad u \in \mathbb{R}.$$

To simplify notation, for any positive real $\tau > 0$ and for all $u \in \mathbb{R}$ we set

$$\log \tilde{\varphi}_{Z_\tau}^J(u) = \log \hat{\varphi}_{Z_\tau}^J(u) \cdot \mathbb{1}_{|\log \hat{\varphi}_{Z_\tau}^J(u)| \leq \ln(J)}.$$

In particular, with this notation

$$\tilde{\varphi}_X^J(u) = 1 + \frac{1}{t_2 - t_1} \left[\log \tilde{\varphi}_{Z_{t_2}}^J(u) - \log \tilde{\varphi}_{Z_{t_1}}^J(u) \right], \quad u \in \mathbb{R}. \quad (3.5)$$

Now, we find an estimator of f by performing an inverse Fourier transformation. However, $\tilde{\varphi}_X^J$ is not necessarily integrable. Hence, we eliminate frequencies above a threshold m before applying an inverse Fourier transform, we obtain

$$\hat{f}_{m,J}(x) = \frac{1}{2\pi} \int_{-m}^m e^{-iux} \tilde{\varphi}_X^J(u) du, \quad x \in \mathbb{R}, \quad m \in (0, \infty). \quad (3.6)$$

In the same way, we perform an truncated inverse Fourier transformation from the true characteristic function of X and we define

$$f_m(x) = \frac{1}{2\pi} \int_{-m}^m e^{-iux} \varphi_X(u) du, \quad x \in \mathbb{R}, \quad m \in (0, \infty).$$

3.2.3 Risk bounds

In this section, we study the mean integrated squared error (MISE) of the estimator $\hat{f}_{m,J}$.

For any $0 < t_1 < t_2$, we define

$$C_{t_1, t_2}^J = \min \left\{ m \geq 0 \mid 3t_2 - t_1 + \sup_{[-m, m]} |\log \varphi_\varepsilon(\cdot)| > \ln(J) \right\}. \quad (3.7)$$

Théorème 1. *Suppose that the assumptions ?? – ?? holds. Let $0 < t_1 < t_2$. We suppose that J is enough large such that for all $i = 1, 2$:*

1. $t_i < \frac{1}{4} \log(Jt_i)$,
2. $\sqrt{\log(Jt_i)}(Jt_i)^{2\delta_i - 1/2} < 1$ with $\delta_i = t_i / \log(Jt_i)$.

Then for any $m < C_{t_1, t_2}^J$ with C_{t_1, t_2}^J defined by (??), we have

$$\mathbb{E} \left(\left\| \widehat{f}_{m, J} - f \right\|^2 \right) \leq \|f_m - f\|^{2 + \sum_{i=1}^2 \frac{4e^{4t_i}}{J(t_2 - t_1)^2}} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2} \\ + \frac{4K_{J, t_1, t_2}}{(t_2 - t_1)^2} \cdot \left(\frac{e[X_i^2]}{Jt_i} + \frac{e[\varepsilon^2]}{Jt_i^2} + 4 \frac{m}{(Jt_i)^2} \right).$$

where $K_{J, t_1, t_2} = m \ln^2(J) + 8mt_2^2 + \int_{-m}^m |\log \varphi_\varepsilon(u)|^2 du$.

Remarque 2. Theorem ?? is asymptotic and ensures that the variance term vanishes when $J \rightarrow \infty$. The upper bound obtained is the sum of a bias term and of a variance term.

In the variance, there is a term $V \sim \frac{4e^{4t_i}}{J(t_2 - t_1)^2}$. On one side, the presence of e^{4t_i} means that the estimator is more and more imprecise as we look the sample at a very large time t_2 . On the other side, the presence of $(t_2 - t_1)$ at the denominator means that we cannot take t_1 and t_2 too close to each other. We illustrate this trade-off in Section ?? with numerical simulations.

3.2.4 Speed of convergence

In this section, we study the optimal choice of m based on the regularity of the jump density and regularity of the noise density, *i.e.* that when $J \rightarrow \infty$, we look at the asymptotic behaviour of m when it minimizes the upper term in Theorem in ?. We organize the discussion according to different regularities of f and f_ε . More specifically, we consider:

- ordinary smooth densities: The characteristic function decays as $|u|^{-2a}$ (example: Gamma distribution)
- super smooth densities: The characteristic function decays as $e^{-|u|^s}$ (example: Cauchy distribution)

When f is ordinary smooth and f_ε is ordinary smooth

In this case, we assume that $f \in \mathcal{S}(\beta, L)$ for some $\beta > 0$, $L > 0$, with

$$\mathcal{S}(\beta, L) = \left\{ f \in L^2(\mathbb{R}) \mid \int_{\mathbb{R}} (1 + |u|^2)^\beta |\varphi_X(u)|^2 du \leq L \right\}.$$

We also assume that f_ε is *ordinary smooth*, *i.e.* there exists two reals $a > \frac{1}{2}$ and $d > 0$, such that

$$\forall u \in \mathbb{R}, \quad d \leq (1 + u^2)^a |\varphi_\varepsilon(u)|^2 \leq \frac{1}{d}.$$

It follows that $\|f - f_m\|^2$ is of order $m^{-2\beta}$ and that the variance term $\frac{1}{J} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2}$ is of order $\frac{1}{J} m^{2a+1}$.

Moreover, the variance term $\frac{4K_{J, t_1, t_2}}{(t_2 - t_1)^2} \cdot \left(\frac{e[X_i^2]}{Jt_i} + \frac{e[\varepsilon^2]}{Jt_i^2} + 4 \frac{m}{(Jt_i)^2} \right)$ is of order $\frac{m \ln^2(J)}{J}$. Therefore, if the Assumptions ?? – ?? hold, then

$$\mathbb{E} \left[\left\| \widehat{f}_{m, J} - f \right\|^2 \right] \lesssim m^{-2\beta + \frac{1}{J} m^{2a+1} + m \cdot \frac{\ln^2(J)}{J}}, \quad (3.8)$$

up to a multiplicative constant that depends on t_1 and t_2 .

To obtain a value m^* that reaches the bias-variance compromise, we minimize the upper bound of Equation (??) by differentiating with respect to m . It follows that $m^* \sim J^{\frac{1}{2a+2\beta+1}}$ when $J \rightarrow \infty$ and that

$$\mathbb{E} \left[\left\| \hat{f}_{m,J} - f \right\|^2 \right] = O \left(J^{\frac{-2\beta}{2\beta+2a+1}} \right), \quad J \rightarrow \infty.$$

When f is ordinary smooth and f_ε is super smooth

In this case, we assume that $f \in \mathcal{S}(\beta, L)$ for some reals $\beta > 0$, $L > 0$, *i.e.*

$$f \in \mathcal{S}(\beta, L) = \left\{ f \in L^2(\mathbb{R}) \mid \int_{\mathbb{R}} (1 + |u|^2)^\beta |\varphi_X(u)|^2 du \leq L \right\}.$$

We also assume that f_ε is *super smooth*, *i.e.* there exists $a > 0$, $s > 0$ and $d_1, d_2 > 0$ such that

$$\forall u \in \mathbb{R}, \quad d_1 \leq \exp(b \cdot |u|^s) \cdot |\varphi_\varepsilon(u)|^2 \leq d_2.$$

It follows that $\|f - f_m\|^2$ is of order $m^{-2\beta}$ and that the variance term $\frac{1}{J} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2}$ is of order $\frac{1}{J} m \cdot e^{b \cdot m^s}$.

Moreover, the variance term $\frac{4K_{J,t_1,t_2}}{(t_2-t_1)^2} \cdot \left(\frac{e[X_i^2]}{Jt_i^2} + \frac{e[\varepsilon^2]}{Jt_i^2} + 4 \frac{m}{(Jt_i)^2} \right)$ is of order $m \cdot \frac{\ln^2(J)}{J}$.

Therefore, if the Assumptions of Theorem ?? holds, then

$$\mathbb{E} \left\| \hat{f}_{m,J} - f \right\|^2 \lesssim m^{-2\beta + \frac{1}{J} m \cdot e^{b \cdot m^s} + m \cdot \frac{\ln^2(J)}{J}}, \quad (3.9)$$

up to a multiplicative constant that depends on t_1 and t_2 .

To obtain a value m^* that reaches the bias-variance compromise, we minimize the upper bound of Equation (??) by differentiating with respect to m . It follows that $m^* \sim \left(\frac{\ln(J)}{b} \right)^{1/s}$ when $J \rightarrow \infty$ and that

$$\mathbb{E} \left\| \hat{f}_{m,J} - f \right\|^2 = O \left(\left(\frac{\ln(J)}{b} \right)^{\frac{-2\beta}{s}} \right).$$

When f is super smooth

In this section, we assume that f belongs to the class of super smooth densities,

$$\mathcal{A}_{c,s}(L) = \left\{ f \in L^2(\mathbb{R}) \mid \int_{\mathbb{R}} \exp(c \cdot |u|^s) |\varphi_X(u)|^2 du \leq L \right\}.$$

It follows that $\|f - f_m\|^2$ is of order $\exp(-c \cdot |m|^s)$.

We apply the same strategy that in `sec:smoothsmooth` and `sec:smoothsuper` to compute m^* when f_ε is ordinary smooth or super smooth.

In `table:speed`, we resumes the different results that we obtained with respect to the regularity of f and f_ε . For each case, we give the optimal cut-off m^* and the speed of convergence of the

ff_ε	ordinary smooth: $f_\varepsilon \in \mathcal{S}(a, L)$	super smooth: $f_\varepsilon \in \mathcal{A}_{b,s}(L)$
ordinary smooth: $f \in \mathcal{S}(\beta, L)$	$m^* \sim J^{\frac{1}{2a+2\beta+1}}$ $\mathbb{E} \ \hat{f}_{m,J} - f\ ^{2=O\left(J^{\frac{-2\beta}{2\beta+2a+1}}\right)}$	$m^* \sim \frac{\ln(J)}{b}$ $\mathbb{E} \ \hat{f}_{m,J} - f\ ^{2=O\left(\left(\frac{\ln(J)}{b}\right)^{\frac{-2\beta}{s}}\right)}$
super smooth $f \in \mathcal{A}_{c,s}(L)$	$m^* \sim \left(\frac{\ln(J)}{c}\right)^{\frac{1}{s}}$ $\mathbb{E} \ \hat{f}_{m,J} - f\ ^{2=O(J^{-1})}$	$m^* \sim \left(\frac{\ln(J)}{b+c}\right)^{\frac{1}{s}}$ $\mathbb{E} \ \hat{f}_{m,J} - f\ ^{2=O\left(J^{\frac{-c}{b+c}}\right)}$

Table 3.1 – Speed of the optimal upper bound with respect to the regularity of f and f_ε .

estimator.

3.2.5 The adaptive procedure

We have constructed in `sec:riskdecompoundastatisticalesimulator` $\hat{f}_{m,J}$ to estimate the jump size density f . However, as this estimator strongly depends on the choice of the parameter m , we would like to be able to select a value of m that depends only on the available data, without a priori knowledge on the regularity of the density f .

To do this, we aim to select the parameter m that minimizes the bound obtained in the Theorem ??.

For this, we need to make some further regularity assumptions on the measurement noise, which is to say that there is a function $g \in L^1(\mathbb{R})$ and a real $d > 0$ such that

$$\forall u \in \mathbb{R}, \quad d \cdot g(u) < |\varphi_\varepsilon(u)|^2 < \frac{g(u)}{d}.$$

In the rest of this section, we assume that the noise is ordinary smooth, *i.e.* there exists two reals $a > \frac{1}{2}$ and $d \in (0, 1)$, such that

$$\forall u \in \mathbb{R}, \quad d \leq (1 + u^2)^a \cdot |\varphi_\varepsilon(u)|^2 \leq \frac{1}{d}.$$

Nevertheless, the methods used can be adapted for any function g .

In the following, we simplify the computations by considering the particular case $a = 1$, *i.e.* we assume that

$$\forall u \in \mathbb{R}, \quad d \leq (1 + u^2) \cdot |\varphi_\varepsilon(u)|^2 \leq \frac{1}{d}.$$

How to select $\hat{m}_{J,t_2,t_1}$

As we said above, we aim to select m so as to minimize the bound obtained in the theorem ??.

The dominant terms in this bound are the bias term $\int_{u \in [-m,m]} |\varphi_X(u)|^2 du$ and the variance

term $\frac{4e^{4t_2}}{J(t_2-t_1)^2} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2}$. Through differentiation, the optimal $\overline{m}_J$ satisfies

$$|\varphi_X(\overline{m}_J)|^2 = \frac{4ae^{4t_2}}{J(t_2-t_1)^2} (1 + \overline{m}_J^2),$$

then

$$\left| \frac{\varphi_X(\overline{m}_J)}{\sqrt{1 + \overline{m}_J^2}} \right|^2 = \frac{4ae^{4t_2}}{J(t_2-t_1)^2}.$$

However we do not know φ_X , so it is impossible to calculate directly $\overline{m}_J$. Following the strategy developed by Duval and Kappus [6], we consider

$$\overline{\varphi}_X^J(u) = \tilde{\varphi}_X^J(u) \cdot \mathbf{1}_{\left| \frac{\tilde{\varphi}_X^J(u)}{\sqrt{1+u^2}} \right| \geq \frac{\kappa_{J,t_1,t_2}}{\sqrt{J(t_2-t_1)^2}}},$$

where $\kappa_{J,t_1,t_2} = 2e^{2t_2} + \kappa\sqrt{\ln(J(t_2-t_1)^2)}$, $\kappa > 0$.

It leads us to define the empirical cutoff parameter

$$\hat{m}_J = \max \left\{ u \geq 0 : \left| \frac{\overline{\varphi}_X(u)}{\sqrt{1+u^2}} \right| \geq \frac{\kappa_{J,t_1,t_2}}{\sqrt{J(t_2-t_1)^2}} \right\} \wedge (J(t_2-t_1)^2)^\alpha, \quad \alpha \in (0, 1).$$

For simplicity, we note $m_{\max} = (J(t_2-t_1)^2)$, keeping in mind that it depends on J, t_1 and t_2 .

We define a new estimator

$$\overline{f}_{m,J}(x) = \frac{1}{2\pi} \int_{-m}^m e^{-iux} \overline{\varphi}_X^J(u) du, \quad x \in \mathbb{R}.$$

Théorème 3. Assume ??-??. For all reals $0 < t_1 < t_2$ such that $t_2 \leq \frac{1}{4} \log(Jt_2)$ and $(m_{\max})^\alpha < C_{t_1,t_2}^J$, $Jt_1 \rightarrow \infty$, $Jt_2 \rightarrow \infty$ as $J \rightarrow \infty$. Then,

$$\begin{aligned} \mathbb{E}[\|\overline{f}_{\hat{m}_J} - f\|^2] &\leq \inf_{m \in [0, (m_{\max})^\alpha]} \left\{ \|f_m - f\|^{2+C \frac{\ln(J(t_2-t_1)^2) \cdot m \cdot (1+m^2)}{J(t_2-t_1)^2}} + \tilde{C}A \right\} \\ &\quad + \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2 \cdot T_J \end{aligned}$$

where

$$A = \sum_{i=1}^2 \frac{4e^{4t_i}}{J(t_2-t_1)^2} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2} + \frac{4K_{J,t_1,t_2}}{(t_2-t_1)^2} \cdot \left(\frac{e[X_i^2]}{Jt_i} + \frac{e[\varepsilon^2]}{rJt_i^2} + 4 \frac{m}{(Jt_i)^2} \right)$$

and

$$T_J \leq C_0(J(t_2-t_1)^2)^{\alpha-c(\theta)^2} + \frac{C_1}{J(t_2-t_1)^2} + \frac{C_2}{J(t_2-t_1)^4} \quad (3.10)$$

and $c(\theta) = \kappa(t_2-t_1)e^{2t_2} \cdot \frac{d}{\sqrt{1+(m_{\max})^2}}$ and where C_0, C_1 and C_2 depends on $\mathbb{E}[X_1^2], \mathbb{E}[\varepsilon^2]$ and where C and $\tilde{C}$ are two constants.

3.3 Multiplicative compounding

3.3.1 Preliminaries on Mellin transform

In this section, we first recall some classical results on Mellin transform. We define the multiplicative compounding problem and we define a good non-parametric estimator.

Let μ be a probability measure on $(0, \infty)$. Its Mellin transform is

$$\mathcal{M}[\mu](s) = \int_0^\infty x^{s-1} \mu(dx) , \quad s \in \mathbb{C} , \quad (3.11)$$

for those values of s for which this integral is well defined.

Remarque 4. If $\mathcal{M}[\mu](u)$ is well defined for some $u \in \mathbb{R}$, then $\mathcal{M}[\mu]$ converges on the vertical line $u + i\mathbb{R}$. Furthermore, if the integral exists for u and v in $\mathbb{R}$ ($u < v$), then $\mathcal{M}[\mu](w)$ exists for $w \in (u, v)$. It follows that the Mellin transform of a probability measure is well defined on a vertical band Ξ_μ of the complex plane. The region Ξ_μ is called the fundamental strip of $\mathcal{M}[\mu]$.

Assume that μ has a density $f \in L^1_{\text{loc}}([0, \infty))$, then we denote

$$\mathcal{M}[f](s) := \mathcal{M}[f dx](s) = \int_0^\infty x^{s-1} f(x) dx , \quad s \in \mathbb{C} ,$$

for those values of s for which this integral is well defined. We denote Ξ_f the fundamental strip of $\mathcal{M}[f dx]$

Let $f \in L^1_{\text{loc}}([0, \infty))$ and $c \in \mathbb{R}$ that belongs to Ξ_f . Then we define

$$\mathcal{M}_c[f](t) := \mathcal{M}[f](c + it) = \int_0^\infty x^{c-1+it} f(x) dx , \quad t \in \mathbb{R} .$$

We observe that the function $x \mapsto x^{c-1} f(x)$ belongs to $L^1([0, \infty))$.

We define the norm

$$\|f\|_{\omega_c} = \left(\int_0^\infty |f(x)|^2 x^{2c-1} dx \right)^{1/2} .$$

and $L^2_{[0, \infty)}(\omega_c)$ denotes the space of function $f : [0, \infty) \rightarrow \mathbb{R}$ such that $\|f\|_{\omega_c} < \infty$

Let $f, g \in L^1_{\text{loc}}([0, \infty))$. The multiplicative convolution of f and g is the function defined by

$$f \star g(x) = \int_0^\infty f(y) g\left(\frac{x}{y}\right) dy , \quad x \in [0, \infty) .$$

Proposition 5 (Multiplicative convolution and Mellin transform). *Let μ and ν be two probability measures on $[0, \infty)$. We have*

$$\mathcal{M}[\mu \star \nu](s) = \mathcal{M}[\mu](s) \mathcal{M}[\nu](s) , \quad s \in \mathbb{C} ,$$

whenever this integral is well defined.

Proposition 6 (Parseval's theorem for Mellin transform).

$$\int_0^\infty \frac{|f(x)|^2}{x} dx = \frac{1}{2\pi} \int_0^\infty \mathcal{M}[f](it) dt . \quad (3.12)$$

Proposition 7. (*Inversion formula*) Let $f \in L^1([0, \infty))$ and $c \in \Xi_f$. Then

$$f(x) = \mathcal{M}^{-1} \mathcal{M}[f](x) = \frac{1}{2\pi i} \int_{\nu-i\infty}^{\nu+i\infty} x^{-s} \mathcal{M}[f](s) ds , \quad x \in [0, \infty) .$$

3.3.2 The Mellin estimator

Let $f \in L^2_{[0,\infty)}(\omega_c)$ be a probability distribution. Let us assume that we have a sample $X_1, \dots, X_N$ independent and identically distributed according to f .

We define the Mellin estimator of f

$$\widehat{\mathcal{M}}_c(t) = \frac{1}{n} \sum_{k=1}^N X_k^{c-1+it} , \quad t \in \mathbb{R} .$$

We use Proposition ?? to define the estimator.

$$\widehat{f}_m(x) = \frac{1}{2\pi} \int_{-m}^m x^{-c-it} \widehat{\mathcal{M}}_c(t) dt ,$$

which is an estimator of

$$f_m(x) = \frac{1}{2\pi} \int_{-m}^m x^{-c-it} \mathcal{M}_1(t) dt .$$

The next proposition give a bound of the $L^2_{[0,\infty)}(\omega_c)$ -risk of $\widehat{f}_k$.

Proposition 8 (Miguel et al. [miguel2021spectral] Proposition 2.1). *If $f \in L^2_{[0,\infty)}(\omega_c)$ and $\sigma_c^2 := \mathbb{E}_f(X^{2(c-1)}) < \infty$, then for all $k \in [0, \infty)$,*

$$\mathbb{E}_f(\|f - \widehat{f}_k\|_{\omega_c}^2) \leq \|f - f_k\|_{\omega_c}^2 + \frac{\sigma_c^2 k}{\pi n} .$$

By choosing $k = k_n$ such that $n^{-1}k_n \rightarrow 0$ and $k_n \rightarrow \infty$, $\widehat{f}_{n_k}$ is a consistent estimator of f .

If X and Y are two independant random variables, then

$$\mathcal{M}[X \cdot Y](s) = \mathcal{M}[X](s) \cdot \mathcal{M}[Y](s) , \quad s \in \mathbb{R} .$$

3.3.3 Statistical setting: estimation procedure

Consider the multiplicative compound process

$$Y_t = \prod_{i=1}^{N_t} X_i , \quad (3.13)$$

where $(N_t)_{t \geq 0}$ is a Poisson process with constant intensity $\lambda > 0$, independent of the *i.i.d.* random variables $(X_j)_{j \in \mathbb{N}}$ with common density $f \in L^1([0, \infty)) \cap L^2([0, \infty))$.

We suppose that we observe one trajectory of (Y_t) over $[0, T]$ at equidistant time $\Delta, 2\Delta, \dots, n\Delta$, $T = n\Delta$. We denote $Z_{k\Delta} = \frac{Y_{k\Delta}}{Y_{(k-1)\Delta}}$ the k -th increment of Y .

Let $c \in \Xi_X \cap \Xi_\Delta$. We have $1 \in \Xi_X \cap \Xi_\Delta$. Let $\mathcal{M}_1[f]$ be the Mellin transform of X_1 and $\mathcal{M}_1[\Delta]$ be the Mellin transform of Z_Δ .

Lemme 9. *For any $s \in \mathbb{C}$, $\mathcal{M}_1[\Delta](s) = \exp(\Delta \lambda (\mathcal{M}_1[f](s) - 1))$.*

From Lemma ??, we have

$$\mathcal{M}_1[f](s) = 1 + \frac{1}{\lambda \Delta} \log \mathcal{M}_1[\Delta](s),$$

where $\log \mathcal{M}_1[\Delta]$ is the distinguished logarithm of $\mathcal{M}_1[\Delta]$ that we define in Appendix ??.

This last equation leads us to define the estimator

$$\widehat{\mathcal{M}_1[f]}(s) = 1 + \frac{1}{\lambda \Delta} \log \widehat{\mathcal{M}_1[\Delta]}(s), \quad s \in \mathbb{R}.$$

where

$$\widehat{\mathcal{M}_1[\Delta]}(t) = \frac{1}{n} \sum_{k=1}^n Z_{k\Delta}^{c-1+it}, \quad \widehat{\mathcal{M}'_c[\Delta]}(t) = \frac{1}{n} \sum_{k=1}^n i \log(Z_{k\Delta}) Z_{k\Delta}^{c-1+it},$$

and

$$\log \widehat{\mathcal{M}_1[\Delta]}(s) = \int_0^u \frac{\widehat{\mathcal{M}'_c[\Delta]}(t)}{\widehat{\mathcal{M}_1[\Delta]}(t)} ds.$$

However, the quantity $\log \widehat{\mathcal{M}_1[\Delta]}(s)$ may explode. Then we define

$$\widetilde{\widehat{\mathcal{M}_1[f]}}(s) = \widehat{\mathcal{M}_1[f]}(s) \cdot \mathbb{1}_{|\widehat{\mathcal{M}_1[f]}(s)| \leq 4}, \quad s \in \mathbb{R}.$$

Finally, we apply a Mellin inversion

$$\widehat{f_{m,\Delta}}(x) = \frac{1}{2\pi} \int_{-m}^m x^{-c-it} \widetilde{\widehat{\mathcal{M}_1[f]}}(s) dt, \quad x \in (0, \infty), \quad (3.14)$$

3.3.4 Risk bounds

Théorème 10. *Assume that $\mathbb{E}[X_1^2] < \infty$, $\delta \leq \frac{1}{4} \log(n\Delta)$ and $n\Delta \rightarrow \infty$ as $J \rightarrow \infty$. Then, for any $m \geq 0$, it holds*

$$\mathbb{E}[\|\widehat{f_{m,\Delta}} - f\|_{\omega_1}^2] \leq \|f_m - f\|_{\omega_1}^2 + \frac{1}{2\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + \frac{25}{\pi} \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right). \quad (3.15)$$

We prove this result in Appendix ??. In the demonstration, we need `lm:Mellin1` and `lm:Mellin2`.

3.3.5 The adaptative procedure

We want to minimize the right side term in Theorem ?? . The optimal cutoff $\overline{m}_n$ is defined by

$$\overline{m}_n \in \arg \min_{m > 0} \left(\|f_m - f\|_{\omega_1}^2 + \frac{1}{2\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + \frac{25}{\pi} \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right) \right).$$

In the variance term, the leading term is in $\frac{me^{4\Delta}}{n\Delta}$. Therefore, the optimal cutoff is such that

$$|\mathcal{M}_1[f](\overline{m}_n)|^2 = \frac{e^{4\Delta}}{n\Delta}.$$

We can define the empirical optimal cutoff by

$$|\overline{\mathcal{M}_1[f]}(\overline{m}_n)|^2 = \frac{e^{4\Delta}}{n\Delta},$$

where

$$\overline{\mathcal{M}_1[f]}(s) = \widetilde{\mathcal{M}_1[f]}(s) \mathbb{1}_{|\widetilde{\mathcal{M}_1[f]}(s)| \geq \kappa_{n,\Delta}/\sqrt{n\Delta}},$$

with

$$\kappa_{n,\Delta} = (e^{2\Delta+\kappa} \sqrt{\log(n\Delta)}), \kappa > 0.$$

We consider the empirical cutoff

$$\widehat{m}_n = \max \{m \geq 0 : |\overline{\mathcal{M}_1[f]}(m)| = \kappa_{n,\Delta}/\sqrt{n\Delta}\} \wedge (n\Delta)^\alpha.$$

Théorème 11 (Adaptative method). *Assume that $\mathbb{E}[X_1^2] < \infty$, $\delta \leq \frac{1}{4} \log(n\Delta)$ and $J\Delta \rightarrow \infty$ as $n \rightarrow \infty$. Then, there exist two positive constants A and B depending on κ and $\mathbb{E}[\ln(X_1)^2]$ such that*

$$\begin{aligned} & \mathbb{E}[\|\overline{f_{\widehat{m}_n, \Delta}} - f\|_{\omega_1}^2] \\ & \leq \inf_{m \in [0, (n\Delta)^\alpha]} A \left(\|f_m - f\|_{\omega_1}^2 + \frac{\log(n\Delta)m}{n\Delta} + \frac{1}{n\Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + 4 \frac{m^2}{(n\Delta)^2} \right) \\ & \quad + B \left((n\Delta)^{\alpha-c(\Delta)^2} + \frac{1}{n\Delta} \right). \end{aligned}$$

3.4 Numerical experiments

3.4.1 Decompounding with noise

Implementation details

For a given estimator $\widehat{f}_{m,J}$ of f , we numerically evaluate the efficiency of $\widehat{f}_{m,J}$. More specifically, we compute the L^2 -risks $\mathbb{E}(\|\widehat{f}_{m,J} - f\|^2)$ by averaging the result obtained for 1000 independent simulations.

We test our method for three different jump distribution densities f :

— the mixture density $0.3\mathcal{N}(-3.5, 1) + 0.7\mathcal{N}(3.5, 1)$.

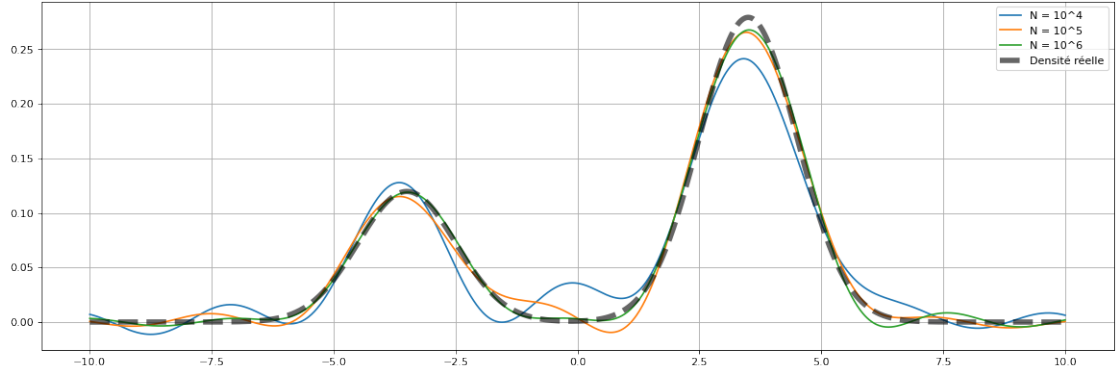


Figure 3.1 – Illustration of the estimator.

Input: Jump distribution $X \sim 0.3\mathcal{N}(-3.5, 1) + 0.7\mathcal{N}(3.5, 1)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.5, t_2 = 1, J = 10^5$.

— the Gamma density $\Gamma(2)$.

— the Cauchy density $\mathcal{C}(0, 1)$.

and the noise density $\mathcal{N}(0, 1)$.

Efficiency of the estimation

The estimator is expected to be progressively more accurate as J increases. We compute the estimator for different values of J for mixed density (Figure ??) and Gamma density (Figure ??).

Dependence of the L^2 -risk to the cut-off m and efficiency of the adaptive procedure

The value of m has a strong impact on the quality of the estimator $\hat{f}_{m,J}$. When m is too low or too high, we expect that the estimator should deteriorate. The adaptive procedure gives an automatic way to select a "good" m based on observations.

We illustrate the efficiency of the adaptive procedure for the mixture density (Figure ??), Gamma density (Figure ??) and the Cauchy density (Figure ??).

The results of the simulations are satisfactory, as we can see that the value of m chosen seems to minimize our error.

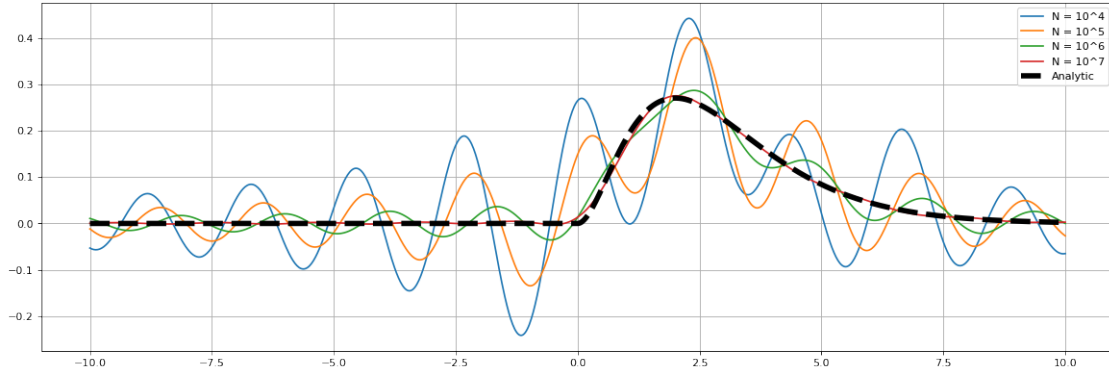


Figure 3.2 – Illustration of the estimator.

Input: Jump distribution $X \sim \Gamma(3)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.5, t_2 = 1, J = 10^5$.

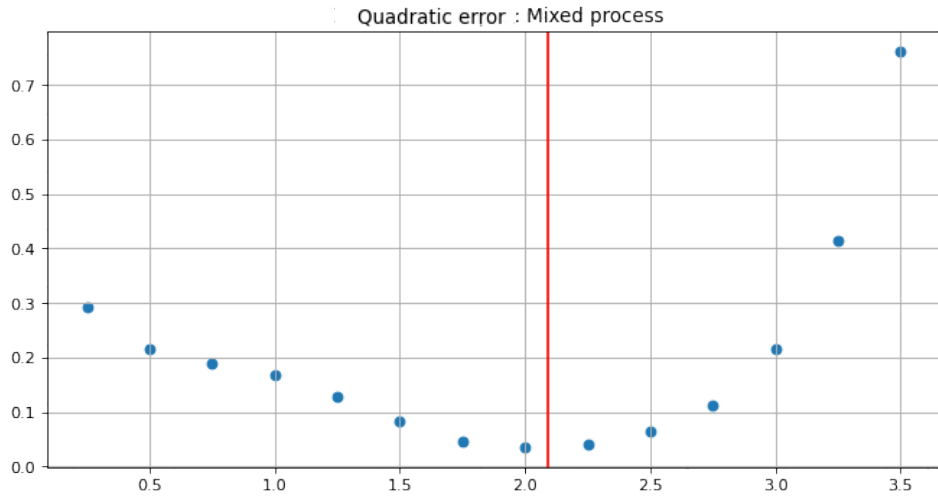


Figure 3.3 – Computations of the L^2 -risks $\mathbb{E}(\|\hat{f}_{m,J} - f\|^2)$ (y axis) for different values of m (x axis). The vertical line represents the value of $\hat{m}$ chosen automatically by the adaptive procedure. *Input: Jump distribution $X \sim 0.3\mathcal{N}(-2, 1) + 0.7\mathcal{N}(2, 1)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.5, t_2 = 1, J = 10^5$.*

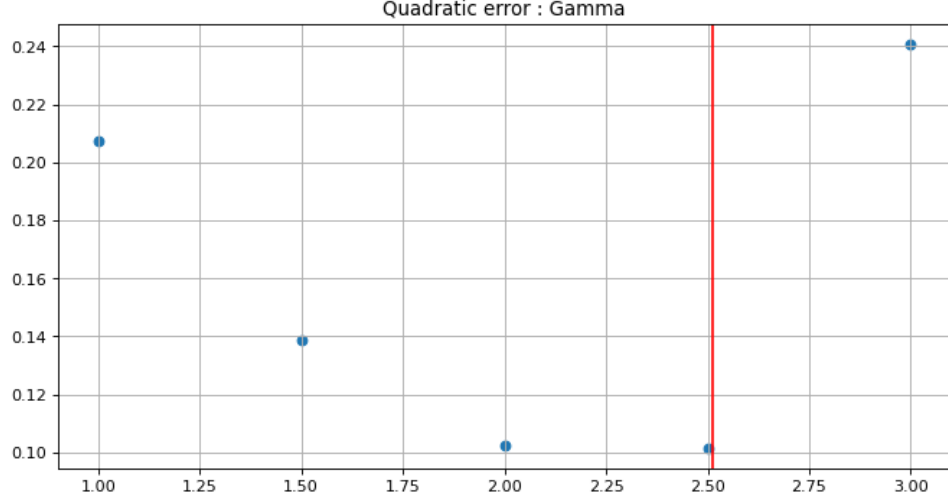


Figure 3.4 – Computations of the L^2 -risks $\mathbb{E}(\|\hat{f}_{m,J} - f\|^2)$ (y axis) for different values of m (x axis). The vertical line represents the value of $\hat{m}$ chosen automatically by the adaptive procedure. *Input: Jump distribution $X \sim \Gamma(2)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.5, t_2 = 1, J = 10^5$.*

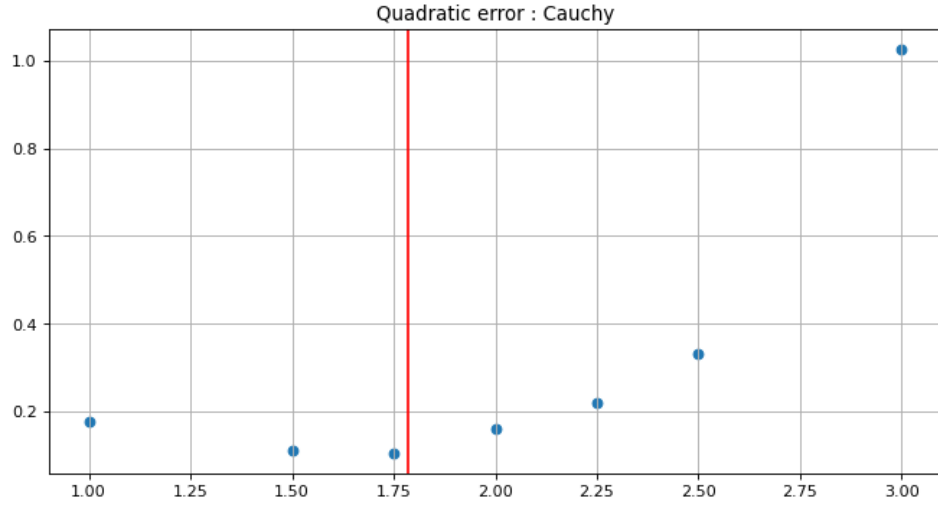


Figure 3.5 – Computations of the L^2 -risks $\mathbb{E}(\|\hat{f}_{m,J} - f\|^2)$ (y axis) for different values of m (x axis). The vertical line represents the value of $\hat{m}$ chosen automatically by the adaptive procedure. *Input: Jump distribution $X \sim \mathcal{C}(0, 1)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.5, t_2 = 1, J = 10^5$.*

How to select an optimal time t_2

To define our estimator, we have chosen two times t_1 and t_2 in an arbitrary way. Therefore, we would like to know how to select them in an optimal way.

Theorem ?? states that t_2 must be neither too large nor too close to t_1 , which we can illustrate with numerical simulation. Figure ?? shows the value of the L^2 -risks $\mathbb{E}(\|\hat{f}_{m,J} - f\|^2)$ of the estimator for different values of t_2 . Each of these values is the average of the results of 1000 different simulations. Here, it seems that the estimate is better when t_2 is between 1 and 1.5, which seems to correspond roughly to the intensity of the number of jumps.

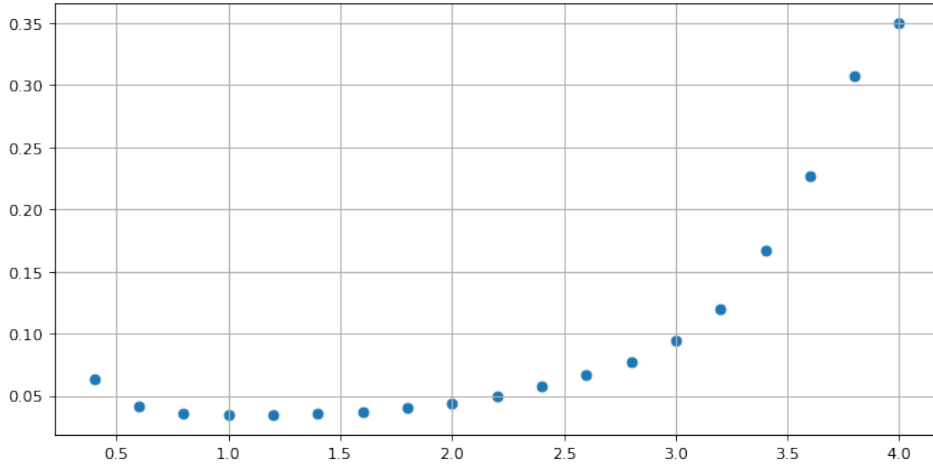


Figure 3.6 – Computations of the L^2 -risks $\mathbb{E}(\|\hat{f}_{m,J} - f\|^2)$ (y axis) for different values of t_2 (x axis).

Input: Jump distribution $X \sim 0.3\mathcal{N}(-2, 1) + 0.7\mathcal{N}(2, 1)$. Observations are corrupted by a Gaussian noise $\mathcal{N}(0, 1)$. $t_1 = 0.2$, $m = 2$, $J = 10^5$.

3.4.2 Multiplicative decompounding

In this section, we illustrate the efficiency of the Mellin estimator we built in `sec:mellin_estimator` on a simulated sample of $n = 5000$.

Here, we assume that the jump distribution densities f is the Beta law $\beta(200, 30)$. The bandwidth $m = 83.7$ was selected by the adaptive procedure described in `sec:mellin_adaptive`. The resulting estimate is plotted in `fig3_4_2`.

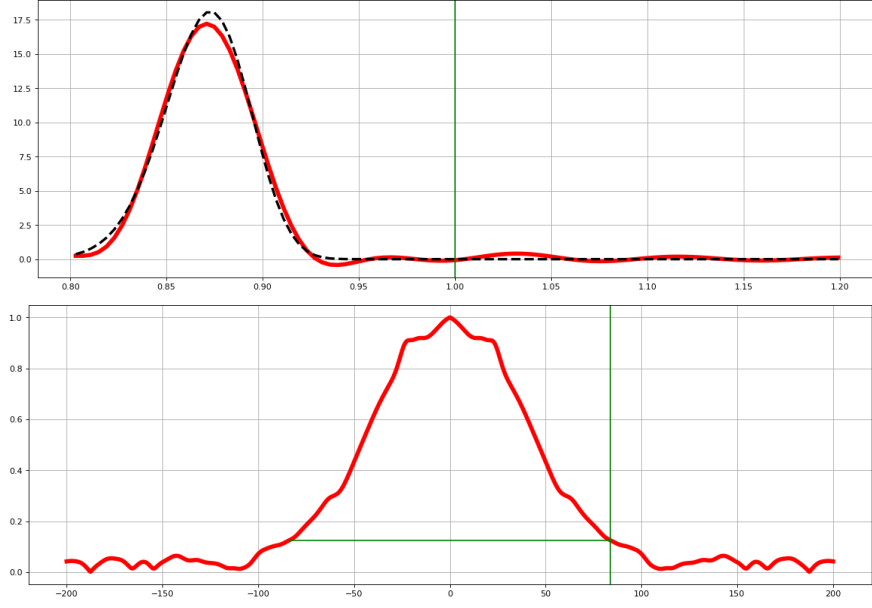


Figure 3.7 – **Top:** Estimation of the Beta density $\beta(200, 30)$ for 5000 observations. The real density is the black dotted-line and the estimator is plotted in red. **Bottom:** Illustration of the adaptive procedure. The green lines show the cutoff on the Mellin transform.

3.5 Proofs

3.5.1 Proof of Theorem ??

This proof is an adaptation of the proof of Theorem 3.1 of Duval and Kappus [6]. Nevertheless, we have in addition a noise term that must be taken into account in our majorations.

From the triangle inequality and Parseval's equality we have

$$\begin{aligned} \left\| \hat{f}_{m,J} - f \right\|^2 &\leq \|f_m - f\|^2 + \|\hat{f}_{m,J} - f_m\|^2 \\ &= \|f_m - f\|^{2 + \frac{1}{2\pi} \int_{-m}^m |\tilde{\varphi}_X^J(u) - \varphi_X(u)|^2 du} . \end{aligned}$$

By recalling that for any reals $(a, b) \in \mathbb{R}^2$, $|a + b|^2 \leq 2(a^2 + b^2)$, and by using Definition ??

and Definition ??, then we have

$$\begin{aligned}
& \int_{-m}^m |\tilde{\varphi}_X^J - \varphi_X(u)|^2 du \\
&= \frac{1}{(t_2 - t_1)^2} \int_{-m}^m |\log \tilde{\varphi}_{Z_{t_2}}^J(u) - \log \tilde{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_2}}(u) + \log \varphi_{Z_{t_1}}(u)|^2 du \\
&\leq \frac{2}{(t_2 - t_1)^2} \int_{-m}^m |\log \tilde{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_1}}(u)|^2 du + \frac{2}{(t_2 - t_1)^2} \int_{-m}^m |\log \tilde{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)|^2 du .
\end{aligned}$$

In the rest of the proof we establish a majorization of the two terms on the right.

Let $\tau \in \{t_1, t_2\}$ and $I_{\tau, J, m}$ denote the quantity

$$I_{\tau, J, m} = \frac{1}{(t_2 - t_1)^2} \int_{-m}^m |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du .$$

Let $(\gamma, \zeta) \in (0, \infty)^2$ be two positive reals such that $\gamma c_m > \zeta$.
In particular, we take γ such that $\gamma c_m = \frac{2\zeta}{1 \wedge (t_2 - t_1)} > \zeta$.

Consider the events

$$\Omega_{\zeta, \tau}(m) = \left\{ \forall u \in [-m, m], |\tilde{\varphi}_{Z_\tau}^J(u) - \varphi_{Z_\tau}(u)| \leq \zeta \sqrt{\frac{\log(J\tau)}{J\tau}} \right\}$$

and

$$M_{J, t}^{(\gamma c_m)} = \min \left\{ u \geq 0 : |\varphi_{Z_t}(u)| = \gamma c_m \sqrt{\log(Jt)/(Jt)} \right\} ,$$

as defined in Lemma ?? and Lemma ??.

As the events

$$\{|u| \leq m \wedge M_{J, \tau}^{(\gamma c_m)}\} \cap \Omega_{\zeta, \tau}(m) , \quad \{|u| \in [M_{J, \tau}^{(\gamma c_m)}, m]\} \cap \Omega_{\zeta, \tau}(m) \quad \text{and} \quad \Omega_{\zeta, \tau}(m)^c$$

form a partition of the sample set Ω , we only need to control $I_{\tau, J, m}$ on each of them. This leads us to define the quantities

$$\begin{aligned}
I_{\tau, J, m}^{(1)} &= \frac{1}{(t_2 - t_1)^2} \int_{-m \wedge M_{J, \tau}^{(\gamma c_m)}}^{m \wedge M_{J, \tau}^{(\gamma c_m)}} \mathbb{1}_{\Omega_{\zeta, \tau}(m)} |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du , \\
I_{\tau, J, m}^{(2)} &= \frac{1}{(t_2 - t_1)^2} \mathbb{1}_{m > M_{J, \tau}^{(\gamma c_m)}} \cdot \mathbb{1}_{\Omega_{\zeta, \tau}(m)} \int_{|u| \in [M_{J, \tau}^{(\gamma c_m)}, m]} |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du , \\
I_{\tau, J, m}^{(3)} &= \frac{1}{(t_2 - t_1)^2} \mathbb{1}_{\Omega_{\zeta, \tau}(m)^c} \int_{-m}^m |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du .
\end{aligned}$$

Obviously, we have the decomposition

$$I_{\tau,J,m} = I_{\tau,J,m}^{(1)} + I_{\tau,J,m}^{(2)} + I_{\tau,J,m}^{(3)} .$$

3.5.1.0.1 Step 1 – Control of $I_{\tau,J,m}^{(1)}$ Define the event $\mathcal{A} = \{|u| \leq m \wedge M_{J,\tau}^{(\gamma c_m)}\} \cap \Omega_{\zeta,\tau}(m)$. The triangle inequality ensures that

$$|\log \widehat{\varphi}_{Z_\tau}^J(u)| \leq |\log \widehat{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)| + |\log \varphi_{Z_\tau}(u)| .$$

Then Lemma ?? applied to $\zeta < \gamma c_m$ ensures that

$$|\log \widehat{\varphi}_{Z_\tau}^J(u)| \leq \frac{\gamma c_m}{\zeta} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) \frac{|\widehat{\varphi}_{Z_\tau}^J(u) - \varphi_{Z_\tau}(u)|}{|\varphi_{Z_\tau}(u)|} + |\log \varphi_{Z_\tau}(u)| .$$

Then we have by definition of $\Omega_{\zeta,\tau}(m)$,

$$|\log \widehat{\varphi}_{Z_\tau}^J(u)| \leq \frac{\gamma c_m}{\zeta} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) \frac{1}{|\varphi_{Z_\tau}(u)|} \zeta \sqrt{\frac{\log(J\tau)}{J\tau}} + |\log \varphi_{Z_\tau}(u)| .$$

For $J\tau$ large enough, we have $\gamma c_m \sqrt{\frac{\log(J\tau)}{J\tau}} < 1$, hence on $\mathcal{A}$ the definition of $M_{J,\tau}^{(\gamma c_m)}$ ensures that

$$\frac{1}{|\varphi_{Z_\tau}(u)|} \leq \frac{1}{\gamma c_m \sqrt{\frac{\log(J\tau)}{J\tau}}} .$$

It follows that

$$\begin{aligned} |\log \widehat{\varphi}_{Z_\tau}^J(u)| &\leq \frac{\gamma c_m}{\zeta} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) \frac{1}{\gamma c_m \sqrt{\frac{\log(J\tau)}{J\tau}}} \zeta \sqrt{\frac{\log(J\tau)}{J\tau}} + |\log \varphi_{Z_\tau}(u)| , \\ &\leq \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + |\log \varphi_{Z_\tau}(u)| . \end{aligned}$$

By definition, this means (see also Proposition ??)

$$|\log \widehat{\varphi}_{Z_\tau}^J(u)| \leq \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + |\log \varphi_{Y_\tau}(u) + \log \varphi_\varepsilon(u)| ,$$

However, we know that for all $u \in \mathbb{R}$

$$\varphi_X(u) = 1 + \frac{1}{\tau} \log \varphi_{Y_\tau}(u) .$$

It follows that

$$\begin{aligned} |\log \widehat{\varphi}_{Z_\tau}^J(u)| &\leq \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + |\tau(\varphi_X(u) + 1) + \log \varphi_\varepsilon(u)| \\ &\leq \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + 2\tau + |\log \varphi_\varepsilon(u)| , \\ &\leq \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + 2\tau + \sup_{[-m,m]} |\log \varphi_\varepsilon(\cdot)| . \end{aligned}$$

Therefore,

$$\left| \frac{\log \widehat{\varphi}_{Z_\tau}^J(u)}{t_2 - t_1} \right| \leq \frac{1}{t_2 - t_1} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) + \frac{2\tau}{t_2 - t_1} + \frac{\sup_{[-m, m]} |\log \varphi_\varepsilon(\cdot)|}{t_2 - t_1}.$$

There are two different cases.

— If $t_2 - t_1 \geq 1$, then $\gamma c_m = \frac{2\zeta}{1 \wedge (t_2 - t_1)} = 2\zeta$. It follows that

$$\frac{1}{t_2 - t_1} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) = \frac{1}{t_2 - t_1} \log(2) < 1.$$

— If $t_2 - t_1 < 1$, then $\gamma c_m = \frac{2\zeta}{1 \wedge (t_2 - t_1)} = \frac{2\zeta}{t_2 - t_1}$. It follows that

$$\begin{aligned} \frac{1}{t_2 - t_1} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) &= \frac{1}{t_2 - t_1} \log \left(\frac{\frac{2\zeta}{t_2 - t_1}}{\frac{2\zeta}{t_2 - t_1} - \frac{\zeta(t_2 - t_1)}{t_2 - t_1}} \right) \\ &= \frac{1}{t_2 - t_1} \log \left(\frac{2}{2 - (t_2 - t_1)} \right) \\ &\leq 1, \end{aligned}$$

because for any real $x \in (0, 1)$ we have

$$0 < \frac{1}{x} \log \left(\frac{2}{2 - x} \right) < 1.$$

It leads to

$$\begin{aligned} \left| \frac{\log \widehat{\varphi}_{Z_\tau}^J(u)}{t_2 - t_1} \right| &\leq 1 + \frac{2\tau}{t_2 - t_1} + \frac{\sup_{[-m, m]} |\log \varphi_\varepsilon(\cdot)|}{t_2 - t_1} \\ &\leq 1 + \frac{2t_2}{t_2 - t_1} + \frac{\sup_{[-m, m]} |\log \varphi_\varepsilon(\cdot)|}{t_2 - t_1}. \end{aligned} \quad (3.23)$$

Since we consider $0 \leq m \leq C_{t_1, t_2}^J$, we have

$$\left| \log \widehat{\varphi}_{Z_\tau}^J(u) \right| \leq 3t_2 - t_1 + \sup_{[-m, m]} |\log \varphi_\varepsilon(\cdot)| \leq \ln(J).$$

Therefore, it comes from Equation (??) and Equation (??) that

$$\log \widetilde{\varphi}_{Z_\tau}^J(u) = \log \widehat{\varphi}_{Z_\tau}^J(u).$$

By using Lemma ?? and the definition of γ , we have

$$\begin{aligned} \mathbb{E}[I_{\tau, J, m}^{(1)}] &= \frac{1}{(t_2 - t_1)^2} \int_{-m \wedge M_{J, \tau}^{(\gamma c_m)}}^{m \wedge M_{J, \tau}^{(\gamma c_m)}} e \left[\mathbb{1}_{\Omega_{\zeta, \tau}(m)} |\log \widehat{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 \right] du \\ &\leq \frac{1}{(t_2 - t_1)^2} \int_{-m}^m \frac{\mathbb{E}[|\widehat{\varphi}_{Z_\tau}^J(u) - \varphi_{Z_\tau}(u)|^2]}{|\varphi_{Z_\tau}(u)|^2} du, \end{aligned}$$

because $\frac{\gamma c_m}{\zeta} \log \left(\frac{\gamma c_m}{\gamma c_m - \zeta} \right) \leq 1$ due to the fact that $\gamma c_m > \zeta$.

Direct computations lead to

$$\begin{aligned} \mathbb{E}[|\hat{\varphi}_{Z_\tau}^J(u) - \varphi_{Z_\tau}(u)|^2] &= \frac{1 - |\varphi_{Z_\tau}(u)|^2}{J} \\ &= \frac{1 - |\varphi_{Y_\tau}(u)|^2 |\varphi_\varepsilon(u)|^2}{J} \\ &= \frac{(1 - |\varphi_{Y_\tau}(u)|^2) |\varphi_\varepsilon(u)|^2 + 1 - |\varphi_\varepsilon(u)|^2}{J} \\ &\leq \frac{(1 - |\varphi_{Y_\tau}(u)|^2) |\varphi_\varepsilon(u)|^2 + 1 - |\varphi_\varepsilon(u)|^2}{J} \\ &\leq \frac{2}{J}. \end{aligned}$$

We deduce that

$$\mathbb{E}[I_{\tau, J, m}^{(1)}] \leq \frac{2}{J(t_2 - t_1)^2} \int_{-m}^m \frac{1}{|\varphi_{Z_\tau}(u)|^2} du.$$

Futhermore, we know that (Y_τ) is a compound Poisson process with intensity $\lambda = 1$. It leads to

$$|\varphi_{Y_\tau}(u)| \geq e^{-2\tau}, \quad u \in \mathbb{R}.$$

Then,

$$|\varphi_{Z_\tau}(u)|^2 = |\varphi_{Y_\tau}(u) \varphi_\varepsilon(u)|^2 \geq e^{-4\tau} |\varphi_\varepsilon(u)|^2.$$

It leads to

$$\mathbb{E}[I_{\tau, J, m}^{(1)}] \leq \frac{2e^{4\tau}}{J(t_2 - t_1)^2} \int_{-m}^m \frac{1}{|\varphi_\varepsilon(u)|^2} du.$$

3.5.1.0.2 Step 2 – Control of $I_{\tau, J, m}^{(3)}$ on the event $\Omega_{\zeta, \tau}(m)^c$, it is more complicated to control the empirical distinguished logarithm. Nevertheless, the cut-off on the estimator allows us to write that

$$\begin{aligned} \int_{-m}^m |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du &\leq 2 \int_{-m}^m |\log \tilde{\varphi}_{Z_\tau}^J(u)|^2 du + 2 \int_{-m}^m |\log \varphi_{Z_\tau}(u)|^2 du \\ &\leq 4m \ln^2(J) + 2 \int_{-m}^m |\log \varphi_{Y_\tau}(u) + \log \varphi_\varepsilon(u)|^2 du \\ &\leq 4m \ln^2(J) + 4 \int_{-m}^m |\log \varphi_{Y_\tau}(u)|^2 du + 4 \int_{-m}^m |\log \varphi_\varepsilon(u)|^2 du \\ &\leq 4m \ln^2(J) + 32m\tau^2 + 4 \int_{-m}^m |\log \varphi_\varepsilon(u)|^2 du. \end{aligned}$$

Indeed, we have that $\varphi_X = 1 + \frac{1}{\tau} \log \varphi_{Y_\tau}$. It follows that $|\log \varphi_{Y_\tau}| \leq 2\tau$.

Recall that we have

$$I_{\tau,J,m}^{(3)} = \frac{1}{(t_2 - t_1)^2} \mathbb{1}_{\Omega_{\zeta,\tau}(m)^c} \int_{-m}^m |\log \tilde{\varphi}_{Z_\tau}^J(u) - \log \varphi_{Z_\tau}(u)|^2 du ,$$

which leads to write

$$\mathbb{E}[I_{\tau,J,m}^{(3)}] \leq \frac{1}{(t_2 - t_1)^2} \cdot \left(4m \ln^2(J) + 32m\tau^2 + 4 \int_{-m}^m |\log \varphi_\varepsilon(u)|^2 du \right) \cdot \mathbb{P}(\Omega_{\zeta,\tau}(m)^c) .$$

We apply Lemma ?? with $\eta = 2$ and $\zeta > \sqrt{5\tau}$. It ensures that

$$\mathbb{E}[I_{\tau,J,m}^{(3)}] \leq \frac{1}{(t_2 - t_1)^2} \cdot \left(4m \ln^2(J) + 32m\tau^2 + 4 \int_{-m}^m |\log \varphi_\varepsilon(u)|^2 du \right) \cdot \left(\frac{\mathbb{E}[X_i^2]}{J\tau} + \frac{\mathbb{E}[\varepsilon^2]}{J\tau^2} + 4 \frac{m}{(J\tau)^2} \right) .$$

3.5.1.0.3 Step 3 – Control of $I_{\tau,J,m}^{(2)}$ By Assumption, $\tau \leq \delta \log(J\tau)$ and $\delta < 1/4$. It follows that for all $u \in [-m, m]$,

$$|\varphi_{Z_\tau}(u)| = |\varphi_{Y_\tau}| \cdot |\varphi_\varepsilon| > e^{-2\tau} c_m > (J\tau)^{-2\delta} c_m > \gamma c_m \sqrt{\log(J\tau)/(J\tau)} \Big\} ,$$

when $J\tau$ is enough high.

Finally, $M_{J,\tau}^{(\gamma c_m)} \geq m$, and $I_{\tau,J,m}^{(2)} = 0$ almost surely.

3.5.2 Proof of Theorem ??

Let $m \geq 0$. The proof of the theorem is divided in two steps: in a first one, we control $\mathbb{E} \left\| \hat{f}_{\hat{m}_n, J} - f \right\|^2$ on the event $\mathcal{E} = \{\hat{m}_J < m\}$, then we control it on the complementary event $\mathcal{E}^c = \{\hat{m}_J \geq m\}$.

As we know, it follows from the triangle inequality and Parseval's equality that

$$\begin{aligned} \left\| \bar{f}_{m,J} - f \right\|^2 &= \|f_m - f\|^{2+\frac{1}{2\pi}} \int_{-m}^m |\tilde{\varphi}_X^J(u) - \varphi_X(u)|^2 du \\ &= \frac{1}{2\pi} \int_{|u| \in [-m, m]^c} |\varphi_X(u)|^2 du + \frac{1}{2\pi} \int_{-m}^m |\tilde{\varphi}_X^J(u) - \varphi_X(u)|^2 du . \end{aligned}$$

Therefore, by replacing the estimator $\hat{f}_{m,J}$ by $\hat{f}_{\hat{m}_n, J}$ produces a surplus of bias on $\mathcal{E}$ and and surplus of variance on $\mathcal{E}^c$. Then, we only need to control these surpluses.

3.5.2.0.1 Step 1 – Control on $\mathcal{E}$

$$\begin{aligned}
\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_J, m]} |\varphi_X(u)|^2 du \right] &\leq 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_J, m]} |\widetilde{\varphi}_X^J(u)|^2 du \right] + 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_J, m]} |\widetilde{\varphi}_X^J(u) - \varphi_X(u)|^2 du \right] \\
&\leq 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_J, m]} \frac{\kappa_{J, t_1, t_2}^2 \cdot (1 + u^2)}{J(t_2 - t_1)^2} du \right] + 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [0, m]} |\widetilde{\varphi}_X^J(u) - \varphi_X(u)|^2 du \right] \\
&\leq 4m \cdot (1 + m^2) \cdot \frac{\kappa_{J, t_1, t_2}^2}{J(t_2 - t_1)^2} + 2A,
\end{aligned}$$

where $A = \sum_{i=1}^2 \frac{4e^{4t_i}}{J(t_2 - t_1)^2} \int_{-m}^m \frac{du}{|\varphi_\varepsilon(u)|^2} + \frac{4K_{J, t_1, t_2}}{(t_2 - t_1)^2} \cdot \left(\frac{\mathbb{E}[X_i^2]}{Jt_i} + \frac{\mathbb{E}[\varepsilon^2]}{Jt_i^2} + 4\frac{m}{(Jt_i)^2} \right)$ is given by Theorem ??.

Given that $\kappa_{J, t_1, t_2} = 2\sqrt{2}e^{2t_2} + \kappa\sqrt{\ln(J(t_2 - t_1)^2)}$, we have that

$$\mathbb{E} \left[\|\bar{f}_{\widehat{m}_J} - f\|^{2\mathbb{1}_{\mathcal{E}}} \right] \leq \|f_m - f\|^{2+C \frac{m \cdot (1+m^2) \cdot \ln(J(t_2 - t_1)^2)}{J(t_2 - t_1)^2} + \bar{C}A},$$

where C and $\bar{C}$ are two constantes.

3.5.2.0.2 Step 2 – Control on $\mathcal{E}^c$ Now, we look at the case where where $\widehat{m}_J > m$. By construction, $\widehat{m}_J$ is bounded by $(m_{\max})^\alpha$. It follows that

$$\mathbb{E} \left[\int_{m < |u| < \widehat{m}_J} |\widetilde{\varphi}_X^J(u) - \varphi(u)|^2 du \mathbb{1}_{\mathcal{E}^c} \right] \leq \mathbb{E} \left[\int_{m < |u| < (m_{\max})^\alpha} |\widetilde{\varphi}_X^J(u) - \varphi(u)|^2 du \right].$$

Let $\eta > 2$, such that $\alpha - \eta < -1$. As for the proof of the theorem ??, Lemma ?? and Lemma ?? leads to

$$\begin{aligned}
\mathbb{E} [|\widetilde{\varphi}_X^J(u) - \varphi(u)|^2] &\leq |\varphi_X(u)|^2 + \mathbb{E} [|\widehat{\varphi}_X^J(u) - \varphi(u)|^2] \\
&\leq |\varphi_X(u)|^2 + \sum_{i=1}^2 \frac{4e^{4t_i}}{J(t_2 - t_1)^2} \frac{du}{|\varphi_\varepsilon(u)|^2} + \frac{\mathbb{E}[X_1^2]}{Jt_1} + \frac{\mathbb{E}[\varepsilon^2]}{Jt_1^2} + 4\frac{m}{(Jt_1)^\eta}.
\end{aligned}$$

On the event

$$\left| \frac{\varphi_X(u)}{\sqrt{1+u^2}} \right| \geq \frac{2e^{2t_2}}{\sqrt{J}(t_2 - t_1)},$$

we see that

$$\begin{aligned}
\mathbb{E} [|\widetilde{\varphi}_X^J(u) - \varphi(u)|^2] &\leq |\varphi_X(u)|^2 + \frac{8}{J(t_2 - t_1)^2} \frac{e^{4t_2}}{|\varphi_\varepsilon(u)|^2} + \frac{\mathbb{E}[X_1^2]}{Jt_1} + \frac{\mathbb{E}[\varepsilon^2]}{Jt_1^2} + 4\frac{m}{(Jt_1)^\eta} \\
&\leq |\varphi_X(u)|^2 + \frac{2|\varphi_X(u)|^2}{1+u^2} \frac{1}{|\varphi_\varepsilon(u)|^2} + \frac{\mathbb{E}[X_1^2]}{Jt_1} + \frac{\mathbb{E}[\varepsilon^2]}{Jt_1^2} + 4\frac{m_{\max}}{(Jt_1)^\eta} \\
&\leq |\varphi_X(u)|^2 + \frac{2|\varphi_X(u)|^2}{(1+u^2)} \frac{(1+u^2)}{d} + \frac{\mathbb{E}[X_1^2]}{Jt_1} + \frac{\mathbb{E}[\varepsilon^2]}{Jt_1^2} + 4\frac{(t_2 - t_1)^{2\alpha}}{J \cdot t_1^\eta} \\
&\leq C_0 |\varphi_X(u)|^2,
\end{aligned}$$

as $J \rightarrow \infty$. From now on, all that remains to be done is to look at the event

$$\mathcal{A} = \left\{ \left| \frac{\varphi_X(u)}{\sqrt{1+u^2}} \right| \leq \frac{2e^{2t_2}}{\sqrt{J}(t_2-t_1)} \right\}.$$

By definition, we know that

$$\bar{\varphi}_X^J(u) \leq 1 + \frac{2 \log(J)}{(t_2-t_1)}.$$

Then, on the event $\mathcal{A}$ we see that

$$\begin{aligned} & \mathbb{E} \left[\int_{[m, \hat{m}_n]} |\bar{\varphi}_X^J(u) - \varphi(u)|^2 \cdot \mathbf{1}_{\mathcal{A}} \right] \\ & \leq \int_{|u| \in [m, (m_{\max})^\alpha]} |\varphi(u)|^2 du + \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2 \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P} \left(\left| \frac{\hat{\varphi}_X(u)}{\sqrt{1+u^2}} \right| \geq \frac{\kappa_{J,t_1,t_2}}{\sqrt{J}(t_2-t_1)} \right) \mathbf{1}_{\mathcal{A}} du \\ & \leq \int_{|u| \in [m, (m_{\max})^\alpha]} |\varphi(u)|^2 du + \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2 \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P} \left(\left| \frac{\hat{\varphi}_X(u)}{\sqrt{1+u^2}} \right| \geq \frac{\kappa_{J,t_1,t_2}}{\sqrt{J}(t_2-t_1)} \right) \mathbf{1}_{\mathcal{A}} du \\ & \leq \int_{|u| \in [m, (m_{\max})^\alpha]} |\varphi(u)|^2 du + \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2 \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P} \left(\left| \frac{\hat{\varphi}_X(u) - \varphi(u)}{\sqrt{1+u^2}} \right| \geq \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) du \\ & \leq \int_{|u| \in [m, (m_{\max})^\alpha]} |\varphi(u)|^2 du + \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2 \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P} \left(|\hat{\varphi}_X(u) - \varphi(u)| \geq \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) du \\ & \leq \int_{|u| \in [m, (m_{\max})^\alpha]} |\varphi(u)|^2 du + T_J \left(2 + \frac{2 \log(J)}{(t_2-t_1)} \right)^2. \end{aligned}$$

Finally, it only remains to bound T_J where

$$T_J = \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P} \left(|\hat{\varphi}_X(u) - \varphi(u)| \geq \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) du.$$

By definition, we have

$$\begin{aligned} & \mathbb{P} \left(|\hat{\varphi}_X(u) - \varphi(u)| \geq \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) \\ & = \mathbb{P} \left(\left| \frac{\log \hat{\varphi}_{Z_{t_2}}^J(u) - \log \hat{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_2}}(u) + \log \varphi_{Z_{t_1}}(u)}{t_2-t_1} \right| \geq \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) \\ & \leq \mathbb{P} \left(|\log \hat{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_1}}(u)| + |\log \hat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)| \geq (t_2-t_1) \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) \\ & \leq \mathbb{P} \left(|\log \hat{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_1}}(u)| \geq (t_2-t_1) \frac{1}{2} \cdot \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right) \\ & \quad + \mathbb{P} \left(|\log \hat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)| \geq (t_2-t_1) \frac{1}{2} \cdot \frac{\kappa \sqrt{\ln(J(t_2-t_1)^2)}}{\sqrt{J}(t_2-t_1)} \right). \end{aligned}$$

Without loss of generality, we only consider the term

$$\mathbb{P}\left(\left|\log \widehat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)\right| \geq (t_2 - t_1) \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right).$$

By taking γ_Δ and $\zeta > 0$, like in the proof of Theorem ??, we have that

$$\begin{aligned} & \mathbb{P}\left(\left|\log \widehat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)\right| \geq (t_2 - t_1) \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right) \\ & \leq \mathbb{P}\left(\left|\widehat{\varphi}_{Z_{t_2}}^J(u) - \varphi_{Z_{t_2}}(u)\right| \geq (t_2 - t_1) |\varphi_{Z_{t_2}}(u)| \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right) + \mathbb{P}(\Omega_{\zeta, (t_2 - t_1)^2}^c). \end{aligned}$$

Using hypothesis on the regularity of $|\varphi_\varepsilon|$, we have that

$$|\varphi_{Z_{t_2}}| = |\varphi_{Y_{t_2}}| \cdot |\varphi_\varepsilon| \geq |\varphi_{Y_{t_2}}| \cdot \frac{d}{\sqrt{1+u^2}} \geq e^{2t_2} \cdot \frac{d}{\sqrt{1+(m_{\max})^2}}.$$

Let $c(\theta) = \kappa(t_2 - t_1)e^{2t_2} \cdot \frac{d}{\sqrt{1+(m_{\max})^2}}$.

It follows from Hoeffding inequality and Lemma ?? that

$$\begin{aligned} & \mathbb{P}\left(\left|\log \widehat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)\right| \geq \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right) \\ & \leq 2(J(t_2 - t_1)^2)^{-c(\theta)^2} + \frac{\mathbb{E}[X_1^2]}{J(t_2 - t_1)^2} + \frac{\mathbb{E}[\varepsilon^2]}{J(t_2 - t_1)^4} + 4 \frac{m_{\max}}{(J(t_2 - t_1)^2)^\eta}. \end{aligned}$$

If we take $\eta > 3$ such that $2\alpha - \eta < -1$ and $\zeta > \sqrt{(t_2 - t_1)^2(1 + 2\eta)}$, then we have

$$\begin{aligned} & \int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P}\left(\left|\log \widehat{\varphi}_{Z_{t_2}}^J(u) - \log \varphi_{Z_{t_2}}(u)\right| \geq \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right) \\ & \leq 4(J(t_2 - t_1)^2)^{\alpha - c(\theta)^2} + \frac{C'}{J(t_2 - t_1)^2} + \frac{C'}{J(t_2 - t_1)^4} \end{aligned}$$

where C' depends on $\mathbb{E}[X_1^2], \mathbb{E}[\varepsilon^2]$. A similar reasoning on

$$\int_{|u| \in [m, (m_{\max})^\alpha]} \mathbb{P}\left(\left|\log \widehat{\varphi}_{Z_{t_1}}^J(u) - \log \varphi_{Z_{t_1}}(u)\right| \geq \frac{\kappa \sqrt{\ln(J(t_2 - t_1)^2)}}{2\sqrt{J}(t_2 - t_1)}\right),$$

leads to

$$T_J \leq C_0(J(t_2 - t_1)^2)^{\alpha - c(\theta)^2} + \frac{C_1}{J(t_2 - t_1)^2} + \frac{C_2}{J(t_2 - t_1)^4}$$

where C_0, C_1 and C_2 depends on $\mathbb{E}[X_1^2], \mathbb{E}[\varepsilon^2]$.

3.5.3 Proof of Theorem ??

This proof is an adaptation of the proof of Theorem 3.1 of Duval and Kappus [6] for the settings of the multiplicative decomposing.

We have the decomposition

$$\|\widehat{f_m} - f\|_{\omega_1}^2 \leq \|f_m - f\|_{\omega_1}^2 + \|\widehat{f_m} - f_m\|_{\omega_1}^2.$$

To prove the theorem, we have to bound the variance term $\|\widehat{f_m} - f_m\|_{\omega_1}^2$. Using the isometry equation, we get

$$\|\widehat{f_m} - f_m\|_{\omega_1}^2 = \frac{1}{2\pi} \int_{-m}^m |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds.$$

To get a majorization of the right side term, we consider different events on which it can be control.

Fix $\gamma > \zeta$ and set $\gamma_\Delta = \frac{2\zeta}{1 \wedge \Delta}$. We consider the events

$$\Omega_{\zeta, \Delta}(m) = \left\{ \forall u \in [-m, m], |\widehat{\mathcal{M}_1[\Delta]}(u) - \mathcal{M}_1[\Delta](u)| \leq \zeta \sqrt{\frac{\log(n\Delta)}{n\Delta}} \right\}$$

and

$$\{m \leq M_{n, \Delta}^{(\gamma_\Delta)}\} \quad \text{where} \quad M_{n, \Delta}^{(\gamma_\Delta)} = \min \left\{ u \geq 0 : |\mathcal{M}_1[\Delta](u)| = \gamma c_m \sqrt{\log(Jt)/(Jt)} \right\}.$$

We have

$$\begin{aligned} \frac{1}{2\pi} \int_{-m}^m |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds &= \frac{1}{2\pi} \int_{-m \wedge M_{n, \Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n, \Delta}^{(\gamma_\Delta)}} \mathbb{1}_{\Omega_{\zeta, \Delta}(m)} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \\ &\quad + \frac{1}{2\pi} \mathbb{1}_{m > M_{n, \Delta}^{(\gamma_\Delta)}} \cdot \mathbb{1}_{\Omega_{\zeta, \Delta}(m)} \int_{|s| \in [M_{n, \Delta}^{(\gamma_\Delta)}, m]} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \\ &\quad + \mathbb{1}_{\Omega_{\zeta, \Delta}^c(m)} \frac{1}{2\pi} \int_{-m}^m |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \end{aligned}$$

Focus on the event $\mathcal{A} = \{|u| \leq m \wedge M_{n, \Delta}^{(\gamma_\Delta)}\} \cap \Omega_{\zeta, \Delta}(m)$. First, we prove that

$$\widehat{\mathcal{M}_1[f]}(s) = \widetilde{\mathcal{M}_1[f]}(s).$$

The triangle inequality ensures that

$$\widehat{\mathcal{M}_1[f]}(s) \leq 1 + \frac{|\log \widehat{\mathcal{M}_1[\Delta]}(s) - \log \mathcal{M}_1[\Delta](s)| + |\log \mathcal{M}_1[\Delta](s)|}{\Delta}.$$

and Lemma ?? ensures that

$$\widehat{\mathcal{M}_1[f]}(s) \leq 1 + \frac{1}{\Delta} \log \left(\frac{\gamma}{\gamma - \zeta} \right) \frac{|\widehat{\mathcal{M}_1[\Delta]}(s) - \mathcal{M}_1[\Delta](s)|}{|\mathcal{M}_1[\Delta](s)|} + \frac{|\log \mathcal{M}_1[\Delta](s)|}{\Delta}.$$

Furthermore, since

$$\mathcal{M}_1[f](s) = 1 + \frac{1}{\Delta} \log \mathcal{M}_1[\Delta](s),$$

it follows that

$$\widehat{\mathcal{M}_1[f]}(s) \leq 3 + \frac{1}{\Delta} \log \left(\frac{\gamma}{\gamma - \zeta} \right) \leq 4.$$

Then $\widehat{\mathcal{M}_1[f]}(s) = \widetilde{\mathcal{M}_1[f]}(s)$.

Therefore,

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2\pi} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \mathbb{1}_{\Omega_{\zeta,\Delta}(m)} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \right] \\ = \frac{1}{2\pi} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \mathbb{E} \left[\mathbb{1}_{\Omega_{\zeta,\Delta}(m)} |\widehat{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 \right] ds \\ = \frac{1}{2\pi\Delta^2} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \mathbb{E} \left[\mathbb{1}_{\Omega_{\zeta,\Delta}(m)} |\log \widehat{\mathcal{M}_1[\Delta]}(s) - \log \mathcal{M}_1[\Delta](s)|^2 \right] ds. \end{aligned}$$

Lemma ?? ensures that

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2\pi} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \mathbb{1}_{\Omega_{\zeta,\Delta}(m)} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \right] \\ \leq \frac{1}{2\pi\Delta^2} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \frac{\mathbb{E} [|\widehat{\mathcal{M}_1[\Delta]}(s) - \mathcal{M}_1[\Delta](s)|^2]}{|\mathcal{M}_1[\Delta](s)|^2} ds. \end{aligned}$$

We have

$$\mathbb{E} [|\widehat{\mathcal{M}_1[\Delta]}(s) - \mathcal{M}_1[\Delta](s)|^2] = \text{Var}(\widehat{\mathcal{M}_1[\Delta]}(s)) \leq \frac{1}{n} \mathbb{E} (|Z_\Delta^{c-1+it}|^2).$$

Then we have

$$\mathbb{E} \left[\frac{1}{2\pi} \int_{-m \wedge M_{n,\Delta}^{(\gamma_\Delta)}}^{m \wedge M_{n,\Delta}^{(\gamma_\Delta)}} \mathbb{1}_{\Omega_{\zeta,\Delta}(m)} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \right] \leq \frac{1}{2\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds.$$

Then, set $\zeta > \sqrt{5}\Delta$. Using Lemma ?? with $\eta = 2$ to get

$$\mathbb{E} \left[\mathbb{1}_{\Omega_{\zeta,\Delta}^c(m)} \frac{1}{2\pi} \int_{-m}^m |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \right] \leq \frac{50m}{2\pi} \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right).$$

Finally, we observe that $|\mathcal{M}_1(\Delta)(s)| > e^{-2\Delta}$. Following the strategy describes in Duval Kappus [6], we obtain that $M_{n,\Delta}^{\gamma_\Delta} = \infty$ and that

$$\mathbb{E} \left[\frac{1}{2\pi} \mathbb{1}_{m > M_{n,\Delta}^{(\gamma_\Delta)}} \cdot \mathbb{1}_{\Omega_{\zeta,\Delta}(m)} \int_{|s| \in [M_{n,\Delta}^{(\gamma_\Delta)}, m]} |\widetilde{\mathcal{M}_1[f]}(s) - \mathcal{M}_1[f](s)|^2 ds \right] = 0.$$

3.5.4 Proof of Theorem ??

Let $0 < m < (n\Delta)^\alpha$. As for the thm:oracle, the proof is divided in two steps: in the first one, we control $\mathbb{E} [\|f_{\widehat{m}_{n,\Delta}} - f\|_{\omega_1}^2]$ on the event $\mathcal{E} = \{\widehat{m}_n < m\}$, then we control it on the complementary event $\mathcal{E}^c = \{\widehat{m}_n \geq m\}$.

3.5.4.0.1 Step 1 – The first event $\mathcal{E}$ The triangle inequality and the isometry equality leads to

$$\begin{aligned} \mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_n, m]} |\mathcal{M}_1[f](u)|^2 du \right] &\leq 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_n, m]} |\widetilde{\mathcal{M}}_1[f](u)|^2 du \right] + 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_n, m]} |\widetilde{\mathcal{M}}_1[f](u) - \mathcal{M}_1[f](u)|^2 du \right] \\ &\leq 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [\widehat{m}_n, m]} \frac{\kappa_{n,\Delta}^2}{n\Delta} du \right] + 2\mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \int_{|u| \in [0, m]} |\widetilde{\mathcal{M}}_1[f](u) - \mathcal{M}_1[f](u)|^2 du \right] \\ &\leq 4 \frac{\kappa_{n,\Delta}^2 m}{n\Delta} + \frac{1}{\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + \frac{50}{\pi} m \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right). \end{aligned}$$

By definition, $\kappa_{n,\Delta} = e^{2\Delta} + \kappa \sqrt{\log(n\Delta)}$. It follows from T:risk that

$$\begin{aligned} \mathbb{E} \left[\mathbb{1}_{\mathcal{E}} \|\widehat{f_{m_n, \Delta}} - f\|_{\omega_1}^2 \right] &\leq \|f_m - f\|_{\omega_1}^2 + C \frac{\log(n\Delta)m}{n\Delta} \\ &\quad + \frac{1}{\pi n \Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + \frac{50}{\pi} m \left(\frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^2} \right). \end{aligned}$$

3.5.4.0.2 Step 2 – The second event $\mathcal{E}^c$ Henceforth, we consider the case where $\widehat{m}_n \geq m$. It remains to control the surplus in the variance of $f_{\widehat{m}_n}$.

Since $m \leq \widehat{m}_n \leq (n\Delta)^\alpha$, it follows that

$$\mathbb{E} \left[\mathbb{1}_{\mathcal{E}^c} \int_{|u| \in [m, \widehat{m}_n]} |\overline{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2 \right] \leq \left[\int_{|u| \in [m, (n\Delta)^\alpha]} \mathbb{E} [|\overline{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2] \right].$$

Let $\eta > 2$ such that $\alpha - \eta < -1$. lm:Mellin1 and lm:Mellin2 ensure that

$$\begin{aligned} \mathbb{E} [|\overline{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2] &\leq |\mathcal{M}_1[f](s)|^2 + \mathbb{E} [|\widehat{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2] \\ &\leq |\mathcal{M}_1[f](s)|^2 + \frac{2}{n\Delta^2 |\mathcal{M}_1[\Delta](s)|^2} + \frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + \frac{4}{n\Delta}. \end{aligned}$$

First, we assume that $\{|\mathcal{M}_1[f](s)|^2 > \frac{e^{2\Delta}}{\sqrt{n\Delta}}\}$. Given that $\mathcal{M}_1[\Delta](u) > e^{-2\Delta}$, it follows in this case that

$$\mathbb{E} [|\overline{\mathcal{M}}_1[f](u) - \mathcal{M}_1[f](u)|^2] \leq |\mathcal{M}_1[f](u)|^2 (6 + \mathbb{E}[\ln(X_1)^2]).$$

Therefore,

$$\begin{aligned} \mathbb{E} \left[\int_{|u| \in [m, \widehat{m}_n]} |\overline{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2 \cdot \mathbb{1}_{\{|\mathcal{M}_1[f](s)|^2 > \frac{e^{2\Delta}}{\sqrt{n\Delta}}\}} \cdot \mathbb{1}_{\mathcal{E}^c} \right] du &\leq A \int_{|u| \in [m, \widehat{m}_n]} |\mathcal{M}_1[f](u)|^2 du \\ &\leq A \int_{[-m, m]^c} |\mathcal{M}_1[f](u)|^2 du, \end{aligned}$$

where $A = 6 + \mathbb{E}[\ln(X_1)^2]$.

Secondly, we assume that $\{|\mathcal{M}_1[f](s)|^2 \leq \frac{e^{2\Delta}}{\sqrt{n\Delta}}\}$. By construction, $|\overline{\mathcal{M}}_1[f](s)| \leq 4$. It follows that

$$\begin{aligned} & \mathbb{E} \left[\int_{|u| \in [m, \widehat{m}_n]} |\overline{\mathcal{M}}_1[f](u) - \mathcal{M}[f](u)|^2 \cdot \mathbf{1}_{\{|\mathcal{M}_1[f](s)|^2 \leq \frac{e^{2\Delta}}{\sqrt{n\Delta}}\}} \cdot \mathbf{1}_{\mathcal{E}^c} \right] du \\ & \leq \int_{|u| \in [m, (n\Delta)^\alpha]} |\mathcal{M}_1[f](u)|^2 du + 25 \int_{|u| \in [m, (n\Delta)^\alpha]} \mathbb{P} \left(|\widehat{\mathcal{M}}_1[f]| \geq \frac{\kappa_{n,\Delta}}{\sqrt{n\Delta}} \right) \cdot \mathbf{1}_{\{|\mathcal{M}_1[f](s)|^2 \leq \frac{e^{2\Delta}}{\sqrt{n\Delta}}\}} du \\ & \leq \int_{|u| \in [m, (n\Delta)^\alpha]} |\mathcal{M}_1[f](u)|^2 du + 25 \int_{|u| \in [m, (n\Delta)^\alpha]} \mathbb{P} \left(|\widehat{\mathcal{M}}_1[f] - \mathcal{M}_1[f]| \geq \kappa \sqrt{\log(n\Delta/(n\Delta))} \right) du. \end{aligned}$$

It remains to bound the last term. For that, we use `lm:Mellin1` and `lm:Mellin2` by taking γ_Δ and $\zeta > 0$, like in the proof of `T:risk`. By definition, we have that

$$\begin{aligned} & \mathbb{P} \left(|\widehat{\mathcal{M}}_1[f] - \mathcal{M}_1[f]| \geq \kappa \sqrt{\log(n\Delta/(n\Delta))} \right) \\ & = \mathbb{P} \left(|\log \widehat{\mathcal{M}}_1[\Delta] - \log \mathcal{M}_1[\Delta]| \geq \kappa \Delta \sqrt{\log(n\Delta/(n\Delta))} \right) \\ & \leq \mathbb{P} \left(|\widehat{\mathcal{M}}_1[\Delta] - \mathcal{M}_1[\Delta]| \geq |\mathcal{M}_1[\Delta]| \kappa \Delta \sqrt{\log(n\Delta/(n\Delta))} \right) + \mathbb{P}(\Omega_{\zeta,\Delta}^c((n\Delta)^\alpha)). \end{aligned}$$

Let $c(\Delta) = k\Delta e^{-2\Delta}$. The Hoeffding inequality and `lm:Mellin1` ensure that

$$\mathbb{P} \left(|\widehat{\mathcal{M}}_1[f] - \mathcal{M}_1[f]| \geq \kappa \sqrt{\log(n\Delta/(n\Delta))} \right) \leq 2(n\Delta)^{\alpha-c(\Delta)^2} + \frac{\mathbb{E}[\ln(X_1)^2]}{n\Delta} + 4(n\Delta)^{\alpha-\eta}.$$

If we take $\eta > 3$ such that $2\alpha - \eta < -1$ and $\zeta > \sqrt{\Delta(1+2\eta)}$, then

$$\int_{|u| \in [m, (n\Delta)^\alpha]} \mathbb{P} \left(|\widehat{\mathcal{M}}_1[f] - \mathcal{M}_1[f]| \geq \kappa \sqrt{\log(n\Delta/(n\Delta))} \right) du \leq 4(n\Delta)^{\alpha-c(\Delta)^2} + \frac{B}{n\Delta}$$

where B depends on $\mathbb{E}[\ln(X_1)^2]$.

It follows that there exist two constant A, B such that

$$\begin{aligned} & \mathbb{E} [\|\overline{f_{\widehat{m}_n, \Delta}} - f\|_{\omega_1}^2] \\ & \leq A \left(\|f_m - f\|_{\omega_1}^2 + \frac{\log(n\Delta)m}{n\Delta} + \frac{1}{n\Delta^2} \int_{-m}^m \frac{1}{|\mathcal{M}_1[\Delta](s)|^2} ds + 4 \frac{m^2}{(n\Delta)^2} \right) \\ & \quad + B \left((n\Delta)^{\alpha-c(\Delta)^2} + \frac{1}{n\Delta} \right). \end{aligned}$$

We conclude by taking the infimum in m .

3.6 Conclusion and perspectives

In this article, we develop an adaptive procedure to estimate the jump density f in a noisy compound Poisson process. In our case, we observe several noisy compound Poisson processes. We have shown that by looking at what happens at two different times, it is possible to reconstruct the density f .

In practice, the experimental data allow us to observe the process at different times. A discussion could be conducted later to understand if it is possible to combine estimators built with different t_1 and t_2 to improve our estimator.

In a future work, we plan to apply our statistical method to biological data from the article by Robert et al. [19].

3.7 The distinguished logarithm

In the section, we recall the definition and some properties of the distinguished logarithm. The reader who would like to have more information can refer to the articles of Duval and Kappus [duval2017nonparametric, 6] or to the article of Finkelstein et al. [finkelstein1999extinguishing].

Théorème 12. *Let $d \in \mathbb{N} - \{0\}$ and let $\varphi \in \mathcal{C}^0(\mathbb{R}^d; \mathbb{C}^*)$ be a continuous application which never vanishes and such that $\varphi(0) = 1$.*

Then there exists a unique continuous application $\psi \in \mathcal{C}^0(\mathbb{R}^d; \mathbb{C})$ such that

1. $\psi(0) = 0$,
2. *for all $x \in \mathbb{R}^d$, $\varphi(x) = e^{\psi(x)}$.*

The application ψ is called the Distinguished Logarithm of φ and is denoted by $\log \varphi$.

Remarque 13. In general, the distinguished logarithm does not reduce to a composition by the principal determination of the logarithm. In his book [sato1999levy], Sato remarks to his readers that one can have $\varphi(z_1) = \varphi(z_2)$ and $\log \varphi(z_1) \neq \log \varphi(z_2)$.

Indeed, consider the application $\varphi(t) = e^{it}$, ($t \in \mathbb{R}$). It verifies all the assumptions of Theorem ??, which ensures that

$$\log \varphi(t) = it, \quad (t \in \mathbb{R}),$$

It follows that

$$\varphi(0) = \varphi(2) = 1 \quad \text{and} \quad \log \varphi(0) = 0, \quad \log \varphi(2) = 2i\pi.$$

Proposition 14. *(Theorem 2. of [finkelstein1999extinguishing])*

Let $d > 0$ be a positive integer and $f, (f_n)_{n \in \mathbb{N}} \in \mathcal{C}^0(\mathbb{R}^d; \mathbb{C}^)$ be continuous functions which never vanishes and such that $f(0) = f_n(0) = 1$ for all $n \in \mathbb{N}$. If (f_n) converges uniformly to f on compact subsets of $\mathbb{R}^d$, then the sequence $(\log(f_n))_{n \in \mathbb{N}}$ converges uniformly to $\log(f)$ on compact subsets of $\mathbb{R}^d$.*

Proposition 15. *(Lemma 3. of [duval2017nonparametric])*

Let φ be a characteristic function without zeroes and assume that φ is differentiable. Then, it follows that

$$\log \varphi(u) = \int_0^u \frac{\varphi'(z)}{\varphi(z)} dz.$$

Corollaire 16. *Let $d \in \mathbb{N} - \{0\}$ and $f_1, f_2 \in \mathcal{C}^0(\mathbb{R}^d; \mathbb{C}^*)$ be two continuous functions which never vanishes and such that $f_1(0) = f_2(0) = 1$. Assume that f_1, f_2 are both differentiable and let $h : \mathbb{R}^d \rightarrow \mathbb{C}^*$ denotes the quotient $h = \frac{f_2}{f_1}$. Then,*

$$\forall u \in \mathbb{R}^d, \quad \log h(u) = \log f_2(u) - \log f_1(u) .$$

Proof.

$$\begin{aligned} \log h(u) &= \int_0^u \frac{h'(z)}{h(z)} dz = \int_0^u \frac{\left(\frac{f_2}{f_1}\right)'(z)}{\left(\frac{f_2}{f_1}\right)(z)} dz = \int_0^u \frac{\frac{f_2' f_1 - f_2 f_1'}{f_1^2}}{\left(\frac{f_2}{f_1}\right)} dz = \int_0^u \left(\frac{f_2' f_1 - f_2 f_1'}{f_1 f_2} \right) dz \\ &= \int_0^u \left(\frac{f_1'}{f_1} \right) dz - \int_0^u \left(\frac{f_2'}{f_2} \right) dz = \log f_2(u) - \log f_1(u) . \end{aligned}$$

■

3.8 Useful lemmas

3.8.1 Lemmas for decompounding with unknown noise

Lemme 17. *Let $m, \zeta, t \in [0, \infty)$ be positive reals. We consider the event*

$$\Omega_{\zeta, t}(m) = \left\{ \forall u \in [-m, m], |\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)| \leq \zeta \sqrt{\frac{\log(Jt)}{Jt}} \right\} .$$

If $\mathbb{E}[X_1^2] < \infty$, then

$$\forall \eta > 0, \quad \forall \zeta > \sqrt{t(1+2\eta)}, \quad \mathbb{P}(\Omega_{\zeta, t}(m)^c) \leq \frac{\mathbb{E}[X_1^2]}{Jt} + \frac{\mathbb{E}[\varepsilon^2]}{Jt^2} + 4 \frac{m}{(Jt)^\eta} .$$

Proof. Let $c, h, \tau \in [0, \infty)$ be some positive reals. We define the events

$$\begin{aligned} A(c) &= \left\{ \left| \frac{1}{J} \sum_{j=1}^J |Z_t^j| - \mathbb{E}[|Z_t|] \right| \leq c \right\} , \\ B_{h, \tau}(m) &= \left\{ \forall |k| \leq \left\lceil \frac{m}{h} \right\rceil, |\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| \leq \tau \sqrt{\frac{\log(Jt)}{Jt}} \right\} . \end{aligned}$$

As we know that the function $x \mapsto e^{iux}$ is 1-Lipschitz, then for all $u \in \mathbb{R}$ and for all $h \in [0, \infty)$

we have

$$\begin{aligned}
|\widehat{\varphi}_{Z_t}^J(u) - \widehat{\varphi}_{Z_t}^J(u+h)| &= \left| \frac{1}{J} \sum_{j=1}^J e^{iuZ_t^j} - \frac{1}{J} \sum_{j=1}^J e^{i(u+h)Z_t^j} \right| \\
&\leq \frac{1}{J} \sum_{j=1}^J \left| e^{iuZ_t^j} - e^{i(u+h)Z_t^j} \right| \\
&\leq \frac{h}{J} \sum_{j=1}^J |Z_t^j| \\
&= h \left(\frac{1}{J} \sum_{j=1}^J \left[|Z_t^j| - \mathbb{E}[|Z_t|] \right] + \mathbb{E}[|Z_t|] \right).
\end{aligned}$$

The definition of $A(c)$ ensures that

$$\forall u \in \mathbb{R}, \quad h > 0, \quad |\widehat{\varphi}_{Z_t}^J(u) - \widehat{\varphi}_{Z_t}^J(u+h)| \mathbb{1}_{A(c)} \leq h(c + \mathbb{E}[|Z_t|]). \quad (3.27)$$

Moreover we have

$$\mathbb{E}[|Z_t|] \leq t\mathbb{E}[|X_1|] + \mathbb{E}[\varepsilon^2]. \quad (3.28)$$

If $\mathbb{E}[X_1^2] < \infty$, by applying the Markov inequalities

$$\text{Var}[|Z_t|] \leq \text{Var}[Z_t] = t\mathbb{E}[X_1^2] + \mathbb{E}[\varepsilon^2],$$

we claim that

$$\mathbb{P}(A(c)^c) \leq \frac{t\mathbb{E}[X_1^2]}{c^2 J} + \frac{\mathbb{E}[\varepsilon^2]}{c^2 J}. \quad (3.29)$$

Moreover we have

$$\mathbb{P}(B_{h,\tau(m)}^c) = \mathbb{P}\left(\exists |k| \leq \left\lceil \frac{m}{h} \right\rceil, |\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| > \tau \sqrt{\frac{\log(Jt)}{Jt}}\right) \quad (3.30a)$$

$$\leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} \mathbb{P}\left(|\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| > \tau \sqrt{\frac{\log(Jt)}{Jt}}\right) \quad (3.30b)$$

$$\leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} \mathbb{P}\left(\left| \sum_{j=1}^J e^{ikhZ_t^j} - \mathbb{E}[e^{ikhZ_t^j}] \right| > \tau \sqrt{\frac{J \log(Jt)}{t}}\right). \quad (3.30c)$$

As $|e^{ikhZ_t^j}| \leq 1$ almost surely, Hoeffding's inequality ensures that

$$\leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} 2 \exp\left(-\frac{\tau^2 \log(Jt)}{2t}\right) \quad (3.30d)$$

$$= 4 \left\lceil \frac{m}{h} \right\rceil (Jt)^{-\tau^2/(2t)}. \quad (3.30e)$$

Let $|u| \leq m$. Then there exists k a positive integer such that $u \in [kh - \frac{h}{2}, kh + \frac{h}{2}]$. Then

$$\begin{aligned} \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} |\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)| &\leq \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \left(|\widehat{\varphi}_{Z_t}^J(u) - \widehat{\varphi}_{Z_t}^J(kh)| \right. \\ &\quad \left. + |\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| + |\varphi_{Z_t}(kh) - \varphi_{Z_t}(u)| \right). \end{aligned} \quad (3.31)$$

Since the function $x \mapsto e^{ixt}$ is 1-Lipschitz, we have

$$\begin{aligned} |\varphi_{Z_t}(kh) - \varphi_{Z_t}(u)| &= |e^{ikhZ_t} - e^{iuZ_t}| \\ &\leq he[|Z_t|]. \end{aligned}$$

By applying this last result to Equation (??)

$$\begin{aligned} \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \sup_{|u| \leq m} |\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)| &\leq \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \left(|\widehat{\varphi}_{Z_t}^J(u) - \widehat{\varphi}_{Z_t}^J(kh)| \right. \\ &\quad \left. + |\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| + he[|Z_t|] \right). \end{aligned}$$

It follows from Equation (??) that

$$\begin{aligned} \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \sup_{|u| \leq m} |\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)| &\leq \\ &h(c + e[|Z_t|]) + \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \left(|\widehat{\varphi}_{Z_t}^J(kh) - \varphi_{Z_t}(kh)| \right) + he[|Z_t|]. \end{aligned}$$

By using the definition of $B_{h,\tau}(m)$, we have

$$\begin{aligned} \mathbb{1}_{A(c) \cap B_{h,\tau}(m)} \sup_{|u| \leq m} |\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)| &\leq h(c + e[|Z_t|]) + \tau \sqrt{\frac{\log(Jt)}{Jt}} + he[|Z_t|] \\ &\leq hc + 2he[|Z_t|] + \tau \sqrt{\frac{\log(Jt)}{Jt}} \\ &\leq hc + 2h(te[|X_1|] + e[\varepsilon])) + \tau \sqrt{\frac{\log(Jt)}{Jt}}. \end{aligned} \quad (3.33)$$

In particular, if $c = t$, $h = o(\sqrt{\frac{\log(Jt)}{Jt}})$ such that $h > \frac{1}{\sqrt{Jt}}$ and $\zeta > \tau$, Equation (??), allows us to say that

$$A(c) \cap B_{h,\tau}(m) \subset \Omega_{\zeta,t}(m).$$

In addition, $h > \frac{1}{\sqrt{Jt}}$, (??) et (??), we prove that for all $\eta > 0$,

$$\begin{aligned} \mathbb{P}(\Omega_{\zeta,t}^c(m)) &\leq \mathbb{P}(A^c(t)) + \mathbb{P}(B_{h,\tau}^c(m)) \\ &\leq \frac{e[X_1^2]}{Jt} + \frac{e[\varepsilon^2]}{Jt^2} + 4 \left\lceil \frac{m}{h} \right\rceil (Jt)^{-\frac{\tau^2}{2t}} \\ &\leq \frac{e[X_1^2]}{Jt} + \frac{e[\varepsilon^2]}{Jt^2} + 4m(Jt)^{-\frac{t-\tau^2}{2t}}. \end{aligned}$$

We obtain the result by taking $\tau^2 = t(1 + 2\eta)$. ■

3.8.2 Lemmas for multiplicative decompounding

Lemme 18. (Lemma 5.2. of [6]) Let $\gamma \in [0, \infty)$, $t \in [0, \infty)$ and consider

$$M_{J,t}^{(\gamma)} = \min \left\{ u \geq 0 : |\varphi_{Z_t}(u)| = \gamma \sqrt{\log(Jt)/(Jt)} \right\},$$

with the convention $\inf\{\emptyset\} = \infty$.

Let $\zeta \in [0, \infty)$ be a positive real t.q. $0 < \zeta < \gamma$. Then

$$\mathbb{1}_{|u| \leq M_{J,t}^{(\gamma)} \wedge m, \Omega_{\zeta,t}(m)} \cdot \left| \log(\widehat{\varphi}_{Z_t}^J(u)) - \log \varphi_{Z_t}(u) \right| \leq \frac{\gamma}{\zeta} \log \left(\frac{\gamma}{\gamma - \zeta} \right) \frac{|\widehat{\varphi}_{Z_t}^J(u) - \varphi_{Z_t}(u)|}{|\varphi_{Z_t}(u)|}.$$

Lemme 19. Let $m > 0$ and $\zeta > 0$. We consider the event

$$\Omega_{\zeta,\Delta}(m) = \left\{ \forall u \in [-m, m], |\widehat{\mathcal{M}}_1[\Delta](u) - \mathcal{M}_1[\Delta](u)| \leq \zeta \sqrt{\frac{\log(n\Delta)}{n\Delta}} \right\}.$$

If $\mathbb{E}[\ln(X_1)^2] < \infty$, then

$$\mathbb{P}(\Omega_{\zeta,t}^c(m)) \leq \frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \frac{m}{(n\Delta)^\eta}.$$

Proof. Let $c, h, \tau \in [0, \infty)$ be some positive reals. We define the events

$$A(c) = \left\{ \left| \frac{1}{n} \sum_{k=1}^n |\ln(Z_{k\Delta})| - \mathbb{E}[\ln(Z_\Delta)] \right| \leq c \right\},$$

$$B_{h,\tau(m)} = \left\{ \forall |k| \leq \left\lceil \frac{m}{h} \right\rceil, |\widehat{\mathcal{M}}_1[\Delta](kh) - \mathcal{M}_1[\Delta](kh)| \leq \tau \sqrt{\frac{\log(n\Delta)}{n\Delta}} \right\}.$$

As we know that the function $u \mapsto e^{iux}$ is 1-Lipschitz, then for all $u \in \mathbb{R}$ and for all $h \in [0, \infty)$ we have

$$\begin{aligned} |\widehat{\mathcal{M}}_1[\Delta](u+h) - \widehat{\mathcal{M}}_1[\Delta](u)| &= \left| \frac{1}{n} \sum_{k=1}^n Z_{k\Delta}^{i(u+h)} - \frac{1}{n} \sum_{k=1}^n Z_{k\Delta}^{c-1+iu} \right| \\ &= \frac{1}{n} \sum_{k=1}^n |Z_{k\Delta}^{i(u+h)} - Z_{k\Delta}^{iu}| \\ &= \frac{1}{n} \sum_{k=1}^n |e^{i(u+h)\ln(Z_{k\Delta})} - e^{iu\ln(Z_{k\Delta})}| \\ &\leq \frac{h}{n} \sum_{k=1}^n |\ln(Z_{k\Delta})| \\ &= h \left(\frac{1}{n} \sum_{k=1}^n [|\ln(Z_{k\Delta})| - \mathbb{E}[|\ln(Z_\Delta)|]] + \mathbb{E}[|\ln(Z_\Delta)|] \right). \end{aligned}$$

The definition of $A(c)$ ensures that for all $u \in \mathbb{R}$ and $h > 0$

$$|\widehat{\mathcal{M}}_1[\Delta](u+h) - \widehat{\mathcal{M}}_1[\Delta](u)| \mathbb{1}_{A(c)} \leq h(c + \mathbb{E}[|\ln(Z_\Delta)|]) \leq h(c + \Delta \mathbb{E}[|\ln(X_1)|]). \quad (3.36)$$

If $\mathbb{E}[\ln(X_1)^2]$ is finite, the Markov inequality leads to

$$\mathbb{P}(A(c)^c) \leq \frac{\Delta \mathbb{E}[\ln(X_1)^2]}{c^2 n}. \quad (3.37)$$

Moreover we have

$$\begin{aligned} \mathbb{P}(B_{h,\tau(m)}^c) &= \mathbb{P}\left(\exists |k| \leq \left\lceil \frac{m}{h} \right\rceil, \left| \widehat{\mathcal{M}_1[\Delta]}(kh) - \mathcal{M}_1[\Delta](kh) \right| > \tau \sqrt{\frac{\log(n\Delta)}{n\Delta}}\right) \\ &\leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} \mathbb{P}\left(\left| \widehat{\mathcal{M}_1[\Delta]}(kh) - \mathcal{M}_1[\Delta](kh) \right| > \tau \sqrt{\frac{\log(n\Delta)}{n\Delta}}\right) \\ &\leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} \mathbb{P}\left(\left| \sum_{k=1}^n Z_{k\Delta}^{i(kh)} - \mathbb{E}[Z_{k\Delta}^{i(kh)}] \right| > \tau \sqrt{\frac{n \log(n\Delta)}{\Delta}}\right). \end{aligned}$$

As $|Z_{k\Delta}^{i(kh)}| \leq 1$ almost surely, Hoeffding's inequality ensures that

$$\mathbb{P}(B_{h,\tau(m)}^c) \leq \sum_{k=-\left\lceil \frac{m}{h} \right\rceil}^{\left\lceil \frac{m}{h} \right\rceil} 2 \exp\left(-\frac{\tau^2 \log(n\Delta)}{2t}\right) = 4 \left\lceil \frac{m}{h} \right\rceil (n\Delta)^{-\tau^2/(2\Delta)}.$$

Let $|u| \leq m$. There exists k a positive integer such that $u \in [kh - \frac{h}{2}, kh + \frac{h}{2}]$. It follows that

$$\begin{aligned} \mathbb{1}_{A(c) \cap B_{h,\tau(m)}} |\widehat{\mathcal{M}_1[\Delta]}(u) - \mathcal{M}_1[\Delta](u)| &\leq \mathbb{1}_{A(c) \cap B_{h,\tau(m)}} \left(\left| \widehat{\mathcal{M}_1[\Delta]}(u) - \widehat{\mathcal{M}_1[\Delta]}(kh) \right| \right. \\ &\quad \left. + \left| \widehat{\mathcal{M}_1[\Delta]}(kh) - \mathcal{M}_1[\Delta](kh) \right| + \left| \mathcal{M}_1[\Delta](kh) - \mathcal{M}_1[\Delta](u) \right| \right). \end{aligned}$$

We bound the three right terms by using respectively the Equation (??), the definition of $B_{h,\tau}$ and the fact that $u \rightarrow e^{iux}$ is 1-Lipschitz. It follows that

$$\mathbb{1}_{A(c) \cap B_{h,\tau(m)}} |\widehat{\mathcal{M}_1[\Delta]}(u) - \mathcal{M}_1[\Delta](u)| \leq 2h\Delta \mathbb{E}[|\ln(X_1)|] + hc + \tau \sqrt{\frac{\log(n\Delta)}{n\Delta}}.$$

By fixing $c = \Delta$, $h = o\left(\sqrt{\frac{\log(n\Delta)}{n\Delta}}\right)$ such that $h > \frac{1}{\sqrt{n\Delta}}$ and $\zeta > \tau$, it follows that

$$A(c) \cap B_{h,\tau(m)} \subset \Omega_{\zeta,\Delta}(m).$$

In addition, $h > \frac{1}{\sqrt{n\Delta}}$, (??), we prove that for all $\eta > 0$,

$$\begin{aligned} \mathbb{P}(\Omega_{\zeta,t}^c(m)) &= \mathbb{P}(A(c)^c) + \mathbb{P}(B_{h,\tau(m)}^c) \leq \frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4 \left\lceil \frac{m}{h} \right\rceil (n\Delta)^{-\tau^2/(2\Delta)} \\ &\leq \frac{\mathbb{E}[\ln(X_1)^2]}{\Delta n} + 4m(n\Delta)^{-\frac{t-\tau^2}{2t}}. \end{aligned}$$

We obtain the result by taking $\tau^2 = t(1 + 2\eta)$. ■

Lemme 20. (*Duval, Kappus [6] Lemma 5.2*) Let $\gamma \in [0, \infty)$ and consider

$$M_{n,\Delta}^{(\gamma\Delta)} = \min \left\{ u \geq 0 : |\mathcal{M}_1[\Delta](u)| = \gamma c_m \sqrt{\log(Jt)/(Jt)} \right\},$$

with the convention $\inf\{\emptyset\} = \infty$.

Let $\zeta \in [0, \infty)$ be a positive real t.q. $0 < \zeta < \gamma$. Then

$$\mathbb{1}_{|u| \leq M_{J,t}^{\gamma} \wedge m, \Omega_{\zeta,t}(m)} \cdot \left| \log \widehat{\mathcal{M}_1[\Delta]}(s) - \log \mathcal{M}_1[\Delta](s) \right| \leq \frac{\gamma}{\zeta} \log \left(\frac{\gamma}{\gamma - \zeta} \right) \frac{|\widehat{\mathcal{M}_1[\Delta]}(s) - \mathcal{M}_1[\Delta](s)|}{|\mathcal{M}_1[\Delta](s)|}.$$

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Chapter 4

PDEs modelling

4.1 Introduction

4.1.1 Motivation

In the mathematical literature, many authors have considered models to study the evolution of a population under the effect of mutations. In **[fisher1937wave]**, Ronald A. Fisher models the spreading of an advantageous gene in a population living on the line by the reaction-diffusion equation

$$\begin{cases} \partial_t u(t, x) - \Delta u(t, x) = u(t, x)(1 - u(t, x)) & \text{for } x \in \mathbb{R}, t > 0, \\ u(0, x) = u_0(x) & \text{for } x \in \mathbb{R}, \end{cases}$$

where $u_0 \in \mathcal{C}(\mathbb{R})$ is bounded, non-negative, non-trivial and compactly supported. In this model, $u(t, x)$ represents the frequency of mutants in the population at location x and time t . Since then, diffusion-reaction models have been widely studied to understand mutation phenomena **[fife1979mathematical, lam2022introduction]**. These models were extended to non-local variants where the diffusion is replaced by a convolution operator **[coville2007non, alfaro2023confining]** or an integral operator **[burger1996stationary, bonnefon2017concentration]** and the competition may also occur in a nonlocal way - what is justified, in terms of modelling, when x represents a trait rather than a space variable.

In [19], the authors have considered a microfluidic setting to study the impact of a new mutation on the *individual* fitness of a cell, i.e. its exponential growth rate. Following this modelling framework, we are interested in a population of cells $u(t, x)$ evolving through the time $t \in \mathbb{R}_+$ and structured by the growth rate $x \in \mathbb{R}_+$. Mutations are assumed to happen at a constant rate μ (denoted λ in [19]), which is justified by biological experiments [19]. For a general mutation kernel $\kappa(y, x)$, representing the probability distribution function of growth rate arising when a bacteria of growth rate y mutates, we obtain the equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) = -\mu u(t, x) + \mu \int_0^\infty \kappa(y, x) u(y) dy & \text{for } x \in \mathbb{R}, t > 0 \\ u(0, x) = u_0(x) & \text{for } x \in \mathbb{R}. \end{cases} \quad (4.1)$$

Apart from its biological interest, we note that Equation (??) is very similar to the fragmentation equation, which has many applications. Only a few reference are given here among the huge

number of applications : breakage of proteins [xue2013imaging], cell division [doumic2013estimating], mining industry [kolmogorov1941logarithmic, bertoin2005fragmentation], polymer degradation [montroll1940theory, ziff1985kinetics]. However, a main difference in Equation (??) is that $k(\cdot, y)$ is not necessarily included in $[0, y]$ contrarily to the case of the fragmentation equation.

In this article, we assume - here again following [19] - that $\kappa(y, x) := \frac{1}{y}k\left(\frac{x}{y}\right)$, which leads to the equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) = -\mu u(t, x) + \mu \int_0^\infty k\left(\frac{x}{y}\right) u(y) \frac{dy}{y} & \text{for } x \in \mathbb{R}, t > 0 \\ u(0, x) = u_0(x) & \text{for } x \in \mathbb{R}. \end{cases} \quad (4.2)$$

This assumption is referred as the '*self-similarity property*'. It reflects the fact that the probability to obtain a cell of fitness y out of a particle of size x only depends on the ratio y/x . This is a classical assumption for the fragmentation equation [escobedo2005self], and it means that mutations act as a multiplicative convolution on the growth rate. This assumption appears not to be so classical in the evolutionary dynamics literature, where additive convolution - natural when x represents a space variable - has been more often considered than multiplicative convolution.

We aim to understand the asymptotic behavior of u when $t \rightarrow \infty$. We want to understand how the distribution of mutants evolves over time, first in a lineage population like the microfluidic experiment, then in a growing population. If the fitness of the mutants increases indefinitely, we would like to be able to characterise an escape rate. Conversely, if all the mutants die, we would like to obtain a population collapse rate.

4.1.2 Review on existing results

Many authors have studied mutation - selection models. In most studies, the authors introduce a non-local selection term. This is not the case in our study, since we assume that the cells are not in competition with each other. Another main difference between our work and existing work is that we consider that mutations have a multiplicative effect on fitness, whereas most authors focus on additive effects. This assumption comes directly from the model proposed in [19]. As far as we know, there is no study that focus on multiplicative effects. To study this 'multiplicative' term, we use the Mellin transformation.

In our model, when all mutations are deleterious, the support of k is contained in $(0, 1)$. In this case, Equation (??) is a close variant of the fragmentation equation for which many studies have already been done: existence and uniqueness for the Cauchy problem [melzak1957scalar], asymptotic behaviour in many case studies including our case of a constant mutation/fragmentation rate in [doumic2016time], reconstructing the kernel of fragmentation [hoang2017estimating, hoang2017nonparametric, doumic2018estimating, doumic2021inverse]. For a rather complete list of references the reader may consult [banasiak2019analytic].

4.1.3 Review on existing results

4.1.4 Main results and organization of the article

In the present article,

In Section ?? In Section ??, we seek to see if the growth of the population may compensate the effect of deleterious mutations [Mal Dit].

To summarize, the main novelties brought by this article are:

- An extension of the work done in [doumic2016time] when the support of k is not restricted to $[0, 1]$
-

The organization of the article is as follows.

4.1.5 Notation

4.1.5.1 Spaces of functions

Let $\mathcal{C}(\mathbb{R}_+)$ be the set of continuous real functions on $\mathbb{R}_+$. We define the supremum norm

$$\|\varphi\|_\infty := \sup_{x \in \mathbb{R}_+} |\varphi(x)| .$$

The set $\mathcal{M}(\mathbb{R}_+)$ is the set of Radon measures μ such that $(\mu) \subset \mathbb{R}_+$. The Total Variation norm (TV-norm) of a measure $\mu \in \mathcal{M}(\mathbb{R}_+)$ is given by

$$\|\mu\|_{TV} := \sup \left\{ \int_{\mathbb{R}_+} \varphi(x) \mu(x) ; \varphi \in \mathcal{C}(\mathbb{R}_+) \cap L^1(|\mu|), \|\varphi\|_\infty \leq 1 \right\}$$

We recall that $(\mathcal{M}(\mathbb{R}_+), \|\cdot\|_{TV})$ is a Banach space.

4.1.5.2 The Mellin transform

We recall some basic definitions and facts about the Mellin transform. We refer the reader to the chapter 11 of [misra1986transform] for the proofs and further information.

Définition 21. For a measure $\mu \in \mathcal{M}(\mathbb{R}_+)$, its Mellin transform $\mathcal{M}[\mu]$ is defined as

$$\mathcal{M}[\mu](s) = \int_{\mathbb{R}_+} x^{s-1} d\mu(x) , \quad (4.3)$$

for $s \in \mathbb{C}$ such that (??) is well-defined. If f a measurable function defined on $(0, \infty)$, its Mellin transform is defined as

$$\mathcal{M}[f](s) = \int_{\mathbb{R}_+} x^{s-1} f(x) dx , \quad s \in \mathbb{C} , \quad (4.4)$$

whenever this integral converges.

Lemme 22. Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a measurable function. The set

$$I(f) := \left\{ s \in \mathbb{R} \int_{\mathbb{R}_+} x^{s-1} f(x) dx < \infty \right\}$$

is an interval of $\mathbb{R}$ and the application $s \mapsto \mathcal{M}_f(s)$ is analytic on the interior of the strip $I(f) + i\mathbb{R}$. The set $I(f)$ is designed as the fundamental strip of $\mathcal{M}_f$.

Lemme 23 (Inversion of the Mellin transform). *Let F be a holomorphic function on the vertical strip $(a_1, a_2) + i\mathbb{R}$. If there exists a constant $K \in \mathbb{R}$ such that*

$$\forall s \in (a_1, a_2) + i\mathbb{R}, \quad |F(s)| \leq K|s|^{-2},$$

then the function $f : (0, \infty) \rightarrow \mathbb{C}$ defined by

$$f(x) = \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} F(s)x^{-s}ds, \quad c \in (a_1, a_2) + i\mathbb{R} \quad (4.5)$$

is a continuous function and its Mellin transform is F .

4.2 Mutation Model for lineages

4.2.1 Motivation

In this section, we propose a deterministic model to modeling the microfluidic mutation accumulation (μMA) experiment described in [19]. For that, we consider the Cauchy problem

$$\begin{cases} \frac{\partial}{\partial t}u(t, x) = -\mu u(t, x) + \mu \int_0^\infty k_0\left(\frac{x}{y}\right)u(t, y)\frac{dy}{y} & \text{for } t \in (0, \infty), x \in \mathbb{R}_+, \\ u(0, x) = u_0(x) & \text{for } x \in \mathbb{R}_+, \end{cases} \quad (4.6)$$

with $\mu > 0$. We assume that the initial data u_0 is a function satisfying the following conditions

$$u_0 \geq 0, \quad \int_0^\infty u_0(x)(1+x)dx < \infty \quad (4.7)$$

and $k_0 \in \mathcal{M}(\mathbb{R}_+)$ is a probability measure designed as the kernel of mutation. We suppose that

$$k_0 \geq 0, \quad \int_0^\infty (1+x)k_0(dx) < \infty \quad \text{and} \quad \int_0^\infty k_0(dx) = 1. \quad (4.8)$$

An important tool to study Equation (??) is the Mellin transform that we defined in (??). Indeed, by applying the Mellin transform to Equation (??), we have formally at this stage

$$\begin{cases} \frac{\partial}{\partial t}U(t, s) = -\mu U(t, s) + \mu K(s)U(t, s), \\ U(0, s) = U_0(s) \end{cases}$$

where

$$\begin{aligned} U(t, s) &:= \int_0^\infty x^{s-1}u(t, x)dx, \quad U_0(s) := \int_0^\infty x^{s-1}u_0(x)dx \\ &\text{and} \quad K(s) := \int_0^\infty x^{s-1}k_0(x)dx \end{aligned}$$

are respectively the Mellin transform of u , u_0 and k_0 with respect to x . Under Assumptions (??) and (??), it follows that

$$U(t, s) = U_0(s)e^{-\mu t(1-K(s))}, \quad U_0(s) = \mathcal{M}_{u_0}(s).$$

We note that $\upharpoonright K\mathbb{R} : \mathbb{R} \rightarrow \mathbb{C}$ is a convex function, such that $K(1) = 1$.

4.2.1.1 A heuristic explanation of the deterministic model

4.2.1.2 The choice of the kernel

The choice of the mutation kernel k in Equation (??) has an impact on the solution. Indeed, if the support of k is included in $[0, 1]$, then no beneficial mutations can happen in the population. The fitness of a cell can only decrease with the time.

In [19], the authors estimate the expression of k from experimental observations. However, they assume that all mutations are deleterious. They were able to calculate the first moments of the law of k , and found that it could be close to a beta law.

In the rest of this section, we give different examples of kernels, and for each of them we explain what its biological interpretation would be. We also calculate the Mellin transform of each of these kernels. The reasons for these calculations will become clearer later in the article.

1. *The Dirac Kernel:* The effect of the mutation is always the same. Given $\alpha > 0$, we define $k_0(x) = \delta_\alpha$. For all $s \neq 0$,

$$K(s) = \int_0^\infty x^{s-1} \delta_\alpha(x) = \alpha^{s-1}.$$

2. *The Uniform Kernel:* The effect of the mutation does not depend on the fitness of the cell. We define $k_0(x) = \frac{1}{b-a} \mathbb{1}_{(a,b)}$, $a, b > 0$. Then for all $s \neq 0$,

$$K(s) = \frac{1}{b-a} \int_a^b x^{s-1} dx = \frac{1}{s} \cdot \frac{b^s - a^s}{b-a}.$$

3. *The Log-normal kernel.* $k_0(x) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\log(x)-m)^2}{2\sigma^2}}$. By the change of variables $y = \log(x)$, we have

$$\begin{aligned} K(s) &= \int_0^\infty \frac{x^{s-2}}{\sqrt{2\pi}\sigma} e^{-\frac{(\log(x)-m)^2}{2\sigma^2}} dx \\ &= \int_{-\infty}^\infty \frac{e^{y(s-1)}}{\sqrt{2\pi}\sigma} e^{-\frac{(y-m)^2}{2\sigma^2}} dy \\ &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2 - m^2 + 2my + 2\sigma^2 y(s-1)}{2\sigma^2}} dy \\ &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2 - m^2 + 2y[m + \sigma^2(s-1)]}{2\sigma^2}} dy \\ &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - [m + \sigma^2(s-1)])^2}{2\sigma^2}} e^{\frac{2m\sigma^2(s-1) + \sigma^4(s-1)^2}{2\sigma^2}} dy \\ &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - [m + \sigma^2(s-1)])^2}{2\sigma^2}} e^{m(s-1) + \frac{\sigma^2}{2}(s-1)^2} dy. \end{aligned}$$

Let us denote $:= e^{m(s-1) + \frac{\sigma^2}{2}(s-1)^2}$, then

$$\begin{aligned} K(s) &= C(s, m, \sigma) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - [m + \frac{\sigma^2}{2}(s-1)])^2}{2\sigma^2}} dy \\ &= C(s, m, \sigma) \end{aligned}$$

4.2.2 An asymptotic behavior

In this section, we aim to find a condition on the mutation kernel k_0 that would make possible to determine whether the fitness of all the cells converges to a Dirac function or, on the contrary, escapes towards infinity. On $I(k_0)$, Proposition ?? shows it only depends on the location of the minimum of the Mellin transform of k_0 .

4.2.2.1 Main result

Proposition 24. *Let $k_0 \in \mathcal{M}(\mathbb{R}_+)$ such that (??) and let K be its Mellin transform. Let u be the solution of the Cauchy problem (??). Suppose that K is minimum in $s_* \in (0, \infty)$, then for all $R > 0$*

1. (a) Suppose that $s_* < 1$, then

$$\lim_{t \rightarrow \infty} \int_R^{\infty} u(t, x) dx = \int_0^{\infty} u_0(x) dx .$$

- (b) Suppose that $s_* > 1$, then

$$\lim_{t \rightarrow \infty} \int_0^R u(t, x) dx = \int_0^{\infty} u_0(x) dx .$$

2. Furthermore, let $n > 1$ be an integer.

- (a) Suppose that $s_* > n$, then

$$\lim_{t \rightarrow \infty} \int_0^R x^{n-1} u(t, x) dx = 0 .$$

- (b) Suppose that $s_* < n$, then

- i. If $K(n) < 1$

$$\lim_{t \rightarrow \infty} \int_R^{\infty} x^{n-1} u(t, x) dx = 0 .$$

- ii. If $K(n) > 1$

$$\lim_{t \rightarrow \infty} \int_R^{\infty} x^{n-1} u(t, x) dx = \infty .$$

Proof. First case: $n = 1$

- (a) Suppose that $s_* < 1$. Let $c > 1$. By Mellin inversion, we have that

$$\forall s \in I(k_0) , \quad u(t, x) = \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s} e^{-\mu t(1-K(s))} ds$$

It follows that

$$\begin{aligned}
\forall s \in I(k_0), \quad \int_R^\infty u(t, x) dx &= \frac{1}{2i\pi} \int_R^\infty \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s} e^{-\mu t(1-K(s))} ds dx \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \int_R^\infty U_0(s) x^{-s} e^{-\mu t(1-K(s))} dx ds \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) \frac{R^{-s+1}}{s-1} e^{-\mu t(1-K(s))} ds \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \frac{U_0(s)}{s-1} e^{-\mu t(1-K(s))} e^{(-s+1) \ln R} ds.
\end{aligned}$$

We deform the integration contour towards the left and cross the pole $s = 1$. The Cauchy's residue theorem ensures that

$$\int_R^\infty u(t, x) dx = \text{Res}(1) + \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{s-1} e^{-\mu t(1-K(s))} e^{(-s+1) \ln R} ds \quad (4.9)$$

We define

$$\mathcal{R}(t) = \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{s-1} e^{-\mu t(1-K(s))} e^{(-s+1) \ln R} ds.$$

We have

$$\begin{aligned}
|\mathcal{R}(t)| &\leq \frac{R^{1-s_*}}{2\pi} \int_{s_*-i\infty}^{s_*+i\infty} \left| \frac{U_0(s)}{s-1} \right| \cdot \left| e^{-\mu t(1-K(s))} \right| ds \\
&= \frac{R^{1-s_*}}{2\pi} \int_{s_*-i\infty}^{s_*+i\infty} \left| \frac{U_0(s)}{s-1} \right| e^{\Re[-\mu t(1-K(s))]} ds.
\end{aligned}$$

However, for all $s = s_* + i\nu \in (s_* - i\infty, s_* + i\infty)$

$$\begin{aligned}
\Re[-\mu t(1-K(s))] &= \Re[-\mu t(K(1) - K(s))] \\
&= -\mu t \Re \left(K(1) - \int_0^\infty z^{s-1} k_0(z) dz \right) \\
&= -\mu t \left(K(1) - \int_0^\infty \cos(\nu \log z) z^{s_*-1} k_0(z) dz \right) \\
&\leq -\mu t \left(K(1) - \int_0^\infty z^{s_*-1} k_0(z) dz \right) \\
&= -\mu t (K(1) - K(s_*))
\end{aligned}$$

Since K is analytic on $I(k_0)$, the Taylor's expansion in s_* ensures there exists a real $\xi \in (s_*, 1)$ such that

$$\Re[-\mu t(1-K(s))] = -\mu t \frac{(s_* - 1)^2}{2} K''(\xi)$$

As $K''(\xi) > 0$, it follows that

$$\begin{aligned}
 |\mathcal{R}(t)| &\leq \frac{R^{1-s_*}}{2\pi} \int_{s_*-i\infty}^{s_*+i\infty} \left| \frac{U_0(s)}{s-1} \right| e^{\Re[-\mu t(1-K(s))]} \mathrm{d}s . \\
 &= \frac{R^{1-s_*}}{2\pi} \int_{s_*-i\infty}^{s_*+i\infty} \left| \frac{U_0(s)}{s-1} \right| e^{-\mu t \frac{(s_*-1)^2}{2} K''(\xi)} \mathrm{d}s \\
 &= e^{-\mu t \frac{(s_*-1)^2}{2} K''(s)} \times \frac{R^{1-s_*}}{2\pi} \int_{s_*-i\infty}^{s_*+i\infty} \left| \frac{U_0(s)}{s-1} \right| \mathrm{d}s \\
 &\rightarrow_{t \rightarrow \infty} 0 .
 \end{aligned}$$

Using Equation ??, it follows that

$$\begin{aligned}
 \int_R^\infty u(t, x) \mathrm{d}x &= \text{Res}(1) + \mathcal{R}(t) \\
 &= \lim_{s \rightarrow 1} (s-1) \frac{U_0(s)}{s-1} e^{-\mu t(1-K(s))} e^{(-s+1) \ln R} + \mathcal{R}(t) \\
 &= U_0(1) + \mathcal{R}(t) \\
 &\rightarrow_{t \rightarrow \infty} \int_0^\infty u_0(x) \mathrm{d}x .
 \end{aligned}$$

(b) Suppose that $s_* > 1$. Let $c < 1$.

By Mellin inversion, we have that

$$u(t, x) = \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s} e^{-\mu t(1-K(s))} \mathrm{d}s$$

It follows that

$$\begin{aligned}
 \int_0^R u(t, x) \mathrm{d}x &= \frac{1}{2i\pi} \int_0^R \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s} e^{-\mu t(1-K(s))} \mathrm{d}s \mathrm{d}x \\
 &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \int_0^R U_0(s) x^{-s} e^{-\mu t(1-K(s))} \mathrm{d}x \mathrm{d}s \\
 &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) \frac{R^{-s+1}}{1-s} e^{-\mu t(1-K(s))} \mathrm{d}s \\
 &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \frac{U_0(s)}{1-s} e^{-\mu t(1-K(s))} e^{(1-s) \ln R} \mathrm{d}s
 \end{aligned}$$

We deform the integration contour towards the right and cross the pole $s = 1$. The Cauchy's residue theorem ensures that

$$\int_0^R u(t, x) \mathrm{d}x = -\text{Res}(1) + \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{1-s} e^{-\mu t(1-K(s))} e^{(-s+1) \ln R} \mathrm{d}s \quad (4.10)$$

As previously, we define

$$\mathcal{R}(t) = \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{1-s} e^{-\mu t(1-K(s))} e^{(-s+1)\ln R} ds .$$

and we can show that $\lim_{t \rightarrow \infty} \mathcal{R}(t) = 0$. Using Equation ??, it follows that

$$\begin{aligned} \int_0^R u(t, x) dx &= -\text{Res}(1) + \mathcal{R}(t) \\ &= -\lim_{s \rightarrow 1} (s-1) \frac{U_0(s)}{1-s} e^{-\mu t(1-K(s))} e^{(-s+1)\ln R} + \mathcal{R}(t) \\ &= U_0(1) + \mathcal{R}(t) \\ &\rightarrow_{t \rightarrow \infty} \int_0^\infty u_0(x) dx . \end{aligned}$$

Second case: $n > 1$

(a) Suppose that $s_* > n$. Let $c < n$.

By Mellin inversion, we have that

$$u(t, x) = \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s} e^{-\mu t(1-K(s))} ds$$

It follows that

$$\begin{aligned} \int_0^R x^{n-1} u(t, x) dx &= \frac{1}{2i\pi} \int_0^R \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s+n-1} e^{-\mu t(1-K(s))} ds dx \\ &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \int_0^R U_0(s) x^{-s+n-1} e^{-\mu t(1-K(s))} dx ds \\ &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) \frac{R^{-s+n}}{n-s} e^{-\mu t(1-K(s))} ds \\ &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \frac{U_0(s)}{n-s} e^{-\mu t(1-K(s))} e^{(-s+n)\ln R} ds \end{aligned}$$

We deform the integration contour towards the right and cross the pole $s = n$. The Cauchy's residue theorem ensures that

$$\int_0^R x^{n-1} u(t, x) dx = -\text{Res}(n) + \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{n-s} e^{-\mu t(1-K(s))} e^{(-s+n)\ln R} ds \quad (4.11)$$

As previously, we define

$$\mathcal{R}(t) = \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{n-s} e^{-\mu t(1-K(s))} e^{(-s+n)\ln R} ds .$$

and we can show that $\lim_{t \rightarrow \infty} \mathcal{R}(t) = 0$. Using Equation ??, and the fact that $K(n) < 1$ it follows that

$$\begin{aligned}
\int_0^R x^{n-1} u(t, x) dx &= -\text{Res}(n) + \mathcal{R}(t) \\
&= -\lim_{s \rightarrow n} (s - n) \frac{U_0(s)}{n - s} e^{-\mu t(1-K(s))} e^{(-s+n) \ln R} + \mathcal{R}(t) \\
&= U_0(n) e^{-\mu t(1-K(n))} + \mathcal{R}(t) \\
&\rightarrow_{t \rightarrow \infty} 0.
\end{aligned}$$

(b) Suppose that $s_* < n$. Let $c > n$.

By Mellin inversion, we have that

$$u(t, x) = \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) x^s e^{-\mu t(1-K(s))} ds$$

It follows that

$$\begin{aligned}
\int_0^R x^{n-1} u(t, x) dx &= \frac{1}{2i\pi} \int_R^\infty \int_{c-i\infty}^{c+i\infty} U_0(s) x^{-s+n-1} e^{-\mu t(1-K(s))} ds dx \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \int_R^\infty U_0(s) x^{-s+n-1} e^{-\mu t(1-K(s))} dx ds \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} U_0(s) \frac{R^{-s+n}}{s - n} e^{-\mu t(1-K(s))} ds \\
&= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} \frac{U_0(s)}{s - n} e^{-\mu t(1-K(s))} e^{(-s+n) \ln R} ds
\end{aligned}$$

We deform the integration contour towards the left and cross the pole $s = n$. The Cauchy's residue theorem ensures that

$$\int_R^\infty x^{n-1} u(t, x) dx = \text{Res}(n) + \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{s - n} e^{-\mu t(1-K(s))} e^{(-s+n) \ln R} ds \quad (4.12)$$

As previously, we define

$$\mathcal{R}(t) = \frac{1}{2i\pi} \int_{s_*-i\infty}^{s_*+i\infty} \frac{U_0(s)}{s - n} e^{-\mu t(1-K(s))} e^{(-s+n) \ln R} ds.$$

and we can show that $\lim_{t \rightarrow \infty} \mathcal{R}(t) = 0$ by using Equation ?? ■

4.2.2.2 Biological interpretation of the result

Considering the case $n = 1$, Proposition ?? allows us to determine the fitness distribution of the population. Indeed, when $s_* > 1$, the proposition ensures that for all $R > 0$,

$$\lim_{t \rightarrow \infty} \int_R^\infty u(t, x) dx = \int_0^\infty u_0(x) dx.$$

This ensures that for any compact set $C \subset (0, \infty)$, the set

$$\{x \in C; u(t, x) > 0\}$$

has measure 0 when $t \in \mathbb{R}_+$ is large enough. For any real $A > 0$, this means that the whole population has a fitness greater than A after a sufficiently long time.

Similarly, if $s < s_*$, the proposition ensures that

$$\lim_{t \rightarrow \infty} \int_0^R u(t, x) dx = \int_0^\infty u_0(x) dx .$$

This ensures that for $t \in \mathbb{R}_+$ large enough, the set $\{x \in (R, \infty); u(t, x) > 0\}$ has measure 0. As this is true for all real $R > 0$, it means that after a sufficiently long time, the number of cells with a positive fitness is negligible,

4.2.3 An explicit solution of the Cauchy problem and asymptotic behaviour

In this section, we obtain an explicit solution of the Cauchy problem (??) by using the Mellin transform.

We consider the following assumption on the initial condition :

$$u_0 \geq 0 \quad \text{and} \quad \int_0^\infty u_0(x)(1+x) dx < \infty . \quad (4.13)$$

Assumption (??) ensures that $[1, 2] \subset I(u_0)$. Moreover $I(u_0)$ is an interval of $\mathbb{R}$ (Lemma ??). It leads us to define

$$p_0 = \inf\{p : p \in I(u_0)\} \in [-\infty, 1] , \\ q_0 = \sup\{q : q \in I(u_0)\} \in [2, \infty] .$$

For simplicity, we assume that $p_0 = -\infty$ and $q_0 = \infty$

We also assume that, for all $\nu \in (p_0, q_0)$, u_0 is twice differentiable,

$$\int_0^\infty |u'_0(x)|x^\nu dx < \infty , \quad \lim_{x \rightarrow 0} x^\nu u_0(x) = 0 , \quad \lim_{x \rightarrow \infty} x^\nu u_0(x) = 0 \quad (4.15)$$

and

$$\int_0^\infty |u''_0(x)|x^\nu dx < \infty , \quad \lim_{x \rightarrow 0} x^{\nu+1} u'_0(x) = 0 , \quad \lim_{x \rightarrow \infty} x^{\nu+1} u'_0(x) = 0 \quad (4.16)$$

Théorème 25. (*Existence of an explicit solution*) Suppose that u_0 satisfies (??), (??) and (??) and suppose that k_0 satisfies (??). There exists a unique function $u \in \mathcal{C}([0, \infty); L^1((1+x)dx)) \cap \mathcal{C}^1((0, \infty); L^1((1+x)dx))$ which is solution of the Cauchy problem (??). This solution is given by

$$u(t, x) = \frac{1}{2\pi i} \int_{\nu-i\infty}^{\nu+i\infty} U_0(s) e^{\mu(K(s)-1)t} x^{-s} ds \quad \text{for } (t, x) \in \mathbb{R}_+ \times \mathbb{R}_+ \quad (4.17)$$

for all $\nu \in (p_0, q_0)$.

Proof. See Appendix ?? ■

We want now to characterize the asymptotic behaviour of u when $x \rightarrow \infty$ and $t \rightarrow \infty$. A similar study has been done in [doumic2016time] for the fragmentation equation. We extend this study to the mutation equation where the support of k is not necessarily included in $[0, 1]$.

The key point to study u is the steepest descent method which has been developed by Peter Debye in 1909 [**debye1909naherungsformeln**] to study Bessel functions. We give a quick description of the method in Section ?? . A full description of the method can be found in [**dieudonne1980calcul**].

4.2.3.1 Method of steepest descent

Let $D \subset \mathbb{C}$ be a simply connected open set and $L \subset D$ be a simple closed path. Let $g : D \rightarrow \mathbb{C}$ and $h : D \rightarrow \mathbb{C}$ be two analytic functions. The method of steepest descent is a technique to obtain an asymptotic approximation of the integral

$$I(t) := \int_L g(z) e^{th(z)} dz$$

when $t \rightarrow \infty$. This method is an extension of the Laplace's method [**laplace1986memoir**] which can be used to study functions of the form $t \mapsto \int_a^b \beta(x) e^{t\alpha(x)} dx$ when $a, b \in \mathbb{R}$ and $\alpha : [a, b] \rightarrow \mathbb{R}$, $\beta : [a, b] \rightarrow \mathbb{R}$ are two continuous functions. For any deformation $L' \subset D$ of L , the Cauchy's integral formula ensures that

$$I(t) = \int_{L'} g(z) e^{th(z)} dz .$$

The idea of the method of steepest descent is to choose a good deformation L' to use the Laplace's method. A good deformation L' is a path such the imaginary part $\Im h(z)$ is constant on L' . This choice ensures that

$$I(t) = e^{it\Im h(z)} \int_{L'} g(z) e^{t\Re h(z)} dz$$

and we can study $t \mapsto I(t)$ with the Laplace's method.

The idea of the Laplace's method is the following. Consider the integral $\int_a^b \beta(x) e^{t\alpha(x)} dx$ when $a, b \in \mathbb{R}$ and $\alpha : [a, b] \rightarrow \mathbb{R}$, $\beta : [a, b] \rightarrow \mathbb{R}$ are two continuous functions. Suppose that α reaches its maximum at $x_0 \in (a, b)$. The Taylor's theorem ensures that

$$\alpha(x) = \alpha(x_0) + \alpha'(x_0)(x - x_0) + \frac{1}{2}\alpha''(x_0)(x - x_0)^2 + O((x - x_0)^3) .$$

Since α admits a maximum in x_0 , $\alpha'(x_0) = 0$ and $\alpha''(x_0) < 0$. There is a neighborhood $\mathcal{V}(x_0)$ of x_0 such that

$$\alpha(x) = \alpha(x_0) - \frac{1}{2}|\alpha''(x_0)|(x - x_0)^2 \quad \text{for } x \in \mathcal{V}(x_0) .$$

It can be show that

$$\begin{aligned} \int_a^b \beta(x) e^{t\alpha(x)} dx &\approx \beta(x_0) e^{t\alpha(x_0)} \int_a^b e^{-t|\alpha''(x_0)|(x-x_0)^2} dx && \text{when } t \rightarrow \infty \\ &\approx \beta(x_0) e^{t\alpha(x_0)} \sqrt{\frac{2\pi}{t|\alpha''(x_0)|}} && \text{when } t \rightarrow \infty \end{aligned}$$

4.2.3.2 Main theorem

Rather than studying the asymptotic behaviour of u directly, we study the asymptotic behaviour of the function

$$w(t, x) := e^{\mu t} u(t, x) \quad \text{for } t \in \mathbb{R}_+, x \in \mathbb{R}_+.$$

The explicit expression of u ensures that

$$\omega(t, x) = \int_{c-i\infty}^{c+i\infty} U_0(s) e^{\mu K(s)t} x^{-s} ds = \int_{c-i\infty}^{c+i\infty} U_0(s) e^{\phi(s, t, x)} ds.$$

with $\phi(s, t, x) := \mu K(s)t - s \ln(x)$.

Our aim is to study the asymptotic behaviour of w as t becomes larger and larger. For that, we apply the method of steepest descent which leads us to consider the curve $(t, x) \mapsto s_+(t, x)$ where $\partial_s \phi(s_+, t, x) = 0$. A simple computation ensures that

$$s_+(t, x) := (K')^{-1} \left(\frac{\log x}{\mu t} \right) \quad \text{for } t \in \mathbb{R}_+, x \in \mathbb{R}_+.$$

Théorème 26. *Suppose that u_0 satisfies (??), (??) and (??). Suppose that $k_0 \in \mathcal{M}(\mathbb{R}_+)$ satisfies (??). Let $\varepsilon : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be any function such that $\varepsilon(t) \rightarrow 0$ but $t\varepsilon^2(t) \rightarrow \infty$ as $t \rightarrow \infty$. Let us define*

$$\omega(\varepsilon) := \sup_{|v| \leq \varepsilon(t)} |U_0(s_+ + iv) - U_0(s_+)|.$$

Then for all real $\delta > 0$, $A > 0$, there exists a function ω_δ such that:

$$u(t, x) = x^{-s_+(t, x)} e^{(K(s_+(t, x)) - 1)t} \times \left(\frac{U_0(s_+(t, x) + \omega(\varepsilon))}{\sqrt{2\pi t K''(s_+)}} \Theta_1(t) \right)$$

as $t \rightarrow \infty$, uniformly in x such that $-A < s_+(t, x) < A$ and where

$$\Theta_1(t) = \left(1 + \mathcal{O} \left(\frac{e^{-\frac{t\varepsilon^2(t)}{2} K''(s_+)}}{\sqrt{t\varepsilon^2(t) K''(s_+)}} \right) \right)$$

as $t \rightarrow \infty$.

Proof. See Appendix ?? ■

4.2.4 Biological interpretation of the result

In this section we give some consequences of Theorem ??.

In order to understand the behaviour of the function $u(t, x)$ as t increases we may consider the curves $(C_r)_{r \in \mathbb{R}}$ defined by

$$C_r = \{(t, x) \in \mathbb{R}_+ \times \mathbb{R}_+; s_+(t, x) = r\}, \quad r \in \mathbb{R}.$$

As

$$s_+(t, x) := (K')^{-1}\left(\frac{\log x}{\mu t}\right) \quad \text{for } t \in \mathbb{R}_+, x \in \mathbb{R}_+,$$

the curve C_r ($r \in \mathbb{R}$) corresponds to the set of points $(t, x) \in \mathbb{R}_+ \times \mathbb{R}_+$ such that $x = e^{\mu K'(r)t}$.

Let us denote

$$A(t, r) := \frac{U_0(r + \omega_\delta(\varepsilon))}{\sqrt{2\pi t K''(r)}}$$

and define the function $F : \mathbb{R} \rightarrow \mathbb{R}$,

$$F(s) := K(s) - 1 - \mu s_+ K'(s), \quad s \in \mathbb{R}. \quad (4.18)$$

Theorem ?? ensures that for all $r \in \mathbb{R}$,

$$\begin{aligned} u(t, e^{\mu K'(r)t}) &\underset{t \rightarrow \infty}{\approx} A(t, r) e^{(K(r)-1)t - \mu s_+ t K'(r)} \\ &\underset{t \rightarrow \infty}{\approx} A(t, r) e^{F(r)t}. \end{aligned}$$

We observe that the sign of $F(r)$ implies the exponential growth or decay of $u(t, x)$ on the curve $\mathcal{C}_r$.

Remarque 27. We can easily obtain the behavior of $u(t, x)$ on the curve $\mathcal{C}_{s_*}$ and $\mathcal{C}_1$.

— On the curve $\mathcal{C}_{s_*}$ where s_* is defined as the minimum of K on $(0, \infty)$, we observe that

$$K(s_*) - 1 - \mu s_* K'(s_*) = K(s_*) - 1 < 0.$$

It follows $u(t, x)$ decrease to 0 when $t \rightarrow \infty$ on the curve $\mathcal{C}_{s_*}$.

— On the curve $\mathcal{C}_1$ defined by $s_+(t, x) = 1$ it follows that

$$K(1) - 1 - \mu K'(1) = \mu K'(1).$$

If $s_* < 1$ then $K'(1) > 0$. It follows $u(t, x)$ increase to ∞ when $t \rightarrow \infty$ on the curve $\mathcal{C}_1$. If $s_* > 1$ then $K'(1) < 0$. It follows $u(t, x)$ decrease to 0 when $t \rightarrow \infty$ on the curve $\mathcal{C}_1$.

The next lemmas are intended to give us a better understanding of the sign of the function F .

Lemme 28. *Let k_0 be a probability measure on $\mathbb{R}_+$ such that (??) and let K denote its Mellin transform. Assume that $I(k_0) = (-\infty, \infty)$. Let us consider the function F defined by (??) Then*

$$\lim_{s \rightarrow -\infty} F(s) = -\infty \quad \text{and} \quad \lim_{s \rightarrow +\infty} F(s) = -\infty.$$

and F achieves a global maximum on $\mathbb{R}$.

Proof. By continuity of F , if

$$\lim_{s \rightarrow -\infty} F(s) = -\infty \quad \text{and} \quad \lim_{s \rightarrow +\infty} F(s) = -\infty.$$

then F achieves a global maximum on $\mathbb{R}$.

For $s \in \mathbb{R}$.

$$\begin{aligned} F(s) &= K(s) - 1 - \mu s K'(s) \\ &= \int_0^\infty (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz - 1 \\ &= \int_0^1 (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz + \int_1^\infty (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz - 1 \end{aligned}$$

If $s > 1$ We observe that

$$\begin{aligned} \int_0^1 (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz &= \int_0^1 (1 + \mu s |\ln(z)|) k_0(z) z^{s-1} dz \xrightarrow{s \rightarrow +\infty} 0 \\ \int_1^\infty (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz &\xrightarrow{s \rightarrow +\infty} -\infty \end{aligned}$$

If $s < 0$ We observe that

$$\begin{aligned} \int_0^1 (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz &\xrightarrow{s \rightarrow +\infty} -\infty \\ \int_1^\infty (1 - \mu s \ln(z)) k_0(z) z^{s-1} dz &= \int_1^\infty (1 + \mu |s| \ln(z)) k_0(z) z^{s-1} dz \xrightarrow{s \rightarrow +\infty} 0 \end{aligned}$$

■

Lemma 29. *Let us consider the function F defined by (??). If there exists $s \in \mathbb{R}$ such that $F(s) > 0$, then F has at least two zeros.*

Proof. Lemma ?? ensures that

$$\lim_{s \rightarrow -\infty} F(s) = -\infty \quad \text{and} \quad \lim_{s \rightarrow +\infty} F(s) = -\infty$$

If there exists $s \in \mathbb{R}$ such that $F(s) > 0$, then F has at least two zeros by continuity. ■

Lemma ?? allows us to distinguish between two different cases:

- In the first case, F has at least two zeros. The cells concentrate on the paths $(C_r)_{r \in \mathbb{R}}$ where $F(r) > 0$. This is the case in Examples 1, 2 and 3 below.
- In the second case F has no zeros. In this case the quantities $u(t, x)$ are decreasing on all paths $(C_r)_{r \in \mathbb{R}}$. Biologically, this could mean that the cells are spreading quickly in x . This is the case in Example 4 below.

We want to know where F achieves a global maximum. Indeed this curve should be the curve containing the most cells. For that, we are looking for the solutions

$$F'(s) = K'(s)(1 - \mu) - \mu s K''(s) = 0, \quad s \in \mathbb{R}.$$

Example 1: The sum of Dirac For the uniform kernel $k_0 = \frac{1}{2}\delta_{1-\epsilon} + \frac{1}{2}\delta_{1+\epsilon}$, we have $K(s) = \frac{1}{2}(1 - \epsilon)^{s-1} + \frac{1}{2}(1 + \epsilon)^{s-1}$ with $\epsilon = 0.7$. Let $\mu = 0.7$. We plot F and several curves of the form $s_+ = \gamma$ in Figure ?? . In this example, the curve containing the most cells (the dotted curve) converges to 0 when $t \rightarrow \infty$. Nevertheless, a non-negligible proportion of cells have an increasing fitness.

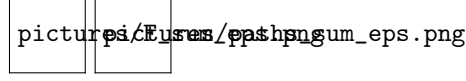


Figure 4.1 – **Left:** Function $F(s)$ for $k_0 = \frac{1}{2}\delta_{1-\varepsilon} + \frac{1}{2}\delta_{1+\varepsilon}$. We see there are two zeros. **Right:** Several curves of the form $s_+ = \gamma$ for different values of γ . Between the black curves, $u(t, x)$ tends to infinity. Outside, $u(t, x)$ tends to zero exponentially fast. The red dotted line correspond to the curve where F achieves a global maximum.

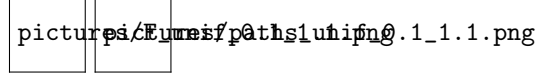


Figure 4.2 – **Left:** Function $F(s)$ for $k_0 = \mathbb{1}_{[0.1;1.1]}$. We see there are two zeros. **Right:** Several curves of the form $s_+ = \gamma$ for different values of γ . Between the black curves, $u(t, x)$ tends to infinity. Outside, $u(t, x)$ tends to zero exponentially fast. The red dotted line correspond to the curve where F achieves a global maximum.

Example 2: The uniform kernel $k_0 = \mathbb{1}_{[0.1;1.1]}$
 For the uniform kernel $k_0 = \mathbb{1}_{[0.1;1.1]}$, we have $K(s) = \frac{1}{s} \cdot \frac{1.1^s - 0.1^s}{1.1 - 0.1}$. Let $\mu = 0.7$. We plot F and several curves of the form $s_+ = \gamma$ in Figure ??

Example 3: The uniform kernel $k_0 = \mathbb{1}_{[0.98;1.1]}$
 For the uniform kernel $k_0 = \mathbb{1}_{[0.98;1.1]}$, we have $K(s) = \frac{1}{s} \cdot \frac{1.1^s - 0.98^s}{1.1 - 0.98}$. Let $\mu = 0.7$. We plot F and several curves of the form $s_+ = \gamma$ in Figure ??

Example 4: The uniform kernel $k_0 = \mathbb{1}_{[0.1;5]}$
 For the uniform kernel $k_0 = \mathbb{1}_{[0.1;5]}$, we have $K(s) = \frac{1}{s} \cdot \frac{1.5^s - 0.1^s}{1.5 - 0.1}$. Let $\mu = 0.7$. We plot F and several curves of the form $s_+ = \gamma$ in Figure ??

4.3 Linear mutation rate

In this section we consider a whole population dynamics, in a constant environment (infinite space and nutrient). Since x represents the individual growth rate, it is well-known that x is also the Malthus parameter of a clonal population which reproduces by fission and grows at this rate x [robert:hal-00981312]. Rather than a constant mutation rate, we now assume that the number of mutations is proportional to the growth rate x . This modelling assumption can be justified by the fact that a mutation corresponds to an error in DNA replication. When a cell grows faster, it replicates its DNA faster. This increases the number of errors and therefore the

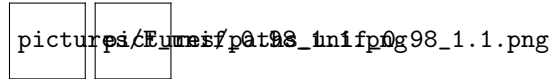


Figure 4.3 – **Left:** Function $F(s)$ for $k_0 = \mathbb{1}_{[0.98;1.1]}$. We see there are two zeros. **Right:** Several curves of the form $s_+ = \gamma$ for different values of γ . Between the black curves, $u(t, x)$ tends to infinity. Outside, $u(t, x)$ tends to zero exponentially fast. The red dotted line correspond to the curve where F achieves a global maximum.

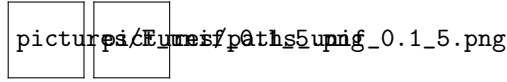


Figure 4.4 – **Left:** Function $F(s)$ for $k_0 = \mathbf{1}_{[0.1;5]}$. We see there is no zero. **Right:** Differents curves of the form $s_+ = \gamma$ for different values of γ . The red dotted line correspond to the curve where F achieves a global maximum.

number of mutations.

We thus consider the following problem:

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) = x(1 - \mu)u(t, x) + \mu \int_0^\infty y k\left(\frac{x}{y}\right) u(t, y) \frac{dy}{y} & \text{for } t \in \mathbb{R}_+, x \in \mathbb{R}_+, \\ u(0, x) = u_0(x) & \text{for } x \in \mathbb{R}_+. \end{cases} \quad (4.19)$$

Here, $u(t, x)$ represents the density of cells with a growth rate x at the time t . The mutation rate is given by μx with $\mu > 0$ a given constant.

We consider several assumptions upon the mutation kernel.

1. We assume that $\mu < 1$ and that the fundamental strip of K contains $[0, \infty)$.
2. We suppose there exists $s_0 \in \mathbb{R}$, $\varepsilon_b > 0$ such that on the vertical strip $\{s \in \mathbb{C} ; \Re(s) \in (s_0 - \varepsilon_b, s_0 + 1 + \varepsilon_b)\}$, the Mellin transform K of the kernel k satisfies

$$|K(\Re(s))| < \frac{1}{\mu} - 1 \quad \text{for } s \in \mathbb{C} \text{ s.t. } \Re(s) \in (s_0 - \varepsilon_b, s_0 + 1 + \varepsilon_b).$$

We note that the first assumption has a biological interpretation: we want the mutation rate to be less than the natural growth rate, i.e., the average division rate. The fact that the fundamental strip of K is $[0, \infty)$ means that k_0 decreases faster than any power law at ∞ and is such that $\int_0^\infty x^{-1} k(x) < \infty$, hence

$$\begin{cases} k(x) = o\left(\frac{1}{x^s}\right) & \text{for every } s \in (1, \infty) \text{ when } x \rightarrow \infty, \\ k(x) = o\left(\frac{1}{x^s}\right) & \text{for every } s \in (0, 1) \text{ when } x \rightarrow 0. \end{cases}$$

The second assumption means that μ is small enough so that such an interval exists. For instance if $\mu < \frac{1}{2}$ we have $K(1) = 1 < \frac{1}{\mu} - 1$. The fact that μ is small means that mutations are (relatively) rare events.

To obtain a solution to Equation (??), our strategy is to transform it by successively applying a Mellin transform and then a Laplace transform. By complex analysis arguments, we can then obtain an explicit expression for the Laplace-Mellin transform, which we can invert and study in order to obtain asymptotic expansions for large times and small or large trait x .

By applying the Mellin transform in the x variable, we have

$$\begin{cases} \frac{\partial}{\partial t} U(t, s) = [-\mu(1 - K(s)) + 1] U(t, s + 1) & \text{for } t \in \mathbb{R}_+, s \in S_M \\ U_0(s) = \mathcal{M}(u_0)(s) & \text{for } s \in S_M, \end{cases} \quad (4.20)$$

where S_M denotes the vertical strip where the Mellin transform of $u(t, \cdot)$ is defined and analytic.

We consider the Laplace transform of $t \mapsto U(t, s)$

$$L(z, s) = \int_0^\infty U(t, s) e^{-zt} dt . \quad (4.21)$$

By applying the Laplace transform to Equation (??) in the t variable we have

$$zL(z, s) - U_0(s) = [-\mu(1 - K(s)) + 1] L(z, s + 1)$$

It follows that

$$\begin{aligned} zL(z, s) &= [-\mu(1 - K(s)) + 1] L(z, s + 1) + U_0(s) \\ &= \Phi(s)L(z, s + 1) + U_0(s) , \end{aligned} \quad (4.22)$$

where $\Phi(s) := [-\mu(1 - K(s)) + 1]$.

We see that (??) links $L(z, s)$ on the left-hand side to $L(z, s + 1)$ on the right-hand side. Such equation can be solved using the Wiener–Hopf method, successfully implemented e.g. in [escobedo2007fundamental] to solve the linearized Uehling-Uhlenbeck equation. For a general description of this method the reader may consult the Chapter 4 of [zakharov2012kolmogorov].

We present briefly the different steps of the Wiener–Hopf method.

- *Step 1 - Change of variables:* We transform by a change of variable Equation (??), which links the solution $L(z, \cdot)$ to $L(z, \cdot + 1)$, where L is defined on a strip $(a, b) + i\mathbb{R}$, into an equation which links $g(z, x + i0)$ to $g(z, x - i0)$ where the unknown is the function $g : \mathbb{C} \setminus \mathbb{R}_+ \rightarrow \mathbb{C}$ (See Figure ??). On the line $\mathbb{R}_+$, $g(z, \cdot)$ satisfies the equation

$$zg(z, x + i0) = \varphi(x)g(z, x - i0) + 1 \quad \text{for all } x \in \mathbb{R}_+ ,$$

with

$$\begin{aligned} g(z, x + i0) &= \lim_{\varepsilon \rightarrow 0^+} g(z, xe^{i\varepsilon}), \\ g(z, x - i0) &= \lim_{\varepsilon \rightarrow 0^+} g(z, xe^{i(2\pi - \varepsilon)}) . \end{aligned}$$

- *Step 2 - Find the good expression:* We aim to obtain an expression on $\mathbb{C} \setminus \mathbb{R}_+$ of the form $\Xi(z, x + i0) = \Xi(z, x - i0)$ where g is contained in the expression of Ξ . The goal is to obtain an expression that only depends on $x + i0$ (resp. $x - i0$) on the left (resp. right) side of the equality. This allows us to extend Ξ on $\mathbb{C}^*$ as an analytic function. To obtain this expression, the key point is the Sokhotski–Plemelj formula [sokhotski1873definite] that split a continuous function f into two part **pas clair pour moi**

$$f(x) = f_-(x - i0) + f_+(x + i0) \quad \text{for } x \in (0, \infty) .$$

In the proof of Theorem ?? below, we obtain the expression

$$\Xi(\cdot) = M(z, \cdot)g(z, \cdot) - W(z, \cdot)$$

where M and W will be defined later.

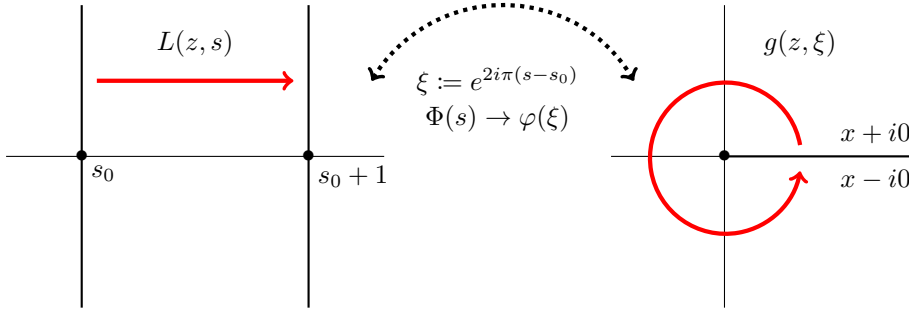


Figure 4.5 – Illustration of the change of variables. The strip $\Re(s) \in (s_0, s_0 + 1)$ on the left is transformed in $\mathbb{C} \setminus \mathbb{R}_+$. Red arrows show the "direction" of the transformation. The line $s_0 + i\mathbb{R}$ is transformed into $\mathbb{R}_+ + i0$ and $s_0 - i\mathbb{R}$ is transformed into $\mathbb{R}_+ - i0$.

- *Step 3 - The majoration:* In this part, we obtain bounds on M , W and g that compose the expression of Ξ . This allows us to apply Liouville's theorem to show that Ξ is constant on $\mathbb{C}^*$. From this, we obtain an explicit formula

$$g = \frac{\Xi - W}{M} \quad ,$$

- *Step 4 - Inversion of the transformation:* We revert the change of variables of Step 1 to obtain an explicit expression of $L(s, z)$ from the explicit expression of g .

In addition, the proof of Theorem ?? is based on the following technical lemma which is proved in [escobedo2007fundamental].

Lemme 30 (Lemma C.2. in [escobedo2007fundamental]). *Suppose that f is analytic in the cone*

$$C(2\varepsilon_0) =: \{\zeta \in \mathbb{C}; \quad \zeta = |\zeta|e^{i\theta}, \theta \in (-2\varepsilon_0, 2\varepsilon_0)\}$$

for some $\varepsilon_0 > 0$. Let us also assume that

$$\int_0^\infty \frac{|f(re^{i\theta})|}{1+r^2} dr < \infty \quad \text{for any } \theta \in (-2\varepsilon_0, 2\varepsilon_0) \quad (4.23)$$

$$\lim_{\lambda \rightarrow 0, \lambda \in C(2\varepsilon_0)} f(\lambda) = L_1, \quad \lim_{\lambda \rightarrow \infty, \lambda \in C(2\varepsilon_0)} f(\lambda) = L_2 \quad (4.24)$$

$$|f'(\lambda)| = o\left(\frac{1}{\lambda}\right) \quad \text{as } \lambda \rightarrow 0, \lambda \rightarrow \infty, \lambda \in C(2\varepsilon_0). \quad (4.25)$$

Then, for any $\lambda_0 \in \mathbb{C} \setminus C(2\varepsilon_0)$, the function

$$F(\zeta) = \frac{1}{2i\pi} \int_0^\infty f(\lambda) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda \quad (4.26)$$

is well-defined and analytic in the domain $D(\varepsilon_0) \subset S$. Moreover

$$F(\zeta) = -\frac{L_1}{2i\pi} \ln(\zeta) + o(\ln|\zeta|) \quad \text{as } \zeta \rightarrow 0, \zeta \in D(\varepsilon_0) \quad (4.27)$$

$$F(\zeta) = -\frac{L_2}{2i\pi} \ln(\zeta) + o(\ln|\zeta|) \quad \text{as } \zeta \rightarrow \infty, \zeta \in D(\varepsilon_0) \quad (4.28)$$

4.3.1 Resolution of the equation

Théorème 31. *Consider Assumption ?? . For any $z \in \mathbb{C} \setminus \mathbb{R}_-$, we denote*

$$V(s) := \exp \left[- \int_{s_0 - i\infty}^{s_0 + i\infty} \log \left(\frac{\Phi(\sigma)}{1 - \mu} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(s-\sigma)}} - \frac{1}{1 - e^{2i\pi(\sigma_0 - \sigma)}} \right) d\sigma \right] \quad (4.29)$$

and

$$L(z, s) = -\frac{1}{z} \int_{s_0 - i\infty}^{s_0 + i\infty} \frac{V(y)}{V(s)} e^{\alpha(z)(y-s)} \frac{dy}{(e^{2i\pi(y-s)} - 1)}, \quad (4.30)$$

with $\alpha(z) := \left(\log \left(\frac{z}{1-\mu} \right) \right)$. Then $L(z, \cdot)$ is the unique solution of Equation (??)

4.3.1.1 Reformulating the equation

The first step is to transform Equation (??) on a strip $[s_0, s_0 + 1] + i\mathbb{R}$ into an equation on $\mathbb{C} \setminus \mathbb{R}_+$. For this, we introduce the variables:

$$\zeta := T(s) := e^{2i\pi(s-s_0)}, \quad (4.31)$$

$$g(z, \zeta) := L(z, s), \quad (4.32)$$

$$\varphi(\zeta) := \Phi(s). \quad (4.33)$$

Notice that the transformation T transforms the complex plane $\mathbb{C}$ in the Riemann surface association to the logarithmic function that we denote from now on as $\mathcal{S}$. The function φ is meromorphic in this Riemann surface. We can characterize uniquely each of the sheets of this Riemann surface by means of the argument θ of ζ , where $\zeta := re^{i\theta}$.

We denote D the following portion of the Riemann surface $\mathcal{S}$:

$$D := \{\zeta \in \mathcal{S} \mid \zeta = re^{i\theta}, r > 0, 0 < \theta < 2\pi\}. \quad (4.34)$$

Here, we consider the equation for the fundamental solution, *i.e.*

$$u_0(x) = \delta_1(x) \quad \text{and} \quad U_0(x) = 1,$$

with $U_0(s) = \mathcal{M}(u_0)(s)$. Equation (??) becomes

$$zg(z, x + i0) = \varphi(x)g(z, x - i0) + 1 \quad \text{for all } x \in \mathbb{R}_+ \quad (4.35)$$

where, for any $x \in \mathbb{R}_+$,

$$\begin{aligned} g(z, x + i0) &= \lim_{\varepsilon \rightarrow 0^+} g(z, xe^{i\varepsilon}), \\ g(z, x - i0) &= \lim_{\varepsilon \rightarrow 0^+} g(z, xe^{i(2\pi - \varepsilon)}), \end{aligned}$$

Proposition 32. *Under Assumption ??, the function $g(z, \cdot)$ is analytic on D defined by (??)*

for any $z \in \mathbb{C} \setminus \mathbb{R}_-$ and

$$g(z, \zeta) = \frac{\zeta}{2i\pi z} \int_0^\infty \frac{M(z, \lambda + i0)}{M(z, \zeta)} \frac{d\lambda}{\lambda(\lambda - \zeta)}$$

where $\lambda_0 \in \mathbb{C} \setminus \mathbb{R}_+$ and

$$M(z, \zeta) := \exp \left(-\frac{1}{2i\pi} \int_0^\infty \ln \left(\frac{\varphi(\lambda)}{z} \right) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda \right)$$

Proof. For any $s \in \mathbb{R}_+$, $K(s) > 0 > 1 - \frac{1}{\mu}$. Then, for all $\zeta \in \mathbb{R}_+^*$, $\varphi(\zeta) > 0$. Since φ is analytic on $\mathbb{C}$, it follows that for all $x \in \mathbb{R}_+$

$$\varphi(x) = \lim_{\varepsilon \rightarrow 0^+} \varphi(xe^{i\varepsilon}) = \lim_{\varepsilon \rightarrow 0^-} \varphi(xe^{-i\varepsilon}) = \varphi(x) > 0.$$

For any complex number $r = |r|e^{i\theta}$ we define the following branch of the complex logarithm

$$\ln(r) := \ln(|r|) + i\theta, \quad -\frac{\pi}{4} \leq \theta < 2\pi - \frac{\pi}{4}. \quad (4.36)$$

Let us denote for $\zeta \in D \subset \mathcal{S}$

$$H(z, \zeta) := \frac{1}{2i\pi} \int_0^\infty \ln \left(\frac{\varphi(\lambda)}{z} \right) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda \quad (4.37)$$

for a given $\lambda_0 \in D$ and $z \in \mathbb{C} \setminus \mathbb{R}_-$. We notice that for $\lambda \in (0, \infty)$ we have $\varphi(\lambda) = 1 - \mu(1 - K(s_0 - \frac{i}{2\pi} \ln(\lambda)))$.

There are three things to verify that H is well-defined for $\zeta \in D$.

- (Step 1) The integrand is integrable in 0
- (Step 2) The integrand is integrable in ∞
- (Step 3) The quantity $\ln \left(\frac{\varphi(\lambda)}{z} \right)$ is bounded on $(0, \infty)$.

Step 1: Existence when $\lambda \rightarrow 0$:

We have that:

$$\lim_{\lambda \rightarrow 0} \frac{\varphi(\lambda)}{z} = \lim_{\nu \rightarrow -\infty} \frac{-\mu(1 - K(s_0 + i\nu)) + 1}{z} = \frac{-\mu + 1}{z}.$$

Step 2: Existence when $\lambda \rightarrow \infty$:

$$\lim_{\lambda \rightarrow \infty} \frac{\varphi(\lambda)}{z} = \lim_{\nu \rightarrow \infty} \frac{-\mu(1 - K(s_0 + i\nu)) + 1}{z} = \frac{-\mu + 1}{z}.$$

Step 3: We show that for all $\lambda \in \mathbb{R}$, $|\varphi(\lambda)| > \varepsilon$:

For any $s \in s_0 + i\mathbb{R}$, we have

$$|-\mu(1 - K(s)) + 1| = \mu \left| K(s) - \left(1 - \frac{1}{\mu} \right) \right|.$$

By Assumption ??, for any $s \in s_0 + i\mathbb{R}$, $|K(s)| \leq K(s_0) < \frac{1}{\mu} - 1$. It follows that

$$\forall s \in s_0 + i\mathbb{R}, \quad \left| K(s) - \left(1 - \frac{1}{\mu} \right) \right| \geq \frac{1}{\mu} - 1 - K(s_0) > 0.$$

It follows that $\lambda \mapsto \ln \left(\frac{\varphi(\lambda)}{z} \right)$ is bounded on $(0, \infty)$

Therefore $H(z, \zeta)$ is well defined.

The formulae of Plemej Sojoltski [sokhotski1873definite] ensure that for $x > 0$

$$\begin{aligned} H(z, x + i0) &= \frac{1}{2} \ln \left(\frac{\varphi(x)}{z} \right) + \frac{1}{2i\pi} PV \int_0^\infty \ln \left(\frac{\varphi(\lambda)}{z} \right) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda \\ H(z, x - i0) &= -\frac{1}{2} \ln \left(\frac{\varphi(x)}{z} \right) + \frac{1}{2i\pi} PV \int_0^\infty \ln \left(\frac{\varphi(\lambda)}{z} \right) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda \end{aligned}$$

where PV denotes the principal value. By substracting these expressions and composing by the exponential, we obtain:

$$\frac{\varphi(x)}{z} = \frac{e^{-H(z, x - i0)}}{e^{-H(z, x + i0)}} = \frac{M(z, x - i0)}{M(z, x + i0)}. \quad (4.38)$$

where we have defined

$$M(z, \zeta) = e^{-H(z, \zeta)} \quad \text{for } \zeta \in D, \quad z \in \mathbb{C} \setminus \mathbb{R}_-. \quad (4.39)$$

Equation (??) gives

$$g(z, x + i0) = \frac{\varphi(x)}{z} g(z, x - i0) + \frac{1}{z} \quad \text{for all } x \in \mathbb{R}_+^*$$

so that together with (??) we have

$$g(z, x + i0) = \frac{M(z, x - i0)}{M(z, x + i0)} g(z, x - i0) + \frac{1}{z} \quad \text{for all } x \in \mathbb{R}_+^*.$$

Then

$$M(z, x + i0)g(z, x + i0) = M(z, x - i0)g(z, x - i0) + \frac{M(z, x + i0)}{z} \quad \text{for all } x \in \mathbb{R}_+^*.$$

We define the function

$$W(z, \zeta) = \frac{1}{2i\pi} \int_0^\infty \frac{M(z, \lambda + i0)}{z} \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda - \lambda_0} \right) d\lambda. \quad (4.40)$$

To verify if this function is well-defined, we study the asymptotic behaviour of M when $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. For this, we verify that the function $f(\zeta) = \ln \left(\frac{\varphi(\zeta)}{z} \right)$ satisfies the hypothesis of the Lemma ??

1. We verify Hypothesis (??): It's straightforward to prove that

$$\int_0^\infty \frac{|f(re^{i\theta})|}{1+r^2} dr < +\infty$$

because due to Assumption ??, $f(re^{i\theta})$ is bounded for any $\theta \in (-2\varepsilon_0, 2\varepsilon_0)$ when $\frac{2\varepsilon_0}{2\pi} < \varepsilon_b$.

2. We verify Hypothesis (??): The Riemann-Lebesgue lemma ensures that

$$\lim_{\lambda \rightarrow 0, \lambda \in C(2\varepsilon_0)} f(\lambda) = \lim_{\lambda \rightarrow \infty, \lambda \in C(2\varepsilon_0)} f(\lambda) = \ln\left(\frac{1-\mu}{z}\right) =: L.$$

3. We verify Hypothesis (??): Denoting $\lambda = e^{2i\pi(s-s_0)}$, we have that $\frac{d\lambda}{ds} = 2i\pi e^{2i\pi(s-s_0)} = 2i\pi\lambda$. Then $\frac{ds}{d\lambda} = \frac{1}{2i\pi\lambda}$. It follows that

$$f'(\lambda) = \frac{\varphi'(\lambda)}{\varphi(\lambda)} = \frac{\mu K'(s)}{1-\mu+\mu K(s)} \frac{ds}{d\lambda} = \frac{\mu K'(s)}{1-\mu+\mu K(s)} \cdot \frac{1}{2i\pi\lambda}.$$

If $\lambda \rightarrow 0$, then $s \rightarrow s_0 + i\infty$ and If $\lambda \rightarrow \infty$, then $s \rightarrow s_0 - i\infty$. By the Riemann-Lebesgue lemma, it follows in both cases that

$$K'(s) \rightarrow 0 \quad \text{and} \quad K(s) \rightarrow 0$$

then $|\lambda f'(\lambda)| \rightarrow 0$ which ensure that $|f'(\lambda)| = o(\frac{1}{\lambda})$.

Therefore, we can apply the Lemma ?? which gives the next lemma.

Lemma 33. *For any $\lambda \in \mathbb{C} \setminus C(2\varepsilon_0)$, the function $H(z, \zeta)$ is analytic in the domain $D(\varepsilon_0) = \{\zeta \in \mathcal{S}; \zeta = |\zeta|e^{i\theta}, \theta \in (-\varepsilon_0, 2\pi + \varepsilon_0)\}$ and*

$$H(z, \zeta) = -\frac{L}{2i\pi} \ln(\zeta) + o(\ln(|\zeta|)) \quad \text{as } \zeta \rightarrow 0, \zeta \in D(\varepsilon_0)$$

$$H(z, \zeta) = -\frac{L}{2i\pi} \ln(\zeta) + o(\ln(|\zeta|)) \quad \text{as } \zeta \rightarrow \infty, \zeta \in D(\varepsilon_0)$$

with $L = \ln(\frac{1-\mu}{z})$.

By Definition, $M(z, \zeta) = \exp(-H(z, \zeta))$. This function is analytic on the domain D . Lemma ?? ensures that

Lemma 34. *The function $\zeta \mapsto M(z, \zeta)$ that was defined and analytic for $\zeta \in D$ has an analytic extension to the domain $D(\varepsilon_0)$ for ε_0 small enough. Moreover, for any $\varepsilon > 0$, there exist $\delta \in (-\frac{1}{2}, \frac{1}{2})$ such that, for any z in the region*

$$\mathcal{Z}_\varepsilon := \{z \in \mathbb{C} : \arg z \in (-\pi - \varepsilon, \pi + \varepsilon)\},$$

we have that

$$|M(z, \zeta)| + |\zeta| \left| \frac{\partial}{\partial \zeta} M(z, \zeta) \right| \leq C(z) |\zeta|^\delta \quad \text{for } |\zeta| < 2, \zeta \in D(\varepsilon_0) \quad (4.41)$$

and

$$|M(z, \zeta)| \leq C(z) |\zeta|^\delta \quad \text{for } |\zeta| > 2, \zeta \in D(\varepsilon_0) \quad (4.42)$$

Proof. We follow the demonstration of Lemma C1 in [escobedo2007fundamental]. From Lemma ?? and from the fact that $M(z, \zeta) = \exp(-H(z, \zeta))$ it follows that M has an analytic extension on $D(\varepsilon_0)$. As $\zeta \rightarrow 0$, Lemma ?? ensures that

$$M(z, \zeta) = \exp\left(\frac{L}{2i\pi} \ln(\zeta) + o(\ln(|\zeta|))\right) \quad \text{as } \zeta \rightarrow 0, \zeta \in D(\varepsilon_0) .$$

hence

$$|M(z, \zeta)| \leq Cst ||\zeta|^{\frac{L}{2i\pi}}| = ||\zeta|^{\frac{\log(\frac{1-\mu}{z})}{2i\pi}}| = |\zeta|^{\frac{\arg(\frac{1-\mu}{z})}{2\pi}} = |\zeta|^\delta \quad \text{for } |\zeta| < 1, \zeta \in D(\varepsilon_0)$$

with $\delta = \frac{\arg(\frac{1}{z})}{2\pi}$. We note that $\delta \in (-\frac{1}{2} - \frac{1}{2\pi}, \frac{1}{2} + \frac{1}{2\pi})$ because $\arg(\frac{1}{z}) \in (-\pi-, \pi+)$. Since $M(z, \cdot)$ is analytic, the Cauchy estimates ensure that

$$|M(z, \zeta)| + |\zeta| \left| \frac{\partial}{\partial \zeta} M(z, \zeta) \right| \leq C(z) |\zeta|^\delta \quad \text{for } |\zeta| < 2, \zeta \in D(\varepsilon_0)$$

Similarly, as $\zeta \rightarrow \infty$, Lemma ?? ensures that

$$|M(z, \zeta)| = \exp\left(\frac{L}{2i\pi} \ln(\zeta) + o(\ln(|\zeta|))\right) \quad \text{as } \zeta \rightarrow \infty, \zeta \in D(\varepsilon_0),$$

which implies

$$|M(z, \zeta)| \leq Cst ||\zeta|^{\frac{L}{2i\pi}}| = ||\zeta|^{\frac{\log(\frac{1-\mu}{z})}{2i\pi}}| = |\zeta|^{\frac{\arg(\frac{1-\mu}{z})}{2\pi}} = |\zeta|^\delta \quad \text{for } |\zeta| > 1, \zeta \in D(\varepsilon_0)$$

■

Lemma 35. For any $z \in \mathcal{Z}_\varepsilon$, the function $W(z, \cdot)$ defined in (??) is analytic on D and there is $\delta \in (0, 1)$ and a constant $C(z)$ such that

$$|W(z, \zeta)| \leq C(z) \quad \text{when } |\zeta| < 1, \zeta \in D$$

and

$$|W(z, \zeta)| \leq C(z) |\zeta|^\delta \quad \text{when } |\zeta| > 1, \zeta \in D$$

Proof. Lemma ?? ensures that $W(z, \cdot)$ is well defined and analytic on D .

(1st case): For any $\zeta \in D$ such that $|\zeta| < \frac{1}{2}$,

$$W(z, \zeta) = \int_0^{2|\zeta|} (\dots) d\lambda + \int_{2|\zeta|}^1 (\dots) d\lambda + \int_1^\infty (\dots) d\lambda := W_1 + W_2 + W_3 .$$

We have from Lemma ?? that

$$|W_3| \leq C(z) \int_1^\infty \lambda^\delta \frac{1 + |\lambda_0|}{|\lambda| |\lambda - \lambda_0|} d\lambda < \infty .$$

Since $\zeta \in D$, there is $\varepsilon_0 > 0$ such that $\arg \zeta \in (\varepsilon_0, 2\pi - \varepsilon_0)$. If $\lambda \in (0, 2|\zeta|)$, then $|\lambda - \zeta| \geq \frac{\varepsilon_0}{2} |\zeta|$.

It follows that

$$W_1 \leq C(z) \int_0^{2|\zeta|} \lambda^\delta \left(\frac{1}{(\lambda - \lambda_0)} + \frac{2}{\varepsilon_0 |\zeta|} \right) \leq \frac{C(z)}{|\zeta|} \int_0^{2|\zeta|} \lambda^\delta d\lambda \leq C(z) |\zeta|^\delta .$$

Similarly, we can prove that $|W_2| < \infty$ by using the fact that for all $\lambda \in (2|\zeta|, 1)$

$$|\lambda - \zeta| \geq \frac{|\lambda|}{2} .$$

(2nd case): For any $\zeta \in D$ such that $|\zeta| > \frac{1}{2}$,

$$W(z, \zeta) = \int_0^1 (\dots) d\lambda + \int_1^\infty (\dots) d\lambda .$$

For all $\lambda \in (0, 1)$,

For all $\lambda \in (1, \infty)$,

$$\left| \int_1^\infty (\dots) d\lambda \right| \leq C \int_1^\infty |\lambda|^\delta \frac{|\zeta|}{|\lambda| \cdot |\lambda - \zeta|} d\lambda = C \int_1^\infty |\lambda|^\delta \frac{1}{|\lambda| \cdot \left| \frac{\lambda}{|\zeta|} - 1 \right|} d\lambda .$$

The change of variables $\lambda = x\zeta$ ensures that

$$\left| \int_1^\infty (\dots) d\lambda \right| \leq C \int_1^\infty |x|^\delta |\zeta|^\delta \frac{1}{|\zeta| |x| \cdot |x - 1|} |\zeta| dx \leq C |\zeta|^\delta \int_1^\infty |x|^\delta \frac{1}{|x| \cdot |x - 1|} dx \leq C |\zeta|^\delta$$

■

The Sokhotski–Plemelj formula ensures that

$$\frac{M(z, x + i0)}{z} = W(z, x + i0) - W(z, x - i0) \quad \text{for any } x > 0 .$$

Therefore, for all $x \in \mathbb{R}_+$,

$$M(z, x + i0)g(z, x + i0) - W(z, x + i0) = M(z, x - i0)g(z, x - i0) - W(z, x - i0) .$$

It follows that the function $C(z, \cdot)$ defined by

$$C(z, \cdot) := M(z, \cdot)g(z, \cdot) - W(z, \cdot)$$

is analytic in $\mathbb{C} \setminus \{0\}$.

Using the fact that g is bounded on $\mathbb{C}$, Lemma ?? and Lemma ?? , the Liouville's theorem ensures that

$$\forall z \in \mathbb{C} \setminus \mathbb{R}_- , \quad C(z, \zeta) = C(z)$$

Then

$$g(z, \zeta) = \frac{C(z) + W(z, \zeta)}{M(z, \zeta)} \tag{4.43}$$

However, Lemma ?? and the boundeness of g ensure that

$$C(z) = \lim_{\zeta \rightarrow 0} [M(z, \cdot)g(z, \cdot) - W(z, \cdot)] = - \lim_{\zeta \rightarrow 0} W(z, \zeta) \quad (4.44)$$

As

$$\lim_{\zeta \rightarrow 0} W(z, \zeta) = \frac{1}{2i\pi z} \int_0^\infty M(z, \lambda + i0) \left(\frac{1}{\lambda} - \frac{1}{\lambda - \lambda_0} \right) d\lambda$$

we have that

$$\begin{aligned} C(z) + W(z, \zeta) &= \frac{1}{2i\pi z} \int_0^\infty M(z, \lambda + i0) \left(\frac{1}{\lambda - \zeta} - \frac{1}{\lambda} \right) d\lambda \\ &= \frac{\zeta}{2i\pi z} \int_0^\infty M(z, \lambda + i0) \frac{1}{\lambda(\lambda - \zeta)} d\lambda \end{aligned}$$

It follows that

$$g(z, \zeta) := \frac{\zeta}{2i\pi z} \int_0^\infty \frac{M(z, \lambda + i0)}{M(z, \zeta)} \frac{d\lambda}{\lambda(\lambda - \zeta)} \quad (4.45)$$

$$(4.46)$$

For any $z \in \mathbb{C} \setminus \mathbb{R}_-$, Equation (??) ensures that $M(z, \cdot)$ does not vanish on D and Lemma ?? ensures that $M(z, \cdot)$ is analytic on $D(\varepsilon_0)$. It follows that $g(z, \cdot)$ is analytic on D for any $z \in \mathbb{C} \setminus \mathbb{R}_-$. ■

4.3.1.2 The solution of the Equation

Now we want to return to the solution of Equation (??). For this purpose, we revert the change of variables (??),(??) and (??) which gives

$$L(z, s) = g(z, T(s)) \quad \text{for all } z \in \mathbb{C} \setminus \mathbb{R}_+, s \in S. \quad (4.47)$$

We define $m(z, s) := M(z, T(s))$. Then

$$\begin{aligned} L(z, s) &= g(z, T(s)) \\ &= \frac{T(s)}{2i\pi z} \int_0^\infty \frac{M(z, \lambda)}{m(z, s)} \frac{d\lambda}{\lambda(\lambda - T(s))} \\ &= \frac{1}{2i\pi z} \int_0^\infty \frac{M(z, \lambda)}{m(z, s)} \frac{d\lambda}{\lambda \left(\frac{\lambda}{T(s)} - 1 \right)} \\ &= \frac{1}{2i\pi z} \int_0^\infty \frac{M(z, \lambda)}{m(z, s)} \frac{d\lambda}{\lambda \left(\frac{\lambda}{e^{2i\pi(s-s_0)}} - 1 \right)} \end{aligned}$$

The change of variable $\lambda = e^{2i\pi(\sigma-s_0)}$ gives

$$\begin{aligned} L(z, s) &= -\frac{2i\pi}{2i\pi z} \int_{s_0-i\infty}^{s_0+i\infty} \frac{m(z, \sigma)}{m(z, s)} \frac{d\sigma}{(e^{2i\pi(\sigma-s)} - 1)} \\ &= -\int_{s_0-i\infty}^{s_0+i\infty} \frac{1}{z} \frac{m(z, \sigma)}{m(z, s)} \frac{d\sigma}{(e^{2i\pi(\sigma-s)} - 1)}. \end{aligned}$$

The change of variable shows that

$$m(z, s) = \exp \left[-\int_{s_0-i\infty}^{s_0+i\infty} \log \left(\frac{\Phi(\sigma)}{z} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(s-\sigma)}} - \frac{1}{1 - e^{2i\pi(\sigma_0-\sigma)}} \right) d\sigma \right] \quad (4.48)$$

with $\lambda_0 = e^{2i\pi(\sigma_0-s_0)}$. We have that

$$L(z, s) = -\int_{s_0-i\infty}^{s_0+i\infty} \frac{1}{z} \frac{m(z, \sigma)}{m(z, s)} \frac{d\sigma}{(e^{2i\pi(\sigma-s)} - 1)} \quad (4.49)$$

We note that the term $\frac{m(z, \sigma)}{m(z, s)}$ depends on z . We have

$$m(z, s) = V(s) \exp \left[-\int_{s_0-i\infty}^{s_0+i\infty} \left[\log \left(\frac{1-\mu}{z} \right) \right] \cdot \left(\frac{1}{1 - e^{2i\pi(s-\sigma)}} - \frac{1}{1 - e^{2i\pi(\sigma_0-\sigma)}} \right) d\sigma \right] \quad (4.50)$$

where

$$V(s) := \exp \left[-\int_{s_0-i\infty}^{s_0+i\infty} \log \left(\frac{\Phi(\sigma)}{1-\mu} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(s-\sigma)}} - \frac{1}{1 - e^{2i\pi(\sigma_0-\sigma)}} \right) d\sigma \right]. \quad (4.51)$$

Then

$$\frac{m(z, y)}{m(z, s)} = \frac{V(y)}{V(s)} \times E(z, y, s)$$

where

$$E(z, y, s) := \exp \left[-\int_{s_0-i\infty}^{s_0+i\infty} \left[\log \left(\frac{1-\mu}{z} \right) \right] \cdot \left(\frac{1}{1 - e^{2i\pi(y-\sigma)}} - \frac{1}{1 - e^{2i\pi(s-\sigma)}} \right) d\sigma \right]$$

We have that

$$E(z, y, s) = \exp \left[\left(\log \left(\frac{1-\mu}{z} \right) \right) \cdot (y-s) \right]$$

Then

$$L(z, s) = -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} \frac{V(y)}{V(s)} e^{\alpha(z)(y-s)} \frac{dy}{(e^{2i\pi(y-s)} - 1)} \quad (4.52)$$

where $\alpha(z) := \left(\log \left(\frac{z}{1-\mu} \right) \right)$

4.3.2 Asymptotic behaviour

In the previous section, we have established an explicit expression for $L(z, s)$. In this section, we want to obtain some information on the asymptotic behaviour of $u(t, x)$ that can be obtained

by applying the inverse Laplace transform and the inverse Mellin transform to $L(z, s)$.

We recall that the inverse Mellin transform of $L(z, \cdot)$ is given by

$$\tilde{L}(z, x) = \mathcal{M}^{-1}(L(z, \cdot))[x] = \frac{1}{2i\pi} \int_{b-i\infty}^{b+i\infty} L(z, s) x^{-s} ds, \quad z \in \mathbb{C} \setminus \mathbb{R}_-, \quad x \in (0, \infty).$$

where b is in $< s_0, s_0 + 1 >$.

The inverse Laplace transform of $\tilde{L}(z, x)$ is given by

$$\mathcal{L}^{-1}(\tilde{L}(\cdot, x))[t] = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{zt} \tilde{L}(z, x) dz.$$

where $c \in \mathbb{R}$ is large enough to have all the singularities of $\tilde{L}(\cdot, x)$ at the left of the line $\Re z = c$.

Since the function $L(z, s)$ is bounded, the operator $\mathcal{L}^{-1}$ and $\mathcal{M}^{-1}$ are defined in the sense of tempered distribution [misra1986transform]. It follows that the fundamental solution of the problem is given by

$$u(t, x) = \mathcal{L}^{-1}(\mathcal{M}^{-1}(G)). \quad (4.53)$$

In order to study the asymptotic behavior of $u(t, x)$ we need the following lemma:

Lemme 36. *The function V defined by Equation (??) satisfies*

$$V(s+1) = V(s) \frac{\Phi(s+1)}{1-\mu} \quad \text{for } s \in (s_0, s_0+1) + i\mathbb{R}. \quad (4.54)$$

It can be extended meromorphically to $\mathbb{C}$ using (??).

Proof. We follow the proof of Lemma C4 in [escobedo2007fundamental] ■

Remarque 37. Equation (??) ensure that $V(s) = (1-\mu) \frac{V(s+1)}{\Phi(s+1)}$. The function V has no zero on the vertical strip $< s_0, s_0 + 1 >$. Let $s_{pole} \in \mathbb{C}$ be a pole of Φ . Then V has a zero for all $s \in \{s_{pole} - \mathbb{N}^*\}$. Let $s_{zero} \in \mathbb{C}$ be a zero of Φ . Then V has a zero for all $s \in \{s_{zero} + \mathbb{N}\}$.

Assumption: We suppose now that the function ϕ satisfies: We assume that the function Φ satisfies

$$|\Im \Phi(\xi) - \Im \Phi(\xi)_\infty| + |\xi| \cdot |\Im \Phi'(\xi) - \Im \Phi(\xi)_\infty| = \mathcal{O}(|\xi|^{-\gamma-1})$$

with $\gamma > 0$, $\alpha > 0$ and

$$\Phi(\xi)_\infty := (1-\mu) + \frac{b_1}{\xi^\alpha} + \frac{b_2}{\xi}.$$

4.3.2.1 Description of $u(t, x)$ as $x \rightarrow \infty$

From the definition of the inverse Mellin transform, we have

$$\mathcal{M}^{-1}[L(z, \cdot)](x) = \frac{1}{2i\pi} \int_{b-i\infty}^{b+i\infty} L(z, s) x^{-s} ds, \quad z \in \mathbb{C} \setminus \mathbb{R}_-. \quad (4.55)$$

The main contribution of $\mathcal{M}^{-1}[L(z, \cdot)](x)$ as $x \rightarrow \infty$ is due to the closest pole s_p of $L(z, \cdot)$ to the line $b + i\mathbb{R}$ and right to this line. The pole of $L(z, \cdot)$ are given by the zeros of V .

By deforming the contour integration to a line $d + i\mathbb{R}$ with $d > \Re(s_p)$, The residue theorem ensures that

$$\mathcal{M}^{-1}[L(z, \cdot)](x) = -x^{-s_p}(L(z, \cdot); s = s_p) + \frac{1}{2i\pi} \int_{d-i\infty}^{d+i\infty} L(z, s)x^{-s}ds \quad (4.56a)$$

$$= -x^{-s_p}(L(z, \cdot); s = s_p) + o(x^{-\Re(s_p)}) . \quad (4.56b)$$

The residue of $L(z, \cdot)$ at the pole s_p is given by

$$(L(z, \cdot); s = s_p) = -\frac{1}{z} \left(\frac{1}{V(s)}; s = s_p \right) \int_{s_0-i\infty}^{s_0+i\infty} V(\sigma) e^{\alpha(z)(\sigma-s_p)} \frac{d\sigma}{(e^{2i\pi(\sigma-s_p)} - 1)} .$$

From Equation (??), we obtain that

$$V(s) = V(s-1) \frac{\Phi(s)}{1-\mu} \quad \text{for } s \in (s_0+1, s_0+2) + i\mathbb{R} .$$

Then

$$\lim_{s \rightarrow s_p} \frac{V(s)}{s-s_p} = \lim_{s \rightarrow s_p} \frac{V(s-1)}{1-\mu} \frac{\Phi(s) - \Phi(s_p)}{s-s_p} = \frac{V(s_p-1)}{1-\mu} \Phi'(s_p) \neq 0 .$$

Therefore, the residue of $1/V$ at the pole $s = s_p$ is given by

$$\left(\frac{1}{V(s)}; s = s_p \right) = \frac{1-\mu}{V(s_p-1)} \frac{1}{\Phi'(s_p)} .$$

$$(L(z, \cdot); s = s_p) = \frac{-(1-\mu)}{z\Phi'(s_p)V(s_p-1)} \int_{s_0-i\infty}^{s_0+i\infty} V(\sigma) e^{\alpha(z)(\sigma-s_p)} \frac{d\sigma}{(e^{2i\pi(\sigma-s_p)} - 1)} . \quad (4.57)$$

By using this expression in Equation (??), we obtain that

$$\mathcal{M}^{-1}[L(z, \cdot)](x) = x^{-s_p} \frac{1-\mu}{z\Phi'(s_p)V(s_p-1)} \int_{s_0-i\infty}^{s_0+i\infty} \frac{V(\sigma) e^{\alpha(z)(\sigma-s_p)}}{(e^{2i\pi(\sigma-s_p)} - 1)} d\sigma + \frac{1}{2i\pi} \int_{d-i\infty}^{d+i\infty} L(z, s)x^{-s}ds .$$

Taking the inverse Laplace transform, we obtain that:

$$\begin{aligned} u(t, x) &= \mathcal{L}^{-1}(\tilde{L}(\cdot, x))[t] \\ &= \frac{1}{2i\pi} \int_{c-i\infty}^{c+i\infty} e^{zt} \mathcal{M}^{-1}[L(z, \cdot)](x) dz . \\ &= A(t)x^{-s_p} + B(t, x) \end{aligned}$$

with

$$A(t) := \frac{1-\mu}{\Phi'(s_p)V(s_p-1)} \int_{s_0-i\infty}^{s_0+i\infty} \frac{V(\sigma)}{(e^{2i\pi(\sigma-s_p)} - 1)} \left[\int_{c-i\infty}^{c+i\infty} \frac{e^{\alpha(z)(\sigma-s_p)} e^{tz}}{z} dz \right] d\sigma \quad (4.58)$$

and

$$B(t, x) := \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} L(z, s) x^{-s} ds \right] (t) \quad (4.59)$$

Let $\ln_m$ denote the main branch of the complex logarithm. There exists $k \in \mathbb{Z}$ such that for all $z \in \mathbb{C} \setminus \{0\}$, $\ln_m(z) = \ln(z) + 2i\pi k$.

We get that

$$\mathcal{L}^{-1} \left[\frac{e^{\alpha(z)(\sigma-s_p)}}{z} \right] = \mathcal{L}^{-1} \left[\frac{e^{\left(\frac{\ln_m(z)+2i\pi k}{1-\mu}\right)(\sigma-s_p)}}{z} \right] \quad (4.60a)$$

$$= e^{\left(\frac{2i\pi k}{1-\mu}\right)(\sigma-s_p)} \int_{c-i\infty}^{c+i\infty} e^{\alpha(z)(\sigma-s_p)} e^{tz} \frac{dz}{z} \quad (4.60b)$$

$$= \beta_p(\sigma) \Gamma(\sigma - s_p) ((1-\mu)t)^{-(\sigma-s_p)} (e^{2i\pi(\sigma-s_p)} - 1), \quad (4.60c)$$

where $\beta_p(\sigma) := e^{\left(\frac{2i\pi k}{1-\mu}\right)(\sigma-s_p)}$.

The two next lemmas give estimates for A and B .

Lemme 38 (An estimate for A). *The following estimates hold:*

$$\begin{aligned} A(t) &= -\frac{t(1-\mu)^2}{\Phi'(s_p)} (\Gamma(z), z = -1) \\ &\quad + \frac{1-\mu}{\Phi'(s_p)V(s_p-1)} \int_{d-i\infty}^{d+i\infty} \beta_p(\sigma) V(\sigma) \Gamma(\sigma - s_p) ((1-\mu)t)^{-(\sigma-s_p)} d\sigma \\ &= \frac{t(1-\mu)^2}{\Phi'(s_p)} + o(t). \end{aligned}$$

as $x \rightarrow \infty$ and $t \rightarrow \infty$.

Proof. We get from Equations (??) and (??) that

$$A(t) = \frac{1-\mu}{\Phi'(s_p)V(s_p-1)} \int_{s_0-i\infty}^{s_0+i\infty} \frac{\beta_p(\sigma) V(\sigma) \Gamma(\sigma - s_p)}{(e^{2i\pi(\sigma-s_p)} - 1)} ((1-\mu)t)^{-(\sigma-s_p)} (e^{2i\pi(\sigma-s_p)} - 1) d\sigma \quad (4.61a)$$

$$= \frac{1-\mu}{\Phi'(s_p)V(s_p-1)} \int_{s_0-i\infty}^{s_0+i\infty} \beta_p(\sigma) V(\sigma) \Gamma(\sigma - s_p) ((1-\mu)t)^{-(\sigma-s_p)} d\sigma. \quad (4.61b)$$

We move the integration contour right to $\Re(\sigma) = s_0$. The pole of the function $\Gamma(\cdot - s_p)$ are located in $\{s_p - \mathbb{N}\}$. Since s_p is the first pole of $L(s, z)$ right to the vertical strip $< s_0, s_0 + 1 >$, it follows that $s_p - 1$ is first pole of $\Gamma(\cdot - s_p)$ right to the vertical line $\Re(\sigma) = s_0$.

Let $d \in (\Re(s_p) - 1, \Re(s_p))$. The Residue Theorem ensures that

$$\begin{aligned} A(t) &= -\frac{t(1-\mu)^2}{\Phi'(s_p)} (\Gamma(z), z = -1) \\ &\quad + \frac{1-\mu}{\Phi'(s_p)V(s_p-1)} \int_{d-i\infty}^{d+i\infty} \beta_p(\sigma) V(\sigma) \Gamma(\sigma - s_p) ((1-\mu)t)^{-(\sigma-s_p)} d\sigma \\ &= \frac{t(1-\mu)^2}{\Phi'(s_p)} + o(t). \end{aligned}$$

■

Lemme 39 (An estimate for B). *The following estimates hold:*

$$\begin{aligned} |B(t, x)| &\leq C e^{-\tilde{b}x} ((1-\mu)t)^{\tilde{d}-\tilde{b}} \text{ for } x > 1, 0 < t < 1. \\ |B(t, x)| &\leq C e^{-\tilde{b}x} ((1-\mu)t)^{r-\tilde{b}} \text{ for } x > 1, t \geq 1. \end{aligned}$$

where $\tilde{b} < s_0$ is an arbitrary real number, $\tilde{d} > s_0 + 1$ is an arbitrary real number and r is an arbitrary number less than s_p .

Proof. Let us recall Equation (??),

$$B(t, x) := \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} L(z, s) x^{-s} ds \right] (t).$$

with

$$L(z, s) = -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} \frac{V(y)}{V(s)} e^{\alpha(z)(y-s)} \frac{dy}{(e^{2i\pi(y-s)} - 1)}$$

We obtain from (??) that

$$L(z, s) = -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} e^{-(y-s) \log(\frac{\Phi(s)}{1-\mu})} e^{h(s, y-s)} \times e^{\alpha(z)(y-s)} \frac{dy}{(e^{2i\pi(y-s)} - 1)}$$

Using the change of variables $\theta := y - s$, we rewrite

$$L(z, s) = -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} e^{-\theta \log(\frac{\Phi(s)}{1-\mu}) + h(s, \theta)} \times e^{\alpha(z)\theta} \frac{d\theta}{(e^{2i\pi\theta} - 1)}$$

Let us define the function

$$\tilde{h}(s, \theta) := h(s, \theta) - \theta \log\left(\frac{\Phi(s)}{1-\mu}\right).$$

We can decompose L as follows:

$$\begin{aligned} L(z, s) &= \mathcal{A}_1 + \mathcal{A}_2 \\ \mathcal{A}_1 &:= -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} e^{-\theta \log(\frac{\Phi(s)}{1-\mu})} \times e^{\alpha(z)\theta} \frac{d\theta}{(e^{2i\pi\theta} - 1)} \\ \mathcal{A}_2 &:= -\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} e^{-\theta \log(\frac{\Phi(s)}{1-\mu})} \times e^{\alpha(z)\theta} \frac{(e^{\tilde{h}(\xi, \theta)} - 1)d\theta}{(e^{2i\pi\theta} - 1)} \end{aligned}$$

Then, we can decompose

$$B(t, x) = B_1(t, x) + B_2(t, x)$$

where

$$\begin{aligned} B_1(t, x) &= \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} \mathcal{A}_1 x^{-s} ds \right] (t) \\ B_2(t, x) &= \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} \mathcal{A}_2 x^{-s} ds \right] (t) \end{aligned}$$

To obtain a bound on B , we start by obtain a bound on B_1 . Then, we obtain a bound on B_2 .

Step 1: Bound on B_1 – We remark that $\mathcal{A}_1$ may be explicitly computed by using residues. To this end, it is enough to replace the integral defining $\mathcal{A}_1$ by the limit of integrations in a sequence of contours Γ_R where these contours are squares. By adding the residues of the integrand, we obtain that

$$\mathcal{A}_1 = \frac{1}{\sqrt{2\pi}z} \sum_{n=0}^{\infty} e^{-2i\pi n\alpha(z)} e^{n \ln(\frac{\Psi(\xi)}{1-\mu})} = \frac{1}{\sqrt{2\pi}z} \left(1 + \frac{\Psi(\xi)}{z} \right)^{-1}.$$

Step 2: Bound on B_2 – We now estimate

$$\begin{aligned} B_2(t, x) &= \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} \mathcal{A}_2 x^{-s} ds \right] (t) \\ &= \frac{1}{2i\pi} \mathcal{L}^{-1} \left[\int_{d-i\infty}^{d+i\infty} \left(-\frac{1}{z} \int_{s_0-i\infty}^{s_0+i\infty} e^{-\theta \log(\frac{\Phi(s)}{1-\mu})} \times e^{\alpha(z)\theta} \frac{(e^{\tilde{h}(\xi, \theta)} - 1)d\theta}{(e^{2i\pi\theta} - 1)} \right) x^{-s} ds \right] (t) \end{aligned}$$

Using (??), we have that

$$\begin{aligned} B_2(t, x) &= \frac{1}{2i\pi} \int_{d-i\infty}^{d+i\infty} \left(-\int_{s_0-i\infty}^{s_0+i\infty} e^{-\theta \log(\frac{\Phi(s)}{1-\mu})} \frac{(e^{\tilde{h}(\xi, \theta)} - 1)d\theta}{(e^{2i\pi\theta} - 1)} \right) x^{-s} ds \\ &\quad \times \beta_p(\theta) \Gamma(\theta) ((1-\mu)t)^{-\theta} (e^{2i\pi\theta} - 1) d\theta \end{aligned}$$

We estimate B_2 when $t \rightarrow 0$ and $x > 1$. Since $\theta = y - s$, we have from Equation (??) that

$$B_2(t, x) = \frac{1}{2i\pi} \int_{d-i\infty}^{d+i\infty} x^{-s} \widehat{B}_2(t, s) ds$$

with

$$\widehat{B}_2(t, s) := - \int_{s_0 - i\infty}^{s_0 + i\infty} \left(\frac{V(y)}{V(s)} - e^{-(y-s) \log(\frac{\Phi(s)}{1-\mu})} \right) \times \beta_p(y-s) \Gamma(y-s) ((1-\mu)t)^{-(y-s)} dy$$

We move the contour of integration. We notice that $\Gamma(y-s)$ has a pole for $y=s$ but this singularity cancels out with the zero of the term in the integrand. We have that

$$\widehat{B}_2(t, s) := - \int_{\bar{s} - i\infty}^{\bar{s} + i\infty} \left(\frac{V(y)}{V(s)} - e^{-(y-s) \log(\frac{\Phi(s)}{1-\mu})} \right) \times \beta_p(y-s) \Gamma(y-s) ((1-\mu)t)^{-(y-s)} dy$$

where $\bar{s}$ is any number in $(s_0, s_0 + 1)$. Using (??), we estimate $\widehat{B}_2(t, s)$

$$e^{-\tilde{b}x} ((1-\mu)t)^{\tilde{d}-\tilde{b}} \int_{\Re \xi = \tilde{b}} |d\xi| \int_{\Re \eta = \tilde{d}} e^{-\pi|\xi-\eta|} \tilde{h}(\xi, \eta - \xi) |d\eta|$$

■

It follows that the integral term is bounded which concludes.

Lemme 40. For any $\xi \in \mathbb{C}$ such that $\Re \xi \in (s_0, s_0 + 1)$ and $\eta \in \mathbb{C}$ such that $\Re \eta = s_0 + 1$,

$$\frac{V(\eta)}{V(\xi)} = e^{-(\eta-\xi) \log(\frac{\Phi(\xi)}{1-\mu})} e^{h(\xi, \eta-\xi)}, \quad (4.62)$$

with

$$h(\xi, \eta - \xi) = - \int_{s_0 + \varepsilon - i\infty}^{s_0 + \varepsilon + i\infty} \log \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(\eta-\sigma)}} - \frac{1}{1 - e^{2i\pi(\xi-\sigma)}} \right) d\sigma$$

which satisfies

$$|h(\xi, \eta - \xi)| \leq C \left[\min \left\{ \frac{|\xi - \eta|^2}{|\xi|^{\alpha+1}}, \frac{|\xi - \eta|}{|\xi|^\alpha} \right\} + \frac{1}{|\xi|^{\alpha+1}} \right]$$

Proof. Equation (??) ensures that for all $\xi \in \mathbb{C}$ such that $\Re \xi \in (s_0, s_0 + 1)$ and $\eta \in \mathbb{C}$ such that $\Re \eta = s_0 + 1$,

$$\frac{V(\eta)}{V(\xi)} = \exp \left[- \int_{s_0 - i\infty}^{s_0 + i\infty} \log \left(\frac{\Phi(\sigma)}{1-\mu} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(\eta-\sigma)}} - \frac{1}{1 - e^{2i\pi(\xi-\sigma)}} \right) d\sigma \right].$$

Let ε such that $s_0 + \varepsilon < \xi$. By deforming and moving the integration contour, we obtain

$$\frac{V(\eta)}{V(\xi)} = \exp \left[- \int_{s_0 + \varepsilon - i\infty}^{s_0 + \varepsilon + i\infty} \log \left(\frac{\Phi(\sigma)}{1-\mu} \right) \cdot \left(\frac{1}{1 - e^{2i\pi(\eta-\sigma)}} - \frac{1}{1 - e^{2i\pi(\xi-\sigma)}} \right) d\sigma \right].$$

It follows that

$$\begin{aligned} \frac{V(\eta)}{V(\xi)} &= \exp \left[- \int_{s_0+\varepsilon-i\infty}^{s_0+\varepsilon+i\infty} \left(\log \left(\frac{\Phi(\xi)}{1-\mu} \right) + \log \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \right) \cdot \left(\frac{1}{1-e^{2i\pi(\eta-\sigma)}} - \frac{1}{1-e^{2i\pi(\xi-\sigma)}} \right) d\sigma \right] \\ &= \exp \left[- \log \left(\frac{\Phi(\xi)}{1-\mu} \right) \int_{s_0+\varepsilon-i\infty}^{s_0+\varepsilon+i\infty} \left(\frac{1}{1-e^{2i\pi(\eta-\sigma)}} - \frac{1}{1-e^{2i\pi(\xi-\sigma)}} \right) d\sigma \right] \times e^{h(\xi, \eta-\xi)} \end{aligned}$$

Elementary calculus yields that

$$\int_{s_0+\varepsilon-i\infty}^{s_0+\varepsilon+i\infty} \left(\frac{1}{1-e^{2i\pi(\eta-\sigma)}} - \frac{1}{1-e^{2i\pi(\xi-\sigma)}} \right) d\sigma = \eta - \xi ,$$

which guarantees (??).

To finish the proof of Lemma ??, it remains to obtain a bound on h . To that, we use the following lemma.

Lemma 41. *Suppose that Φ satisfies Assumption ??. There exists a constant $C > 0$ such that, for all ξ and σ in $\mathbb{C}$ satisfying $\Re \xi \in (s_0, s_0 + 1)$, $\Re \sigma \in (s_0, s_0 + 1)$. There holds*

$$\left| \ln \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \right| \leq C \frac{\sigma - \xi}{|\sigma|^{\alpha+1} + |\xi|^{\alpha+1}} .$$

Proof. We distinguish the two following cases:

1. $|\xi - \sigma| \geq 2|\xi|$;
2. $|\xi - \sigma| < 2|\xi|$.

First case: Consider that $|\xi - \sigma| \geq 2|\xi|$. In this case, $|\sigma| > |\xi|$ and

$$|\Phi(\xi) - \Phi(y)| \leq |\Phi(\xi) - 1| + |\Phi(\sigma) - 1| .$$

Using Assumption ??, it follows that

$$|\Phi(\xi) - \Phi(y)| \leq C \frac{1}{|\xi|^\alpha} \leq C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1}} .$$

Second case: Consider that $|\xi - \sigma| > 2|\xi|$. The Fundamental theorem of calculus ensures that

$$\Phi(\sigma) - \Phi(\xi) = \int_{\xi}^{\sigma} \Phi'(s) ds .$$

Using Assumption ??, we obtain

$$|\Phi(\sigma) - \Phi(\xi)| \leq \int_{\xi}^{\sigma} \frac{1}{|s|^{\alpha+1}} \leq C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1}} ds .$$

Since Φ is bounded from below for and

$$\left| \ln \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \right| \leq C \frac{|\Phi(\sigma) - \Phi(\xi)|}{\Phi(\xi)}$$

we have that

$$\left| \ln \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \right| \leq C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1}}$$

■

End of the proof of Lemma ??. Using Lemma ??, we obtain that

$$\begin{aligned} |h(\xi, \eta - \xi)| &\leq \int_{s_0 + \varepsilon - i\infty}^{s_0 + \varepsilon + i\infty} \left| \ln \left(\frac{\Phi(\sigma)}{\Phi(\xi)} \right) \right| \left| \frac{1}{1 - e^{2i\pi(\eta - \sigma)}} - \frac{1}{1 - e^{2i\pi(\xi - \sigma)}} \right| d\sigma \\ &\leq \int_{s_0 + \varepsilon - i\infty}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \left| \frac{1}{1 - e^{2i\pi(\eta - \sigma)}} - \frac{1}{1 - e^{2i\pi(\xi - \sigma)}} \right| d\sigma \end{aligned}$$

If $\Im \xi \geq \Im \eta$, then

$$|h(\xi, \eta - \xi)| \leq \int_{s_0 + \varepsilon - i\infty}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \cdot \frac{|e^{2i\pi(\eta - \sigma)} - e^{2i\pi(\xi - \sigma)}|}{|1 - e^{2i\pi(\eta - \sigma)}||1 - e^{2i\pi(\xi - \sigma)}|} d\sigma$$

Since $|e^{2i\pi(\eta - \sigma)}| = |e^{-2\pi(\Im \eta - \Im \sigma)}|$ and $|e^{2i\pi(\xi - \sigma)}| = |e^{-2\pi(\Im \xi - \Im \sigma)}|$, we obtain that

$$|e^{2i\pi(\eta - \sigma)} - e^{2i\pi(\xi - \sigma)}| \leq 2|e^{2i\pi(\eta - \sigma)}| = |e^{-2\pi(\Im \eta - \Im \sigma)}|.$$

Therefore,

$$\begin{aligned} |h(\xi, \eta - \xi)| &\leq \int_{\substack{s_0 + \varepsilon - i\infty \\ \Im \sigma \leq \Im \eta}}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \cdot \frac{|e^{-2\pi(\Im \eta - \Im \sigma)}|}{|1 - e^{2i\pi(\eta - \sigma)}||1 - e^{2i\pi(\xi - \sigma)}|} d\sigma \\ &+ \int_{\substack{s_0 + \varepsilon - i\infty \\ \Im \eta \leq \Im \sigma \leq \Im \xi}}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \cdot \frac{|e^{-2\pi(\Im \eta - \Im \sigma)}|}{|1 - e^{2i\pi(\eta - \sigma)}||1 - e^{2i\pi(\xi - \sigma)}|} d\sigma \\ &+ \int_{\substack{s_0 + \varepsilon - i\infty \\ \Im \sigma > \Im \xi}}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \cdot \frac{|e^{-2\pi(\Im \eta - \Im \sigma)}|}{|1 - e^{2i\pi(\eta - \sigma)}||1 - e^{2i\pi(\xi - \sigma)}|} d\sigma \end{aligned}$$

Then

$$\begin{aligned}
|h(\xi, \eta - \xi)| &\leq \int_{\substack{s_0 + \varepsilon - i\infty \\ \Im \sigma \leq \Im \eta}}^{s_0 + \varepsilon + i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} \cdot \frac{1}{|1 - e^{2i\pi(\xi - \sigma)}|} d\sigma \\
&+ \int_{\substack{s_0 + \varepsilon + i\infty \\ \Im \eta \leq \Im \sigma \leq \Im \xi}}^{s_0 + \varepsilon - i\infty} C \frac{|\xi - \sigma|}{|\xi|^{\alpha+1} + |\sigma|^{\alpha+1}} d\sigma \\
&+ \frac{C}{1 + |\xi|^{\alpha+1}} .
\end{aligned}$$

■

As $|\sigma - \xi| \leq |\sigma - \eta| + |\eta - \xi|$, the first term can be estimated as $C(1 + |\eta - \xi|)/|\xi|^{\alpha+1}$. To estimate the second term, we use that $|y - \xi| \leq C|\eta - \xi|$. The remaining integral may be estimated as $C|\xi - \eta||\eta|^{-\alpha}$. This term might be also bounded by $C|\xi - \eta|^2/|\xi|^{\alpha+1}$.

It follows that

$$|h(\xi, \eta - \xi)| \leq C \left[\min \left\{ \frac{|\xi - \eta|^2}{|\xi|^{\alpha+1}}, \frac{|\xi - \eta|}{|\xi|^\alpha} \right\} + \frac{1}{|\xi|^{\alpha+1}} \right] .$$

4.3.3 A discussion about the choice of the branch of the logarithm

In this section, we motivate the choice of the branch of the complex logarithm that we have done in Equation (??). Our purpose is to get a branch of the logarithm that allows to write

$$\ln\left(\frac{\Phi(\sigma)}{z}\right) = \ln(\Phi(\sigma)) + \ln\left(\frac{1}{z}\right)$$

in Equation (??).

Let $z \in \mathbb{C}$ such that $\arg(1/z) \in (\pi/2, 3\pi/2) \bmod 2\pi$. There is $\delta \in (0, 1)$ such that $\mu|K(s)| \leq (1 - \mu)\delta$. It follows that from Definition (??) that

$$\forall s \in S, \quad \begin{cases} \Re(\Phi(s)) = 1 - \mu - \mu\Re(K(s)) , \\ \Im(\Phi(s)) = \mu\Im(K(s)) . \end{cases}$$

Under Assumption ??, we get that for all $s \in S$,

$$-(1 - \mu)\delta \leq \mu|K(s)| \leq (1 - \mu)\delta.$$

It follows that

$$\forall s \in S, \quad \begin{cases} 0 < (1 - \mu)(1 - \delta) \leq \Re(\Phi(s)) \leq (1 - \mu)(1 + \delta) < 2 + 2\mu , \\ \mu - 1 < -(1 - \mu)\delta \leq \Im(\Phi(s)) \leq (1 - \mu)\delta < 1 - \mu . \end{cases}$$

It leads us to consider the following assumption. There exist $\omega \in (0, \frac{\pi}{2})$ such that $\arg \Phi(\sigma) \in (-\omega, \omega) \bmod 2\pi$.

Consider that $\arg \theta \in [-\alpha, 2\pi - \alpha)$, $\alpha > 0$.

— If $\alpha < \pi/2$, then $2\pi - \alpha \geq \frac{3\pi}{2}$. It implies that $\arg(\frac{1}{z}) \in (\frac{\pi}{2}, \frac{3\pi}{2})$. Therefore

$$\arg(\frac{1}{z}) + \arg \phi(\sigma) \in (\frac{\pi}{2} - \omega, \frac{3\pi}{2} + \omega) .$$

We want to get $\begin{cases} \frac{\pi}{2} - \omega \geq -\alpha , \\ \frac{3\pi}{2} + \omega \leq 2\pi - \alpha . \end{cases}$

It implies that $\omega \leq \frac{\pi}{2} - \alpha$

— If $\alpha > \pi/2$, then it doesn't work.

Therefore, we want to have $\omega < \frac{\pi}{2} - \alpha$ and $\omega < \alpha$. It follows that $\omega < \min(\frac{\pi}{2} - \alpha, \alpha)$. Then $\omega \leq \frac{\pi}{4}$.

4.4 Proofs of the theorem of Section ??

In Section ?? we extend the work done in [doumic2016time] to kernels whose support is not necessarily restricted to $[0,1]$. As most of the proofs follow the same scheme as the proofs in the original article, we put them in this appendix.

4.4.1 Proof of Theorem ??

The result is a consequence of the Mellin inversion formula. In order to apply it, we have to prove that for any $\nu \in (p_0, q_0)$,

$$\int_{\nu-i\infty}^{\nu+i\infty} \left| U_0(s) e^{\mu(K(s)-1)t} x^{-s} \right| ds < \infty .$$

We know that for all $s \in (\nu - i\infty, \nu + i\infty)$,

$$\left| U_0(s) e^{\mu(K(s)-1)t} x^{-s} \right| < \left| U_0(s) e^{\mu(K(\nu)-1)t} x^{-\nu} \right|$$

It only remains to prove that

$$\int_{\nu-i\infty}^{\nu+i\infty} |U_0(s)| ds = \int_{-\infty}^{\infty} |U_0(\nu + is)| ds < \infty \quad (4.63)$$

This quantity has already been studied under the same assumptions by Doumic and Escobedo [doumic2016time]. Integration by parts lead to

$$\begin{aligned} U_0(\nu + iv) &= \int_0^\infty u_0(x) x^{\nu-1} x^{iv} dx \\ &= \frac{1}{iv+1} [u_0(x) x^\nu x^{iv}]_0^\infty - \frac{1}{iv+1} \int_0^\infty (u_0(x) x^{\nu-1})' x^{iv+1} dx \\ &\quad . \end{aligned}$$

Assumption (??) guarantees that $[u_0(x)x^\nu x^{iv}]_0^\infty = 0$. Therefore

$$\begin{aligned} U_0(\nu + iv) &= -\frac{1}{iv + 1} \int_0^\infty (u_0(x)x^{\nu-1})' x^{iv+1} dx \\ &= -\frac{\nu-1}{iv+1} \int_0^\infty u_0(x)x^{\nu-2} x^{iv+1} dx - \frac{1}{iv+1} \int_0^\infty u_0'(x)x^{\nu-1} x^{iv+1} dx \\ &= -\frac{\nu-1}{iv+1} \int_0^\infty u_0(x)x^{\nu-1} x^{iv} dx - \frac{1}{iv+1} \int_0^\infty u_0'(x)x^\nu x^{iv} dx \\ &= -\frac{\nu-1}{iv+1} U_0(\nu + iv) - \frac{1}{iv+1} \int_0^\infty u_0'(x)x^\nu x^{iv} dx . \end{aligned}$$

It follows that

$$U_0(\nu + iv) = -\frac{1}{(iv + \nu)} \int_0^\infty u_0'(x)x^\nu x^{iv} dx .$$

The same calculations and Assumption(??) allow us to obtain

$$U_0(\nu + iv) = \frac{1}{(iv + \nu)(iv + \nu + 1)} \int_0^\infty u_0''(x)x^{\nu+1} x^{iv} dx .$$

It follow that

$$|U_0(\nu + iv)| \leq \frac{1}{\sqrt{(\nu^2 + v^2)((1 + \nu)^2 + v^2)}} \int_0^\infty |u_0''(x)|x^{\nu+1} dx .$$

We have that $\int_0^\infty |u_0''(x)|x^{\nu+1} dx < \infty$ from Assumption(??). It allows us to deduce (??)

Conversely, if u_0 satisfies (??), (??) and (??), then

$$u \in \mathcal{C}([0, \infty); L^1((1+x)dx)) \cap \mathcal{C}^1((0, \infty); L^1((1+x)dx))$$

where u is defined by (??). By construction, it is a solution of the equation (??).

4.4.2 Proof of Theorem ??

We want to apply the *steepest descent method* when $t \rightarrow \infty$. For this purpose, we split $w(t, x)$ in two parts :

$$\begin{aligned} \omega(t, x) &= \int_{s_+ - i\infty}^{s_+ + i\infty} U_0(s)e^{\phi(s, t, x)} ds \\ &= \int_{s_+ - i\varepsilon}^{s_+ + i\varepsilon} U_0(s)e^{\phi(s, t, x)} ds + \int_{\Re(s)=s_+, |Im(s_+)| \geq \varepsilon} U_0(s)e^{\phi(s, t, x)} ds \end{aligned}$$

We want to show that the "whole mass" comes from the first term. Close to $s_+(t, x)$, we consider the quantity

$$\begin{aligned} \frac{1}{2i\pi} \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} U_0(s) e^{\phi(s,t,x)} ds &= \frac{1}{2i\pi} U_0(s_+(t,x)) \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} e^{\phi(s,t,x)} ds \\ &+ \frac{1}{2i\pi} \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} (U_0(s) - U_0(s_+(t,x))) e^{\phi(s,t,x)} ds \\ &= I_1 + I_2 \end{aligned}$$

4.4.2.0.1 Step 0 – A first lemma:

We have

$$\begin{aligned} K(s) &= K(s_+) + (s - s_+(t,x)) K'(s_+(t,x)) + \frac{(s - s_+(t,x))^2}{2} K''(s_+(t,x)) \\ &+ \frac{(s - s_+(t,x))^3}{6} K'''(c(t,x)) \end{aligned} \quad (4.64)$$

where $s < c(t,x) < s_+(t,x)$.

$$\begin{aligned} |K'''(c)| &= \left| \int_0^\infty (\log z)^3 k_0(z) e^{(c-1)\log(z)} dz \right| \leq \int_0^\infty |(\log z)^3| k_0(z) e^{\Re(c-1)\log(z)} dz \\ &\leq \int_0^\infty |(\log z)^3| k_0(z) e^{(s_+-1)\log(z)} dz < \infty \end{aligned}$$

Given $\varepsilon > 0$, it follows that for all $t > 0$ and for all $x > 0$ such that $|s - s_+| < \varepsilon$,

$$\Re K(s) = K(s_+) + \frac{(s - s_+(t,x))^2}{2} K''(s_+(t,x)) + O(\varepsilon^3). \quad (4.65)$$

and

$$\Re \phi(s, t, x) = \phi(s_+, t, x) + \mu t \left(\frac{(s - s_+(t,x))^2}{2} K''(s_+(t,x)) + O(\varepsilon^3) \right). \quad (4.66)$$

4.4.2.0.2 First step – Study of I_1 :

For any $s \in (s_+ - i\infty, s_+ + i\infty)$ such that $|s - s_+| < \varepsilon$. We know that

$$\Re \phi(s, t, x) = \phi(s_+, t, x) + t \left(\frac{(s - s_+(t,x))^2}{2} K''(s_+(t,x)) + O(\varepsilon^3) \right). \quad (4.67)$$

Then

$$\begin{aligned} \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} e^{\Re \phi(s,t,x)} ds &= e^{\phi(s_+,t,x)} \cdot \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} e^{t \left(\frac{(s-s_+(t,x))^2}{2} K''(s_+(t,x)) + O(\varepsilon^3) \right)} ds \\ &= i e^{\phi(s_+,t,x)} \cdot \int_{-\varepsilon}^{\varepsilon} e^{t \left(\frac{(s-s_+(t,x))^2}{2} K''(s_+(t,x)) + O(\varepsilon^3) \right)} ds \\ &= i e^{\phi(s_+,t,x)} \cdot \int_{-\varepsilon}^{\varepsilon} e^{-t \left(\frac{v^2}{2} K''(s_+(t,x)) + O(\varepsilon^3) \right)} ds \end{aligned}$$

Now we can see that we are back to the classical method of Laplace [**laplace1986memoir**]. The main idea is to transform the previous integral into a Gauss integral to have

$$\int_{-\varepsilon}^{\varepsilon} e^{-t\left(\frac{v^2}{2}K''(s_+(t,x))+O(\varepsilon^3)\right)} ds \sim_{t \rightarrow \infty} \sqrt{2\pi} \frac{1}{\sqrt{tK''(s_+)}} \quad (4.68)$$

Indeed,

$$\int_{-\varepsilon}^{\varepsilon} e^{-t\left(\frac{v^2}{2}K''(s_+(t,x))+O(\varepsilon^3)\right)} = \frac{1}{\sqrt{tK''(s_+)}} \int_{-\varepsilon\sqrt{tK''(s_+)}}^{\varepsilon\sqrt{tK''(s_+)}} e^{-\frac{y^2}{2}} dy$$

and

$$\int_{-\varepsilon\sqrt{tK''(s_+)}}^{\varepsilon\sqrt{tK''(s_+)}} e^{-\frac{y^2}{2}} dy = \sqrt{2\pi} - 2e^{-\frac{t(\varepsilon^2 K''(s_+)+O(\varepsilon^3))}{2}} \cdot \mathcal{O}\left(\frac{1}{\varepsilon\sqrt{tK''(s_+)}}\right).$$

It follows that

$$\begin{aligned} & \frac{1}{2i\pi} \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} e^{\Re\phi(s,t,x)} ds \\ &= \frac{1}{2i\pi} i e^{\phi(s_+,t,x)} \cdot \int_{-\varepsilon}^{\varepsilon} e^{-t\left(\frac{v^2}{2}K''(s_+(t,x))+O(\varepsilon^3)\right)} ds \\ &= \frac{1}{2\pi} e^{\phi(s_+,t,x)} \cdot \frac{1}{\sqrt{tK''(s_+)}} \int_{-\varepsilon\sqrt{tK''(s_+)}}^{\varepsilon\sqrt{tK''(s_+)}} e^{-\frac{y^2}{2}} dy \\ &= \frac{1}{2\pi} e^{\phi(s_+,t,x)} \cdot \frac{1}{\sqrt{tK''(s_+)}} \left(\sqrt{2\pi} - 2e^{-\frac{t(\varepsilon^2 K''(s_+)+O(\varepsilon^3))}{2}} \cdot \mathcal{O}\left(\frac{1}{\varepsilon\sqrt{tK''(s_+)}}\right) \right) \end{aligned}$$

Therefore,

$$I_1 = \frac{U_0(s_+(t,x))}{\sqrt{2\pi}} e^{\phi(s_+,t,x)} \cdot \frac{1}{\sqrt{t\mu K''(s_+)}} \left(1 + \mathcal{O}\left(\frac{e^{-\frac{t(\varepsilon^2 \mu K''(s_+))}{2}}}{\varepsilon\sqrt{t\mu K''(s_+)}}\right) \right) \quad (4.69)$$

4.4.2.0.3 Second step – Study of I_2 :

We have

$$I_2 = \frac{1}{2i\pi} \int_{s_+(t,x)-i\varepsilon}^{s_+(t,x)+i\varepsilon} (U_0(s) - U_0(s_+(t,x))) e^{\phi(s,t,x)} ds$$

Here, we want to go back more or less to the previous integral

$$\begin{aligned}
|I_2| &= \frac{1}{2i\pi} \sup_{|v| \leq \varepsilon} [U_0(s_+ + iv) - U_0(s_+)] \int_{s_+(t,x) - i\varepsilon}^{s_+(t,x) + i\varepsilon} |e^{\phi(s,t,x)}| ds \\
&= \sup_{|v| \leq \varepsilon} [U_0(s_+ + iv) - U_0(s_+)] \frac{1}{\sqrt{2\pi}} e^{\phi(s_+,t,x)} \cdot \frac{1}{\sqrt{t\mu K''(s_+)}} \left(1 + \mathcal{O}\left(\frac{e^{-\frac{t(\varepsilon^2 \mu K''(s_+))}{2}}}{\varepsilon \sqrt{\mu t K''(s_+)}} \right) \right)
\end{aligned}$$

It only remains to study $\sup_{|v| \leq \varepsilon} [U_0(s_+ + iv) - U_0(s_+)]$

By Definition

$$\begin{aligned}
[U_0(s_+ + iv) - U_0(s_+)] &= \left| \int_0^\infty z^{s_+ - 1} u_0(z) (z^{iv} - 1) dz \right| \\
&\leq \int_0^{\frac{1}{R}} |z^{s_+ - 1} u_0(z) (z^{iv} - 1)| dz + \int_{\frac{1}{R}}^R |z^{s_+ - 1} u_0(z) (z^{iv} - 1)| dz \\
&\quad + \int_R^{+\infty} |z^{s_+ - 1} u_0(z) (z^{iv} - 1)| dz
\end{aligned}$$

If $R \rightarrow +\infty$, then the first and the third terms vanish. Therefore, we only need to consider the integral $\int_{\frac{1}{R}}^R |z^{s_+ - 1} u_0(z) (z^{iv} - 1)| dz$. We have

$$\begin{aligned}
\int_{\frac{1}{R}}^R |z^{s_+ - 1} u_0(z) (z^{iv} - 1)| dz &\leq (\varepsilon \log R) \int_{\frac{1}{R}}^R |z^{s_+ - 1} u_0(z)| dz \\
&\leq (\varepsilon \log R) \int_0^\infty z^{s_+ - 1} u_0(z) dz
\end{aligned}$$

4.4.2.0.4 Third step – A second lemma:

Outside the neighborhood of $s_+(t, x) : |s - s_+(t, x)| < \varepsilon$. We have

$$\begin{aligned}
\Re(\phi(s_+ + i\nu, t, x) - \phi(s_+, t, x)) &= \Re\left(\int_0^\infty x^{s_+ - 1} k_0(x) (x^{i\nu} - 1) dx \right) \\
&= \Re\left(\int_0^\infty x^{s_+ - 1} k_0(x) (e^{i\nu \log(x)} - 1) dx \right) \\
&= \int_0^\infty x^{s_+ - 1} k_0(x) [\cos(\nu \log(x)) - 1] dx \\
&=: G(\nu) \leq 0.
\end{aligned}$$

For any $\nu \in \mathbb{R} \setminus \{0\}$, we have

$$\int_0^\infty x^{s_+ - 1} k_0(x) [\cos(\nu \log(x))] dx < \int_0^\infty x^{s_+ - 1} k_0(x) dx. \quad (4.70)$$

It follows that $G(\nu) < 0$, then there exists $\varepsilon > 0$, $R > 0$ and $\delta > 0$ such that $\sup_{\varepsilon < \nu < R} G(\nu) < -\delta$.

It remains to verify that $\lim_{\nu \rightarrow +\infty} G(\nu) \neq 0$.

$$\begin{aligned} G(\nu) &= \int_0^\infty x^{s_+-1} k_0(x) [\cos(\nu \log(x)) - 1] dx \\ &= \int_{-\infty}^\infty e^{-s_+y} k_0(e^{-y}) (\cos(-\nu y) - 1) dy \end{aligned}$$

From Riemann–Lebesgue lemma, we have that

$$\lim_{\nu \rightarrow +\infty} G(\nu) = - \int_{-\infty}^\infty e^{-s_+y} k_0(e^{-y}) dy = - \int_0^\infty x^{s_+-1} k_0(x) dx < 0 \quad (4.71)$$

Therefore

Lemme 42. *For all $\varepsilon > 0$, there exist $\gamma(\varepsilon)$ such*

$$\sup_{|\nu| \geq \varepsilon} \Re(\phi(s_+ + i\nu, t, x)) \leq \Re(\phi(s_+, t, x)) - \gamma(\varepsilon)t. \quad (4.72)$$

and

$$\sup_{|\nu| \leq \varepsilon} \Re(\phi(s_+ + i\nu, t, x)) \leq \Re(\phi(s_+, t, x)). \quad (4.73)$$

4.4.2.0.5 Fourth step – Control $\int_{\Re(s)=s_+, |Im(s_+)| \geq \varepsilon} U_0(s) e^{\phi(s,t,x)} ds$:

We have that

$$\int_{\Re(s)=s_+, |Im(s_+)| \geq \varepsilon} U_0(s) e^{\phi(s,t,x)} ds \leq e^{\Re(\phi(s_+, t, x)) - \gamma(\varepsilon)t} \int_{\Re(s)=s_+, |Im(s_+)| \geq \varepsilon} U_0(s) ds$$

It follows that

$$\int_{\Re(s)=s_+, |Im(s_+)| \geq \varepsilon} U_0(s) e^{\phi(s,t,x)} ds = \mathcal{O}\left(e^{\Re(\phi(s_+, t, x))}\right)$$

Part II

Data Analysis

Chapter 5

Application of the decomposing method

Chapter 6

Multimodality

Multimodality in a random sample refers to the presence of multiple local maxima or modes in the distribution of the sample data. Detecting multimodality is important because it suggests that the data might be better understood by considering multiple underlying groups or processes rather than a single, homogeneous population. For example, using a single mean to summarize a multimodal dataset might be misleading, as it does not capture the distinct groups within the data.

There is several reasons that may explain multimodality in a dataset. The following is a non-exhaustive list:

- Identifying multimodality can lead to the discovery of subgroups with different characteristics. [\[citer source\]](#)
- Seasonality. [\[citer source\]](#)

In this chapter, we suppose that we observe a n -sample $(X_i, i = 0 \cdot n)$ assume to be *i.i.d.* from an unknown probability density function f . We are interested in non-parametric tests which decide between the two hypothesis:

$$H_0 : f \text{ is unimodal} \quad \text{vs.} \quad H_1 : f \text{ is multimodal.}$$

In Section ??, we present several classical tests for multimodality. In Section ?? we propose a procedure to detect multimodality with observations coming from compound poisson processes.

6.1 Classical tests for multimodality

6.1.1 The Silverman test

6.1.1.1 Presentation of the test

The Silverman's test is a method commonly used to assess the number of modes or peaks in a density estimate [23, 22]. It is based on kernel estimators, due to Rosenblatt [20] and Parzen [17], which are classical estimators for approximating the probability density of a random variable whose n sample (X_i) *i.i.d.* is observed.

The kernel estimator is defined as

$$\hat{f}_n(x, h) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where $K(u) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}u^2)$ and $h \in (0, \infty)$.

The choice of h is a difficult one and must be made by the statistician. The number of modes of the estimator $\hat{f}_n$ depends on the size of h . If h is very large, then the estimator has only one mode, whereas when h is too small, we get a very spiky and over-fit estimate of the PDF. (See Fig ??)

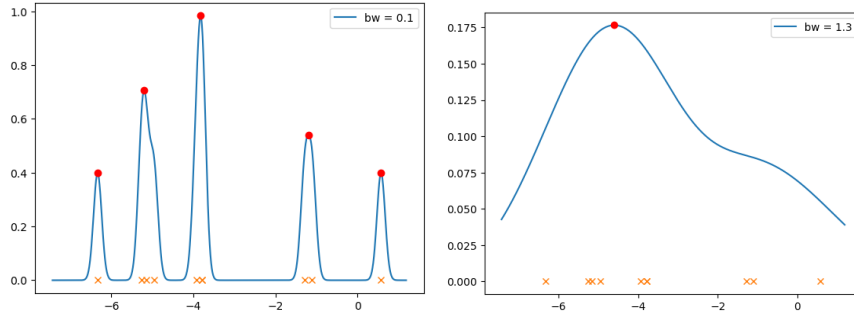


Figure 6.1 – (Left): h is too small (Right): h is too large

In his paper, Silverman proves that using a Gaussian kernel ensures that decreasing h will only introduce new local maxima [23]. Consequently, it is possible to find the minimum width h_k such that $\hat{f}_n(\cdot, h_k)$ has at most k maxima for any integer $k \in \{1, \dots, n\}$.

The estimator $\hat{f}_n(\cdot, h_k)$ form a family of kernel estimators, each with at most k modes. In his paper, Silverman proposes a procedure for determining the significance of each of them.

To do this, he uses a bootstrap method. The principle consists to evaluate the accuracy of the estimator by computing it on observations that are resampled from the original set of observations.

Methodology to evaluate the accuracy of the estimator $\hat{f}(\cdot, h_k)$

1. First, draw a n -sample $X_{I(i)}$ with replacement from the the original set of observations
2. Add some noise

$$y_i = \frac{1}{1 + h_k^2/\sigma^2} (X_{I(i)} + h_k \varepsilon_i)$$

where σ is the standard deviation of $X_1, \dots, X_n$, h_k is the critical width we are testing, and ε_i is a random value sampled from a normal distribution with mean 0 and standard deviation 1.

3. Compute $\hat{f}_n(\cdot, h_k)$ for sample y_i . Check if $\hat{f}_n(\cdot, h_k)$ has less than k -mode.
4. Repeat. By repeating this operation several times, we calculate the p -value as the fraction of simulations where we did not find more than k modes.

Null hypothesis	p-value	bandwidth
Reject 1 mode	p-value 0.99	h_1 1.3595962524414062
Reject 2 mode	p-value 0.48	h_2 0.35981178283691406
Reject 3 mode	p-value 0.75	h_3 0.33596038818359375
Reject 4 mode	p-value 0.45	h_4 0.24213790893554688

Table 6.1 – Result of the Silverman test for the Gaussian mixture with 2 modes

6.1.1.2 Numerical examples

6.1.1.2.1 A Gaussian mixture with 2 modes. We illustrate the Silverman’s test on the Gaussian mixture $0.6\mathcal{N}(-5, 1) + 0.4\mathcal{N}(0, 1)$. In Tab ??, we see that the null hypothesis : *The density is unimodal* can be rejected with a p -value of 0.99.

6.1.2 The DIP test

The DIP test, also known as the Hartigan’s Dip Test, is a statistical test used to assess the unimodality of a data distribution. It measures the departure of the empirical distribution function of a sample from a unimodal distribution. Specifically, the test computes the maximum difference (or "dip") between the empirical distribution and the best-fitting unimodal distribution (defined as the least convex majorant and greatest convex minorant of the empirical distribution). A significant dip value indicates that the sample is not unimodal, suggesting the presence of multiple modes.

A unimodal distribution will have a Probability Density Function (PDF) that increases from 0 to some peak value and then decrease back to 0. When the PDF reaches a local maximum, then the Cumulative Distribution Function (CDF) switches from convex to concave. A multimodal distributions CDF will change from convex to concave several times.

The concept behind the dip statistic is to gauge the extent of adjustment required for the cumulative distribution function (CDF) to become unimodal. Specifically, it represents the greatest distance, observed at any point, between the CDF and the nearest multimodal CDF. To put it differently, we can transform the distribution into a unimodal form by shifting the CDF by no more than the dip value at each instance. Essentially, the dip corresponds to the minimum value needed to achieve this transformation throughout the distribution.

6.1.3 The Excess mass test

6.1.3.1 Presentation of the test

The Excess Mass test is a statistical method used to assess multimodality in a data distribution by comparing the amount of probability mass concentrated in potential modes to what would be expected under unimodality. It has been introduced by Müller and Sawitzki in 1991 [15]. The test evaluates the presence of multiple peaks by measuring the "excess mass," which refers to the extra density that appears around modes beyond what would be expected if the distribution were unimodal.

The Excess Mass test works by identifying regions in the data with significant density peaks and then calculating the total probability mass within these regions. This excess mass is compared to a reference unimodal distribution through optimization, which seeks to maximize the difference between observed density peaks and the unimodal fit. A significant excess mass indicates that the distribution has more than one mode.

The test is flexible and can be adjusted for different modes' widths and shapes, making it suitable for detecting multimodality in complex or irregular distributions. It is particularly useful in settings where traditional methods, like kernel density estimation, might struggle to identify multimodal structures. The results are typically interpreted through p-values or other statistical measures derived from comparisons with reference distributions generated via simulation or other means.

The empirical excess mass for k modes and a constant λ is defined as

$$E_{n,k}(\mathbb{P}, \lambda) = \sup_{C_1(\lambda), \dots, C_k(\lambda)} \left\{ \sum_{m=1}^k \mathbb{P}_n(C_m(\lambda)) - \lambda \|C_m(\lambda)\| \right\}$$

where the supremum is taken over all families $\{C_m(\lambda) : m = 1, \dots, k\}$ of closed intervals with endpoints at data points. $\|C_m\|$ denotes the measure of $C_m(\lambda)$ and

$$\mathbb{P}_n(C_m(\lambda)) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i \in C_m(\lambda)}$$

The key idea is that the difference $D_{n,k+1}(\lambda) = E_{n,k+1}(\mathbb{P}, \lambda) - E_{n,k}(\mathbb{P}, \lambda)$ represents the plausibility of the null hypothesis. The larger $D_{n,k+1}(\lambda)$ is, the more likely we are to reject H_0 .

The excess mass statistic is defined as

$$\Delta_{n,k+1} = \max_{\lambda} \{D_{n,k+1}(\lambda)\}$$

6.1.3.2 Numerical examples

6.2 An application to the biological problem

In this section, we apply

6.2.1 Simulation without noise

6.2.2 Simulation with noise

6.2.3 Real dataset

Part III

Appendix

Appendix A

Mathematic Tools

A.1 Nonparametric estimation

A.2 Statistical tests

A.3 Integral transformations

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Abstract

French version

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Abstract

English version

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